

Physiologic changes during pregnancy

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Learning objectives

1. Discriminating between normal pregnancy symptoms and those related to pathological conditions.
2. Understanding the effect of pregnancy on pre-existing diseases e.g. diabetes and heart diseases.

Genital organs

- A) The Uterus:

? **The Height and Weight :** The height increases to reach 35 cm at ,and weight increases to reach 1000 gm at term, due to:

- Oestrogen and progesterone leading to myometrial hypertrophy and hyperplasia. - The contents of the uterus i.e. foetus, placenta and amniotic fluid ..etc.

? **The Shape:** The shape of the uterus is globular till 14 weeks then it becomes pyriform.

? **The uterine ligaments:** Show hypertrophy.

? **Dextro-rotation:** The uterus is tilted and twisted to the right in 80% of cases.

? **Braxton-Hicks contractions:** Irregular, painless con-tractions that help placental circulation.

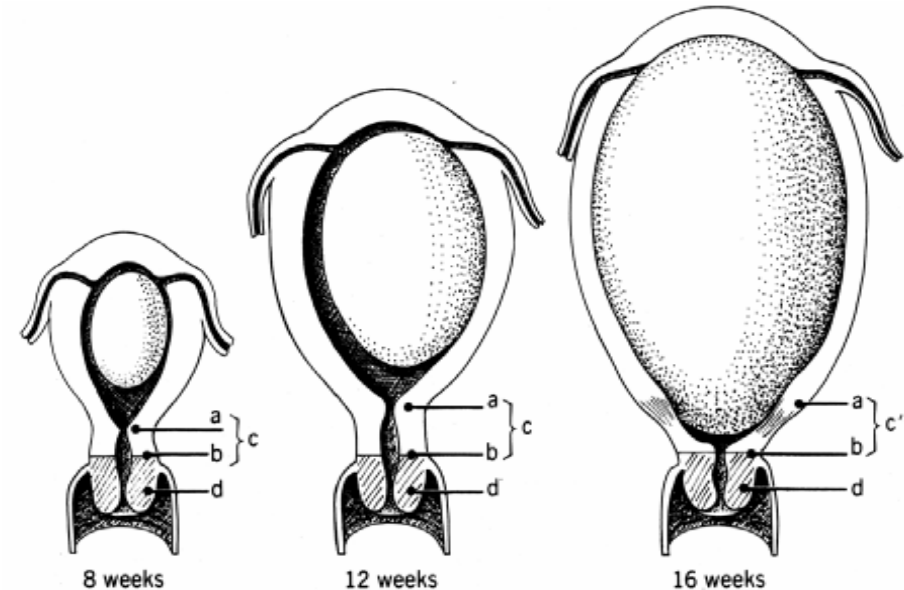
The Lower Uterine Segment (LUS) :

It is formed from the isthmus

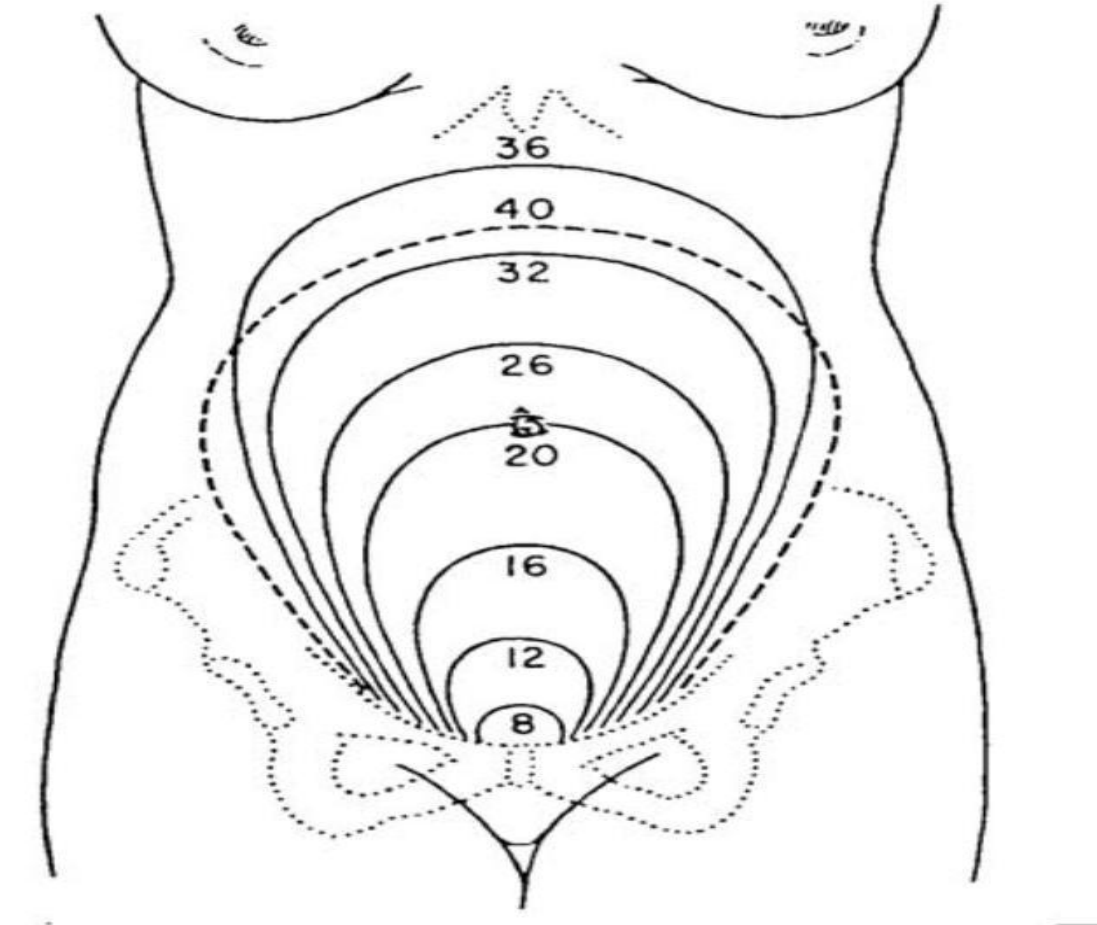
It starts to be formed from the 4th month, to reach 10 cm at full term.

Obstetric significance of the LUS:

1. Site of lower segment cesarean section (LSCS)
2. Site of rupture in obstructed labor.
3. Site of implantation in cases of placenta previa.



- a. anatomical internal os
- b. histological internal os
- c. isthmus
- c'. former isthmus
- d. cervix



Fundus levels during pregnancy

- **DIFFERENCES BETWEEN THE UPPER AND LOWER UTERINE SEGMENTS:**

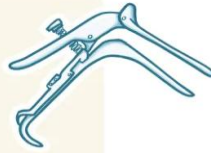
Upper Uterine Segment	Lower uterine segment
- Active during labor - Contracts and retracts to become shorter and thicker. - Thick wall: It consists of 3 muscle layers (outer longitudinal- middle oblique- inner circular) - Covered by adherent peritoneum - The membranes are firmly attached	- Passive during labor - Dilates and stretches to be longer and thinner -thin wall: It consists of 2 muscle layer (outer longitudinal, inner circular) - Covered by loose peritoneum -The membranes are loosely attached

3) Cervix:

- ❑ Oedema and increased vascularity with glandular hypertrophy; it becomes soft "**Goodell's sign**" and bluish "**Chadwick sign**".
- ❑ Thick Mucus plug, which consists of the cervical mucus closing the cervical canal.
- ❑ Hormonal erosion may occur, it disappears spontaneously 3-6 months after labour.
- ❑ Prostaglandins make the cervix softer and more easily dilatable in the last weeks of pregnancy (Cervical ripening).

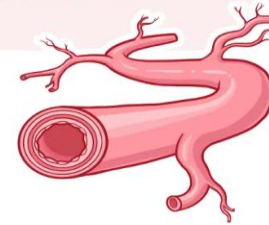
BACKGROUND

- * **CLINICAL INDICATION of EARLY PREGNANCY**
~ NOT DEFINITIVE
- * **SOFTENING of CERVIX** in FIRST 4-8wks



CAUSES

- * **HYPERTROPHY of EXISTING BLOOD VESSELS**
~ ENLARGEMENT
- * **GROWTH of NEW BLOOD VESSELS**
- * **EDEMA** due to ↑↑ BLOOD CIRCULATION



NORMAL



GOODELL SIGN



CHADWICK SIGN
~ BLUE / PURPLE

SIGNS & SYMPTOMS

- * **SOFTENING of CERVIX**
~ MAY PRESENT with CHADWICK SIGN & HEGAR SIGN



HEGAR SIGN

C) The Vulva:

☐ shows increased vascularity and varicosities may develop.

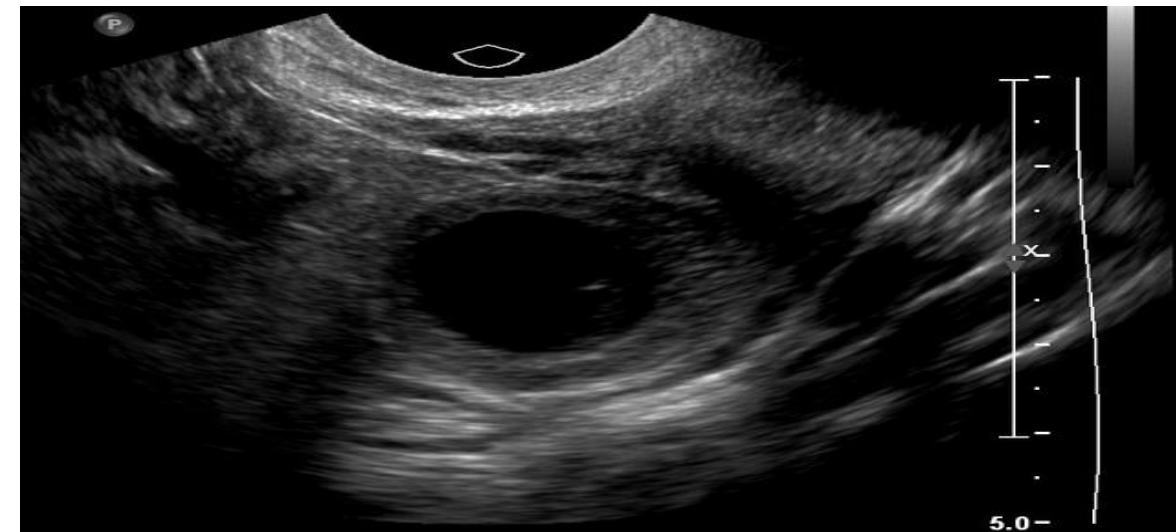
D) The Vagina:

☐ shows increased vascularity which makes it soft, moist and bluish.

C) The ovaries:

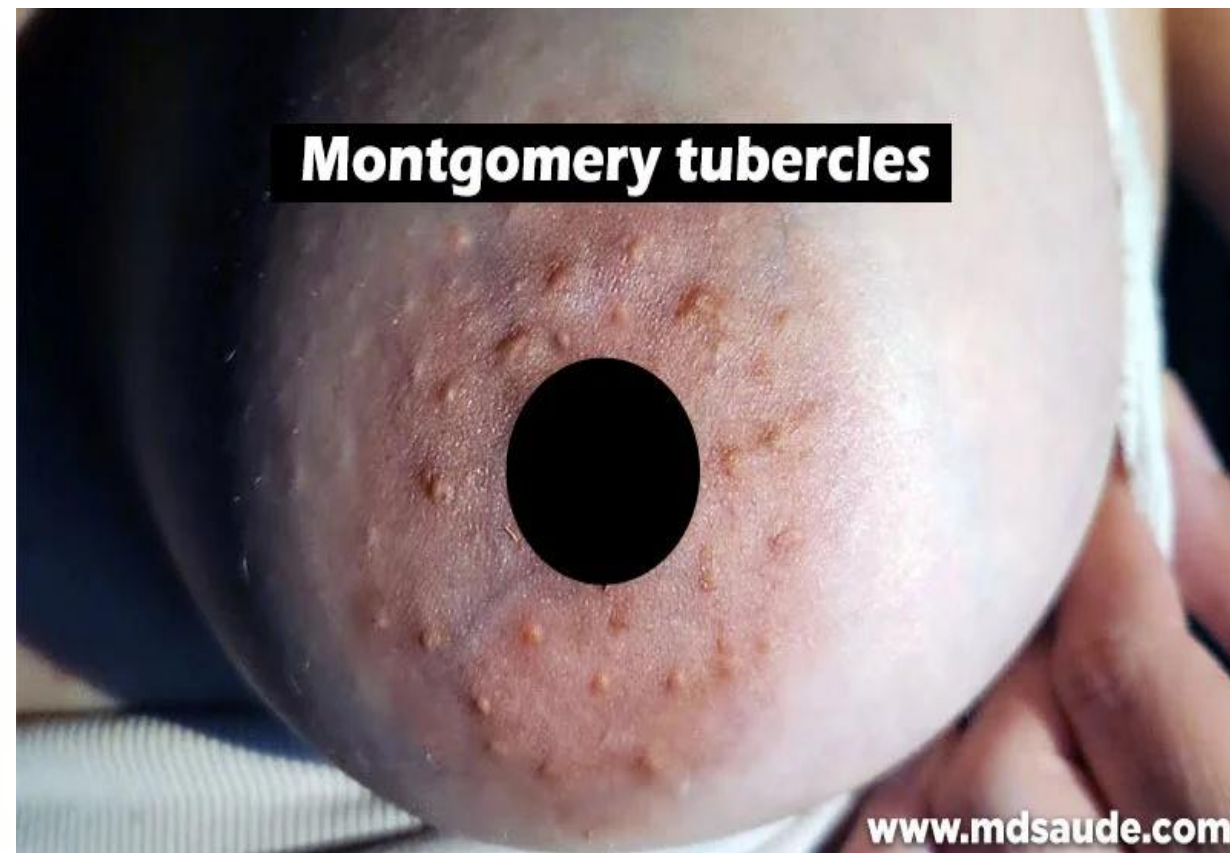
☐ shows increased vascularity and oedema. Theca lutein cysts may occur.

One of the ovaries contains the corpus luteum,
which supports pregnancy till placenta is formed.



Breast changes

- ❑ Breast growth is under the effect of oestrogen (E) and progesterone (P). Breast changes are only diagnostic in primigravidas.
- ❑ Increased size and vascularity (warm, tense and tender).
- ❑ Increased pigmentation of the nipple and areola.
- ❑ 2ry areola appears (light pigmentation around the 1ry areola).
- ❑ Montgomery tubercles appear on the areola (dilated enlarged sebaceous glands but some believe that they aberrant rudimentary lactiferous glands).
- ❑ Colostrum-like fluid can be expressed from the end of the 3rd month.



Breast changes

Cutaneous changes

A. Pigmentation: caused by increased Melanocyte Stimulating Hormone (MSH + E):

☐ Linea nigra: Pigmentation of the Linea alba, more marked below the umbilicus.

☐ Chloasma gravidarum: Butterfly pigmentation of the face

B. Striae gravidarum "stretch marks":

These are pink lines in the flanks due to rupture of the subcutaneous elastic fibres as a result of stretch of the abdominal wall during pregnancy.

After labour, the colour turns white due to fibrosis "striae albicans".



Linea nigra



Stria gravidarum

Metabolic changes

A. Protein metabolism: (anabolic state)

Tendency to nitrogen retention, with increased binding globulins produced by the liver.

B. Carbohydrates metabolism: (diabetogenic state)

? **Progressive increase in insulin requirements** (more marked in 2nd and 3rd trimesters) due to increased insulin antagonists ;

1. Placental hormones; human placental lactogen (HPL), oestrogen (E), and progesterone (P).
2. Production of insulinase enzyme by the placenta

? **Alimentary glycosuria** due to rapid absorption of glucose from the GIT.

? **Renal glycosuria** due to lowering of the renal threshold (increase loss of glucose in urine).

? **Lowered fasting blood glucose** due to feto-placental consumption (fasting hypoglycaemia).

C. Fat metabolism: disturbed secondary to disturbed CHO metabolism

Increased lipolysis to deliver more free fatty acids to the foetus

D. Minerals:

Increased requirements of iron, calcium, phosphorus and iodides.

E. Salt and water retention: occurs due to the effect of oestrogen and progesterone.

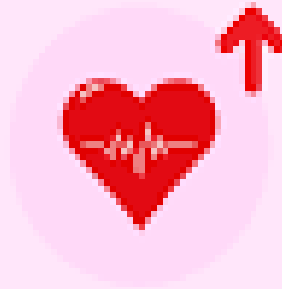
F. Weight:

The average total weight gain is 8-13 kg, most of it occurs in the 3rd trimester.

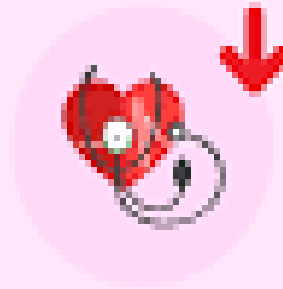
- ❓ **Physiological Haemodilution (physiological anaemia):** Total blood volume increases by 50% maximally between 28-32 weeks. The increase is more in Plasma volume than RBCs.
- ❓ **The blood pressure decreases** (due to opening of arterio-venous shunts at the placenta). This occurs maximally at mid trimester and returns gradually to normal during 3rd trimester.
- ❓ **The Cardiac Output increases** gradually to reach of a peak of 40% increase at about 20 weeks and remains constant. Increased CO is due to an both increased stroke volume (40- 50%) and increase in the heart rate (15%).
- ❓ **The apex of heart is shifted upwards and outwards** (4th intercostal space), due to elevation of the diaphragm by the fundus of the growing uterus.
- ❓ **Functional murmurs** could be detected and are usually systolic.
- ❓ **ECG changes:** Flat T wave and Inverted ST segment in some leads due to change in cardiac axis.



Increased
blood volume



Increased
heart rate



Decrease in
blood pressure



Hypercoagulability

Cardiovascular changes during pregnancy

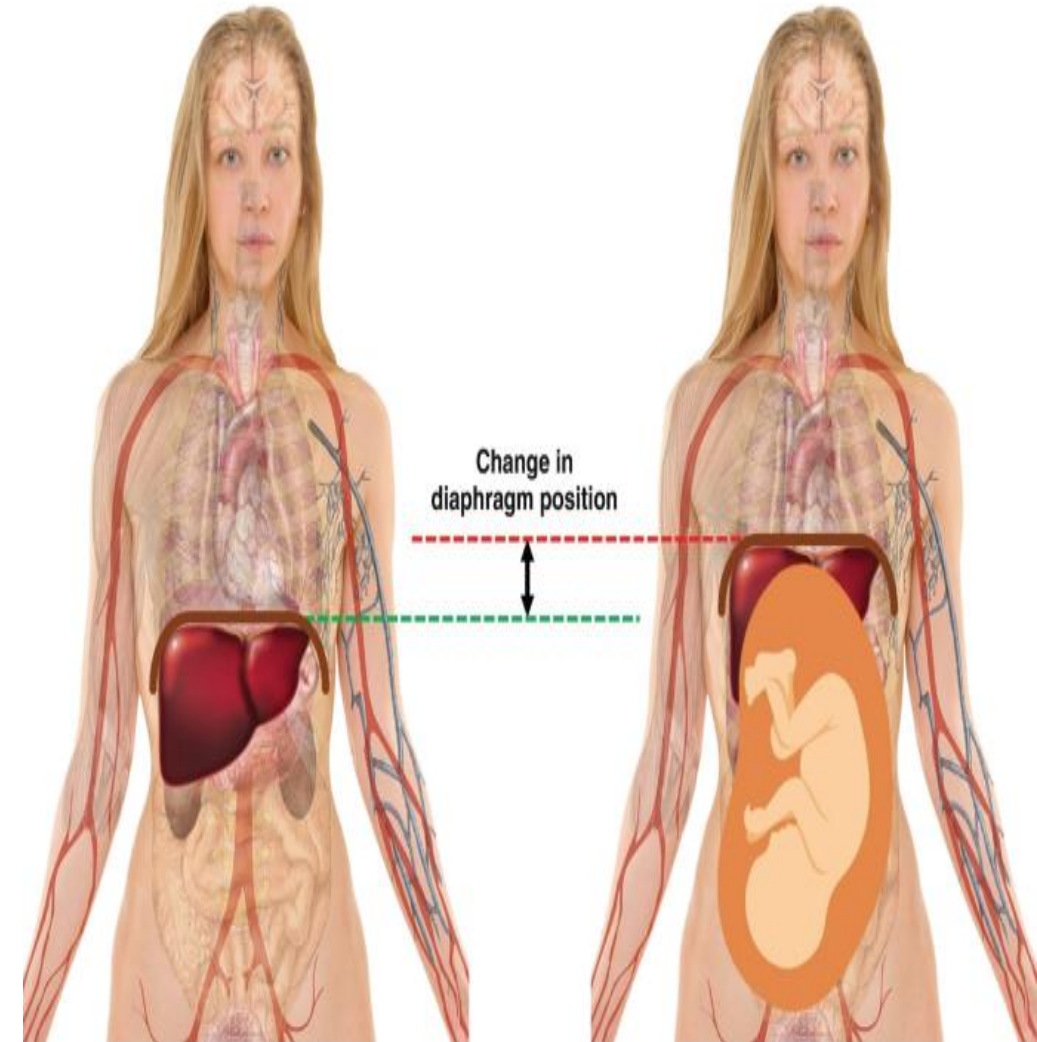
? *Pregnancy is a hypercoagulable state with 5 fold increased risk of thromboembolism due to:*

- Increased fibrinogen level, factor VII, VIII, IX
- Increased platelet activation
- Venous stasis due to pressure of the gravid uterus on the pelvic veins (leading to ankle oedema, varicose veins and piles).

Respiratory changes

❓ **Dyspnoea is common** due to:

- Stimulatory effect of progesterone on respiratory center.
- Elevation of the diaphragm by the gravid uterus.

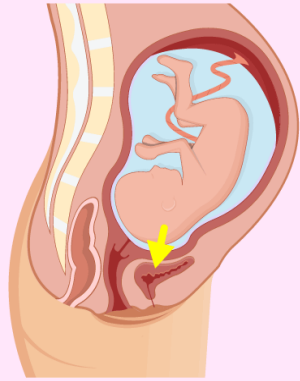


• Urinary changes

❓ **Frequency of micturition:** due to congestion and pressure of the pregnant uterus on the urinary bladder in the 1st trimester and again in late pregnancy due to pressure by fetal head.

❓ **Tendency to urinary stasis with Dilatation of the ureters** above the pelvic brim (mainly right side due to dextrorotation of the uterus), due to:

- Pressure against the pelvic brim by the gravid uterus (lead to stasis of urine that predispose to urinary tract infection)
- Hypotonia of the ureteric muscles caused by progesterone and relaxin.
- Hypertrophy of the wall of the lower end of the ureter caused by oestrogen.



Pressure on
the bladder

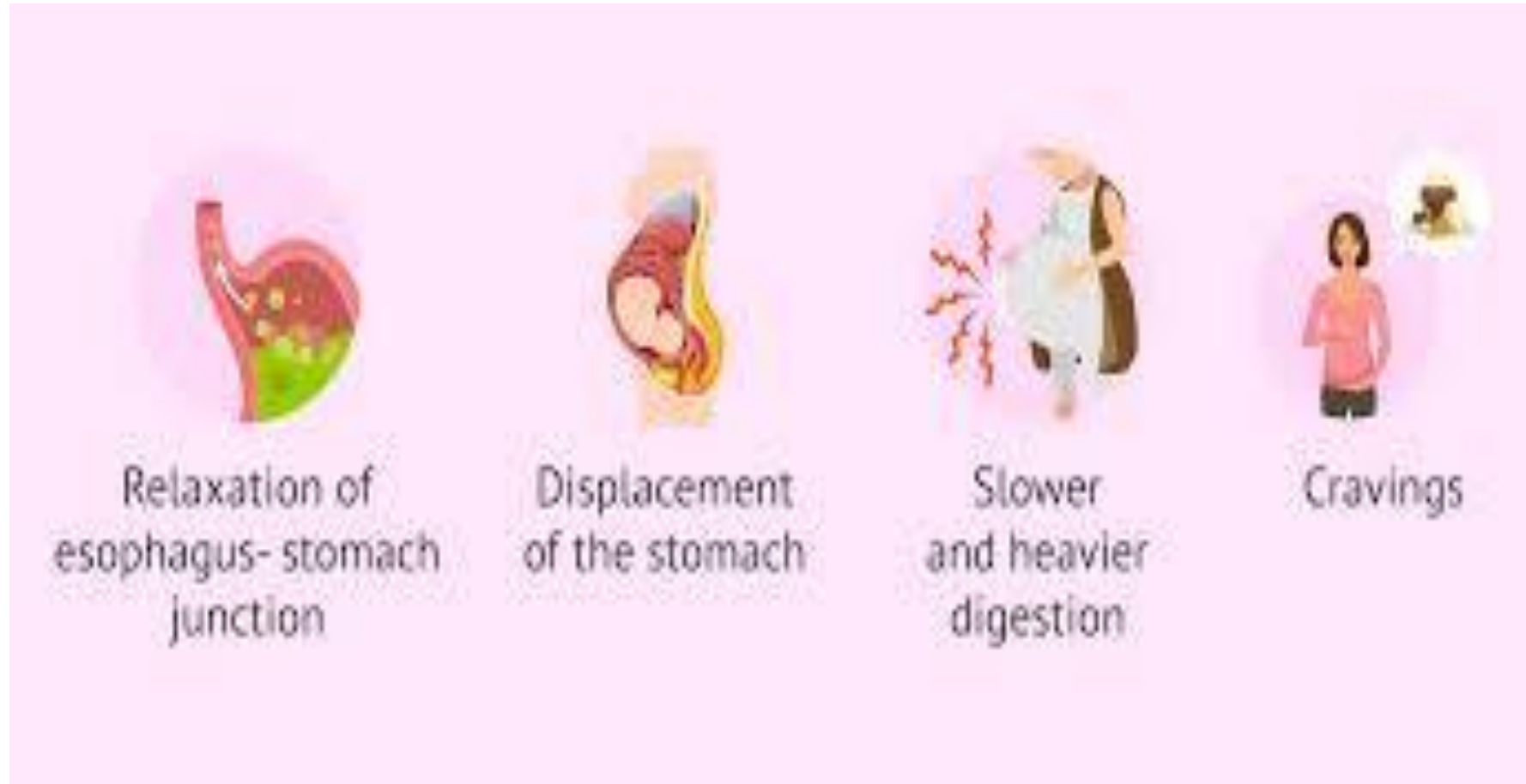


Increased risk
of UTIs

Urinary changes during pregnancy

• Gastrointestinal changes

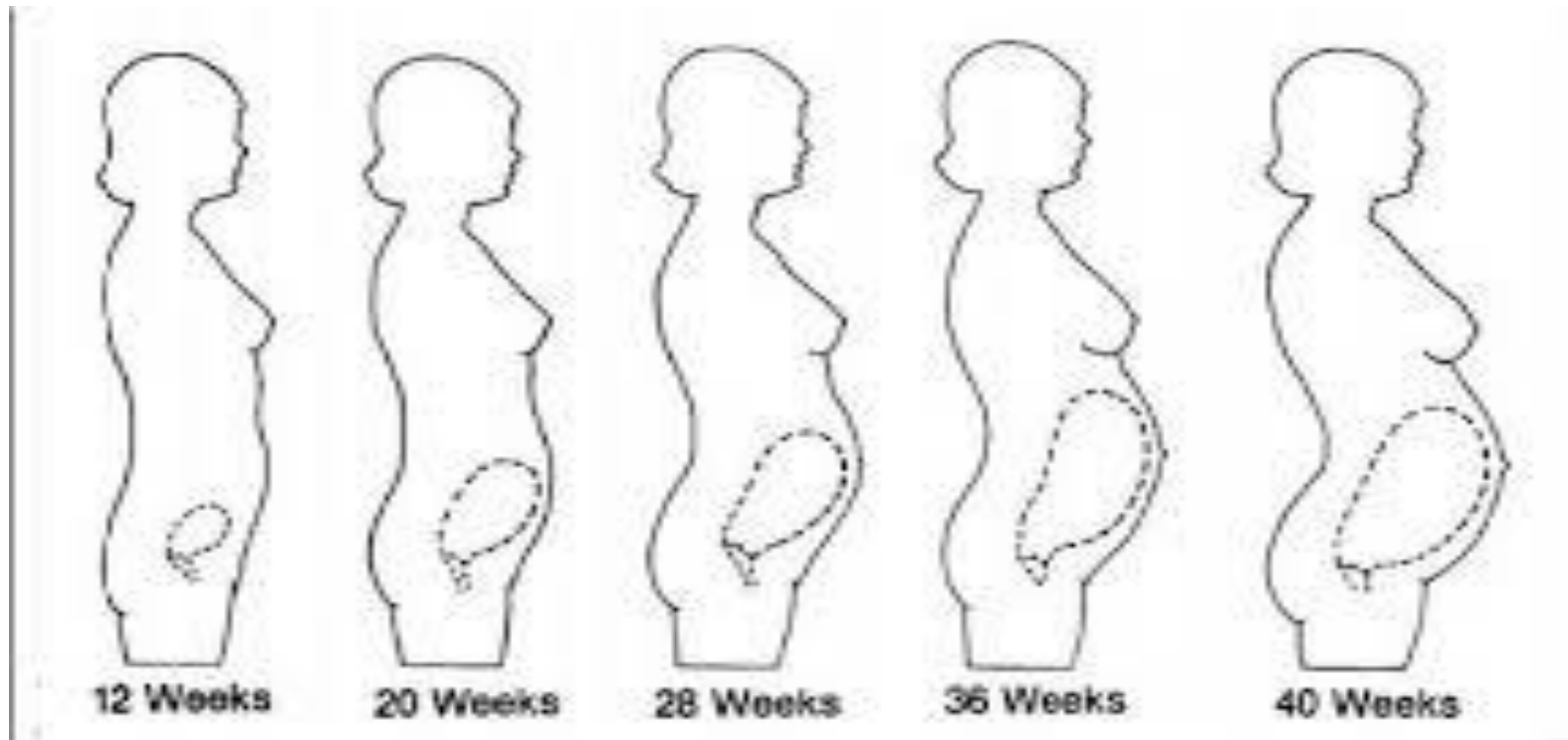
- ❑ Emesis gravidarum "morning sickness" in early pregnancy.
- ❑ Increased salivation (ptyalism), more acidic with increased teeth decay.
- ❑ Heart burn "reflux esophagitis" caused by relaxation of the cardiac sphincter (gastroesophageal regurge) due to the effect of progesterone and relaxin.
- ❑ Decreased gastric acidity.
- ❑ Constipation and flatulence due to reduced gut motility caused by progesterone.



Gastrointestinal changes during pregnancy

• Skeletal changes

- ☐ Increased lumbar lordosis may cause low back pain.
- ☐ Softening of pelvic joints and ligaments due to progesterone and relaxin, may cause pelvic pains.



• Endocrinal changes

? **The Pituitary gland:** increases in size and activity, while the posterior pituitary releases oxytocin at the onset of labour.

? **The Thyroid gland:**

- Slight enlargement with increased activity (physiological goitre).
- Total T3 and T4 increase but the free forms remain unchanged.
- Basal metabolic rate may increase by 30%.
- Thyroid binding globulin increases.

? **The Parathyroid gland:** increases in size and activity to regulate calcium metabolism.

? **The Suprarenal gland:** increases in size and activity with Cortisol increase,



Title: -Obstetric examination & diagnosis of pregnancy

DR: - Marwa Abdalla
OB/GYN Department

- Learning objectives

1. To know and understand obstetric examination for proper assessment of cases.
2. To know how to diagnose pregnancy.



Obstetric Examination

➤ **General Examination:**

- Vitals: pulse, BP, temp
- Gait, weight, height, BMI
- Hair distribution and development of the secondary sexual characters.
- Breast examination
- Inspection of the back: spina bifida

➤ **Abdominal examination:** bladder should be empty

Obstetric Examination

➤ **Abdominal examination:** bladder should be empty

A. Fundal level:

Manual method; using the ulnar border of the left hand, the fundal level correlated with duration of pregnancy in relation to symphysis pubis (SP), umbilicus, and xiphisternum (XS):

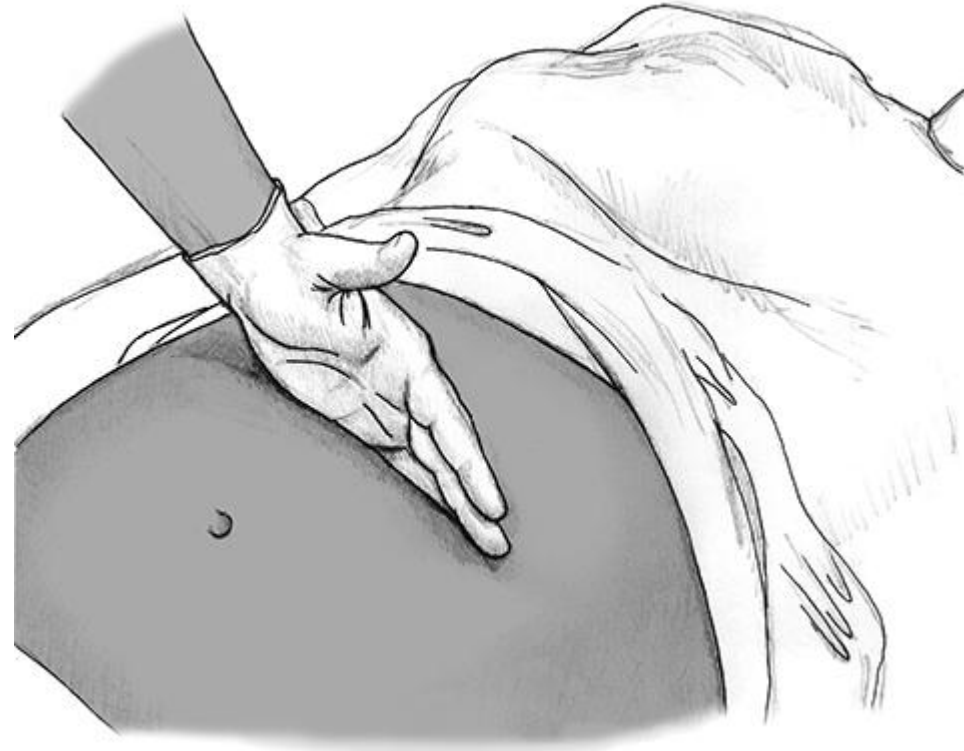
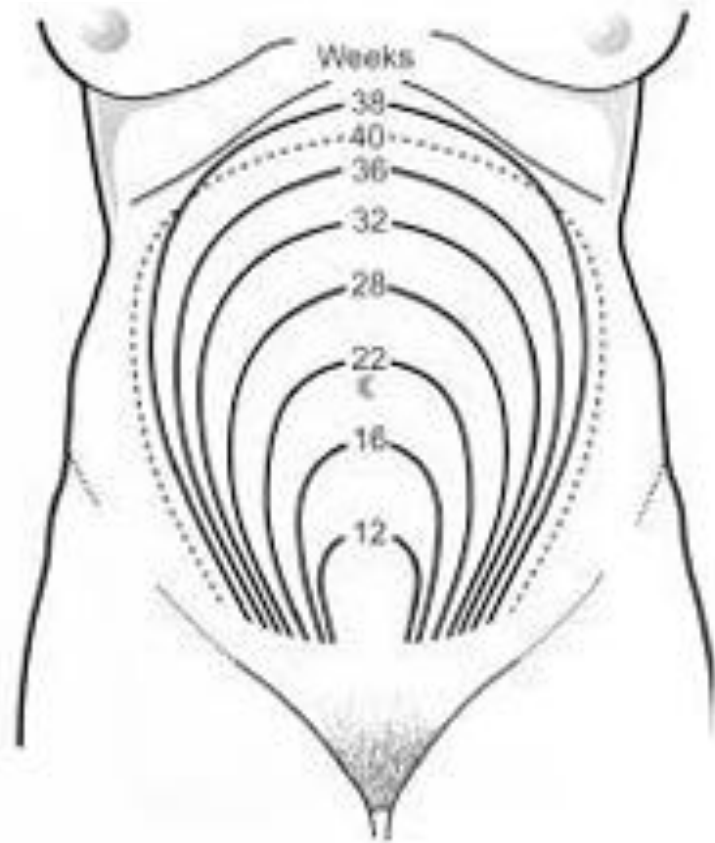
- At 12 weeks; fundal level is at the upper border of the symphysis pubis
- At 20-22 weeks: fundal level is at the umbilicus
- At 36 weeks: fundal level is at the XS.

N.B.: after engagement, the fundal level descends below the XS.



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Obstetric Examination

➤ **Abdominal examination:** bladder should be empty

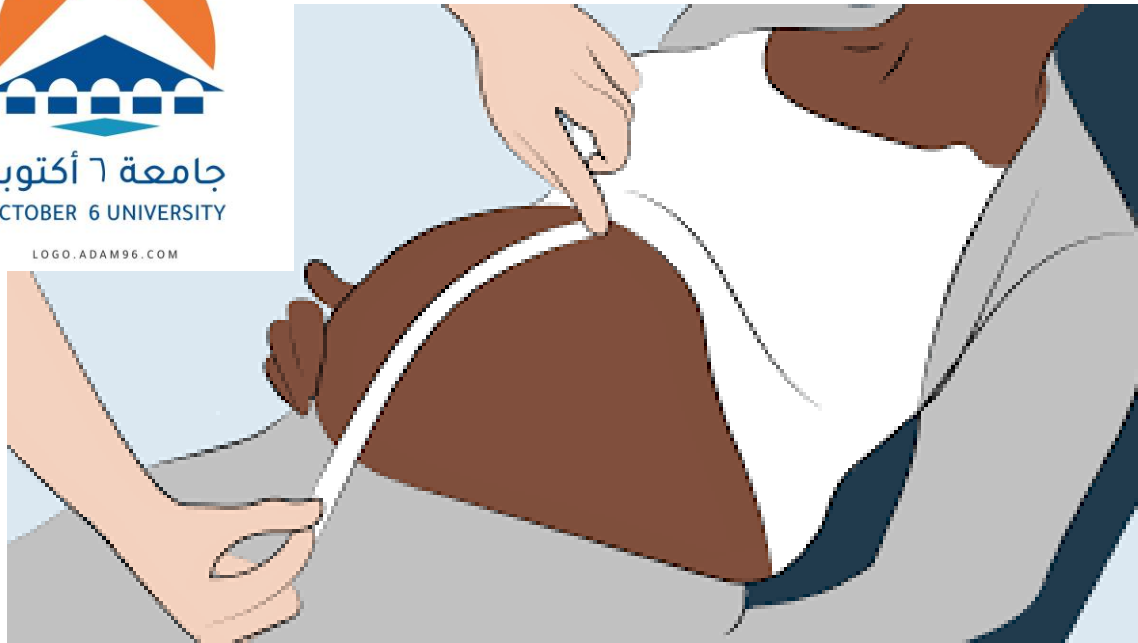
- A. Fundal level:

- ✓ Metric method: the distance between the symphysis pubis and fundal level, in cm, correlates with the duration of gestation.



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Obstetric Examination

B. LEOPOLD MANOEUVRES:

- ✓ Fundal grip: to determine the foetal pole occupying the uterine fundus. Normally the fundus contains the foetal breech near term.
- ✓ Umbilical grip: to determine the direction of the foetal back and spine; - Right or left: in relation to maternal midline - Anterior or posterior: in relation to the back and spine of the mother - Presence of a foetal head at either side denotes transverse lie
- ✓ 1st pelvic grip: gripping the lower uterine segment (LUS) above the SP helps to define; - Cephalic presentations: head is felt small, globular, hard, ballotable and tender. - Breech presentation: breech is felt large, soft, not ballotable, and not tender. - Shoulder presentation: the LUS feels empty.
- ✓ 2nd pelvic grip: Done after 1st pelvic grip to confirm head engagement and flexion.



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What are the 4
maneuver?

- ❑ First maneuver: fundal grip.
- ❑ Second maneuver: lateral grip.
- ❑ Third maneuver: second pelvic grip or Pawlik's grip.
- ❑ Fourth maneuver: Leopold's first pelvic grip.



Fundal grip



umbilical grip



1st pelvic grip



2nd pelvic grip

Obstetric Examination

C. AUSCULTATION OF FOETAL HEART SOUNDS

- ✓ In the 2nd trimester FHS can be audible by sonicaid which will also detect FHR
- ✓ This is the simplest accurate method to ensure foetal life throughout pregnancy
- ✓ Point of maximum intensity will depend upon gestational age, foetal lie, and presentation
- ✓ FHS are most commonly heard below the umbilicus near to the middle line or on either side



Obstetric Examination

D. LOCAL EXAMINATION:

- ✓ Local examination is not routinely performed during pregnancy except when needed
- ✓ It may be performed to ensure diagnosis in certain obstetric conditions as; vaginal bleeding during early or late pregnancy, PROM, diagnosis of the onset of labour, evaluation of cephalopelvic disproportion and contracted pelvis, and associated conditions as uterine leiomyomas.
- ✓ During 2nd trimester: to obtain vaginal swabs for culture in certain cases
- ✓ During 3rd trimester: at 36-38 weeks: to confirm foetal presentation, determine head engagement, pelvic capacity, and cervical status (ripening, effacement, and dilatation).
- ✓ During labour: to confirm onset of labour, cervical status, and station of foetal head

DIAGNOSIS OF PREGNANCY

- Diagnosis of pregnancy is based on the triad of:
 1. Delayed Cycle (missed period)
 2. Positive Pregnancy test (urine or serum).
 3. Pelvic Ultrasound (TAS/TVS) showing intrauterine pregnancy.

DIAGNOSIS OF PREGNANCY

A. DELAYED MENSTRUAL CYCLE (missed period)

A missed period is the earliest symptom of pregnancy.

- ✓ Cause: persistent corpus luteum (CL) function
- ✓ Mechanism: hCG produced by the proliferating trophoblasts stimulates CL for continuous production of both oestrogen (E) and progesterone (P), thus preventing hormone withdrawal and endometrial shedding that normally occur in the absence of a pregnancy.
- ✓ Associated symptoms in 1st trimester: nausea, with or without vomiting, frequency of micturition, fatigue, weakness and lack of concentration, and breast enlargement.
- ✓ Associated symptoms in 2nd trimester: gradual abdominal enlargement and perception of foetal movements (quickening).

DIAGNOSIS OF PREGNANCY



B. POSITIVE PREGNANCY TESTS

- Laboratory tests for diagnosis of pregnancy are based on the detection of hCG in urine or
- blood samples of pregnant women.
- ✓ Urine ELISA tests: Enzyme Linked Immunosorbent Assay slide tests are simple, cheap, accurate, and readily available.
- A positive result will be detectable in the first 48 hours of a missed period.



DIAGNOSIS OF PREGNANCY

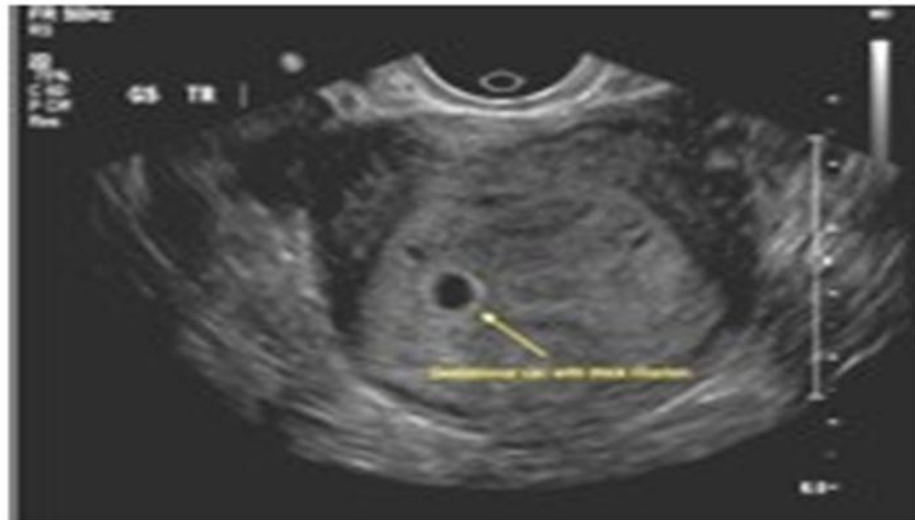
B. POSITIVE PREGNANCY TESTS

- ✓ Serum β -hCG tests : It is the earliest available method for accurate diagnosis of pregnancy.
- The test is positive within 1st 48 hours of a missed period (levels > 50 mIU/ml). - β -hCG levels rise sharply in the first 8 weeks of pregnancy, nearly doubling every 48 to 58 hours .
- -Levels usually > 1500 mIU/ml. by end of the 5th week, and $> 10,000$ mIU/ml by end of the 6th week (calculated from the LMP).
- B-hCG may also serve as an early predictor of abnormal pregnancies where poorly rising levels may be the first indicator of a pregnancy complication, as abortion or ectopic pregnancy.
- Abnormally highly rising levels are associated with vesicular mole and choriocarcinoma.

DIAGNOSIS OF PREGNANCY

C. ULTRASOUND DIAGNOSIS OF PREGNANCY :

- TAS and TVS are the mainstay in the diagnosis of an intrauterine pregnancy.
US is safe, cheap,
- readily available, and easy to interpret.
- ✓ First trimester US: - 5-6 weeks : Gestational sac < 2 cm with no embryonic echoes - 7-9 weeks : Embryo with positive heart pulsations. - 10-13 weeks: Foetus with early trunk movements - Measurements of foetal crown to rump length (CRL) give the most accurate dating of pregnancy, and hence the most acceptable expected delivery date (EDD).



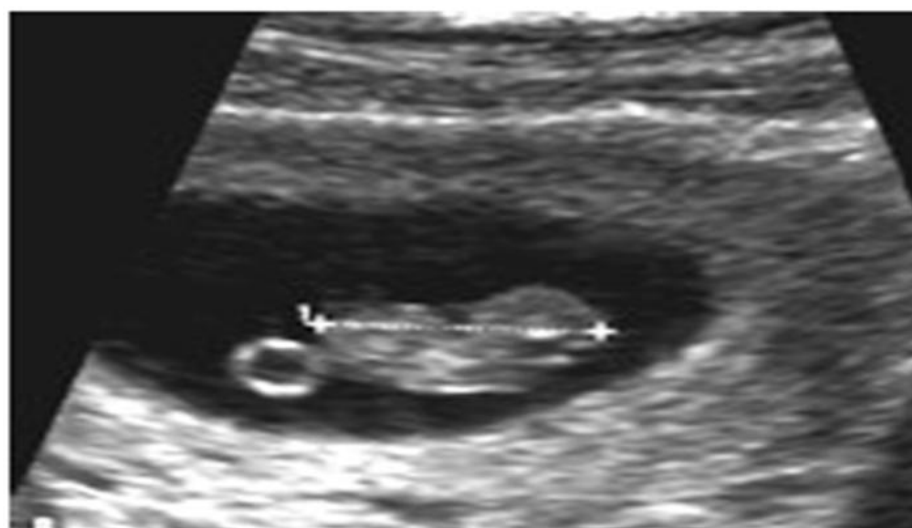
GS 5 wks



GS + YS 6 wks



GS + YS + Embryo 7 wks



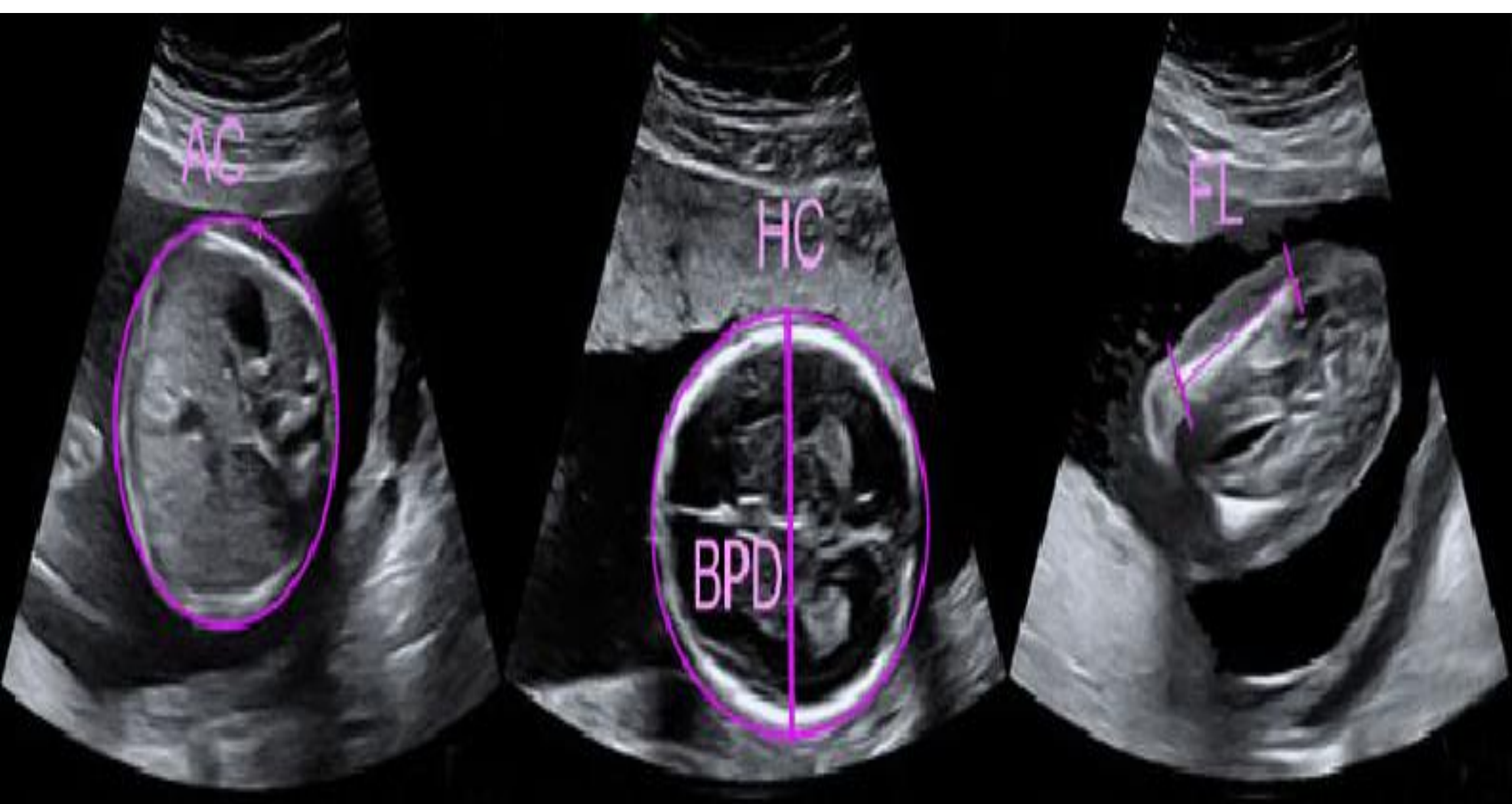
Embryo 8 wks + CRL

Fig 1-5: TVS in diagnosis of early pregnancy

DIAGNOSIS OF PREGNANCY

C. ULTRASOUND DIAGNOSIS OF PREGNANCY :

- Second trimester US: - Confirmation of foetal life (FHR) - Foetal biometry for estimation of gestational age, foetal weight, and growth patterns.
 - Biparietal diameter (BPD)
 - Abdominal circumference (AC)
 - Femur length (FL)



Calculation of the EDD (Naegle's formula)

- - Pregnancy duration is 280 days from first day of LMP.
- - Pregnancy duration = 9 month + 1 week.
- Calculation of EDD: LMP + 9 months + 7 days.
- Every 1 month equals 4 weeks and every 3 months equals 13 weeks.
- e.g.: If LMP at 01/01/2024, then EDD will be at 08/10/2024.



Title: **Antenatal care and high-risk pregnancy**
DR: - Mohamed Shabaan
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Learning objectives

1. Early Finding, and, possibly prevention, of complications specific to pregnancy, as preeclampsia, eclampsia, and obstetric hemorrhage.
2. Detection and management of any medical disorder complicating pregnancy as anemia, diabetes mellitus, cardiac, renal, or endocrine disorders.
3. Finally, classification of patients into normal or high risk to put the plan of proper management.

THE FIRST ANC VISIT

- Aim: History taking and clinical examination to identify important risk factors:
 - **Menstrual History:** To identify LMP, calculate gestational age, and the EDD
 - **Obstetric History:** Previous pregnancies provide important clues to potential problems in the current one.
 - **Medical History:** Medical disorders exacerbated by pregnancy
 - **Surgical History:** e.g. uterine surgery as myomectomy, previous CS.
 - **General examination:** Pulse, temperature and B.P., pallor

THE PRECONCEPTION VISIT

- The ideal first visit should be at a preconception clinic where health education and risk assessment can be directed towards the planned pregnancy.
- Advice can be given regarding the avoidance of harmful and teratogenic factors (drugs, cigarette smoking and alcohol intake...).
- Ensuring an optimal dietary intake and absence or control of chronic medical disorders (as diabetes, hypertension, etc...), in order to allow pregnancy to be started in the optimum conditions.

EXAMINATION

- **Abdominal Examination**: abdominal masses, etc.
- **Fundal level assessment** : in relation to pregnancy duration.
- **Vaginal examination**: done only if necessary, e.g.: suspected pelvic masses, ectopic pregnancy

Investigation

- Blood group and Rh typing, to identify RH negative patients.

<u>Haematology Report</u>	
Test	Result
ABO (Blood Group)	(O)
Rh group	Positive

- Complete blood picture:for Hb, WBCs, and platelets.

Complete Blood Picture (EDTA)			
Test	Result	Unit	Ref.Range
Red cell count	4.49	x10 ⁶ /cmm	4.20 - 5.40
Haemoglobin	12.80	g/dL	12.00 - 15.00
Haematocrit (HCT)	38.7	%	37.0 - 47.0
MCV	86	fl	78 - 99
MCH	29	pg	27 - 31
MCHC	33	g/dL	32 - 36
Platelet count	327	x10 ³ /cmm	140 - 440
White cell count	6.9	10 ³ /cmm	4.0 - 10.5
RDW-CV	14.30	%	11.50 - 14.50
PDW	11.60	fl	9.00 - 14.00
MPV	9.00	fl	7.40 - 10.40

Differential white cell Count

	Relative			Absolute	
	Result			Result/cmm	Ref.Range
Bands	0	%		0.0	
Seg	48.30	%		3.3	
Neutrophils	48.30	%	40.00- 80.00	3.3	x10 ³ /cm 2.0 - 7.0
Lymphocytes	41.50*	%	20.00- 40.00	2.9	x10 ³ /cm 1.0 - 3.0
Monocytes	8.670	%	2.000 - 10.000	0.6	x10 ³ /cm Up to 1.0
Eosinophils	1.020	%	Up to 6.000	0.1	x10 ³ /cm Up to 0.5
Basophils	0.510	%	Up to 2.000	0.0	x10 ³ /cm Up to 0.5

Comments :-

Blood picture is within normal range for age and sex.

Complete Blood Picture (EDTA)			
Test	Result	Unit	Ref.Range
Red cell count	4.94	x10 ⁶ /cmm	4.20 - 5.40
Haemoglobin	*10.50	g/dL	12.00 - 15.00
Haematocrit (HCT)	*32.8	%	37.0 - 47.0
MCV	*67	fl	78 - 99
MCH	*21	pg	27 - 31
MCHC	32	g/dL	32 - 36
Platelet count	282	x10 ³ /cmm	140 - 440
White cell count	5.3	10 ³ /cmm	4.0 - 10.5
RDW-CV	16.90*	%	11.50 - 14.50
PDW	13.10	fl	9.00 - 14.00
MPV	10.60*	fl	7.40 - 10.40

Methods:

(1)Electrical impedance (2)Photometric (3)Conductometry (4)Calculated (5)Light scattering

Differential white cell Count

	Relative			Absolute	
	Result			Result/cmm	Ref.Range
Bands	0	%		0.0	
Seg	64.80	%		3.4	
Neutrophils	64.80	%	40.00- 80.00	3.4	10 ³ /cmm 2.0 - 7.0
Lymphocytes	26.50	%	20.00- 40.00	1.4	10 ³ /cmm 1.0 - 3.0
Monocytes	7.395	%	2.000 - 10.000	0.4	10 ³ /cmm Up to 1.0
Eosinophils	0.870	%	Up to 6.000	0.0	10 ³ /cmm Up to 0.5
Basophils	0.435	%	Up to 2.000	0.0	10 ³ /cmm Up to 0.5

Comments :-

Mild microcytic hypochromic anemia, iron profile or Hb electrophoresis is recommended.

Blood sugar test

Clinical Chemistry Report

Test	Result	Unit	Ref.Range
Glycated Haemoglobin	5.3	%	Non Diabetic 4.5 - 5.6 Pre Diabetic 5.7 - 6.4 Diabetic ≥ 6.5 Controlled Diabetic < 7.0
Estimated Average Glucose (eAG)	105.4	mg/dL	

Complete urine analysis:

URINALYSIS REPORT

Urine Examination:

▪ Physical Examination:

• Volume	:	Random sample
• Color	:	Amber Yellow
• Odor	:	Aromatic
• Aspect	:	Slightly turbid
• Deposit	:	Absent

▪ Chemical Examination:

• Reaction	:	Acidic / pH 5.5
• Protein	:	Absent
• Sugar	:	Absent
• Acetone	:	Absent
• Nitrite	:	Absent
• Specific Gravity	:	1025
• Bilirubin	:	Absent
• Urobilinogen	:	Normal
• Leucocytes	:	Absent
• Blood	:	Absent

▪ Microscopic Examination:

• Pus Cells	:	2 - 4	
• R.B.Cs	:	1 - 2	
• Epithelial Cells	:	Few	
• Crystal	:	Uric Acid	Few
• Amorphous	:	Absent	
• Ova	:	Absent	
• Trophozoite	:	Absent	
• Mucus	:	Absent	
• Urine Artifacts	:	Absent	
• Fungi	:	Absent	
• Bacteria	:	Absent	

Reference Range:

Yellowish
Aromatic
Clear
Absent
4.5 - 6.5 pH
Negligible trace
Absent
Absent
Absent
1015 - 1025
Absent
Normal
Absent
Absent
0 - 5 / HPF
0 - 3 / HPF
Few / HPF
Absent
Absent
Absent
Absent
Absent
Absent
Absent

Doctor's signature:

Thanks

- **Other tests as:** TORCH antibodies IgG and IgM, AIDS, hepatitis B and C if necessary, especially in the first pregnancy

RETURN ANC VISITS

- **Monthly** visits are required in **the first 6 months**, **twice weekly** visits in **7th and 8th months**, then weekly visits in the 9th month until delivery.
- In each visit important data include;
 - **Warning symptoms**, e.g. Bleeding or regular menstrual like colicky pains, persistent vomiting, sudden escape of liquor amnii, severe persistent headache,
 - **Weight gain**: the average weight gain during pregnancy is 11-16 Kg.
 - **Examination for L.L. edema**
 - **Blood pressure measurements**: to detect early gestational hypertension or preeclampsia.
 - **Fundal level**: Should be measured and recorded at each visit after 20 weeks.
 - **Fetal heart sounds**,

Ultrasonography to ensure gestational age, to assure normal fetal growth, to assess fetal well being,



COMMON COMPLAINTS DURING PREGNANCY

- Motion sickness
- sensation of heart burn.
- Constipation.
- Breathlessness.' commonly noticed as early as the first 1-2nd week of pregnancy, due to hyper-ventilation caused by progesterone. In late weeks, the enlarging uterus can cause mechanical pressure.
- Abdominal pain
- Urinary symptoms: frequency, urgency and stress incontinence are quite common in late pregnancy
- Lower limb and Ankle edema is common in late pregnancy.

- Instructions to the patient:

- **Exercise:** Mild to moderate exercise, as walking, and regular daily house work are allowed
- Sleep and rest: Proper night sleep (8 hrs)
- **Breasts:** Regular hygiene by daily washes throughout pregnancy, In the last few weeks, massage of the nipples using lubricant creams might reduce the incidence of cracking. Retracted nipple is withdrawn by the thumb and finger using a lubricant.
- **Sexual Intercourse:** Is better **minimized** in the first trimester to avoid bleeding then gradually allowed
- Routine laboratory tests
- **Bowel habit:** Avoiding constipation by fresh vegetables and mild laxatives if necessary
- **Smoking:** Should be strictly **avoided** as it may result in placental insufficiency

Nutritional Requirements of the pregnant mother should include:.

- A suitable daily diet in pregnancy should thus include: 400 ml. of milk or its derivatives, one egg, fresh fruits and **vegetables** about 120 gm of red meat, fish, or liver.
- **Caloric** requirements average **2300 Kca/day**.
- **Protein**: 80-100 gm/day, Calcium: 1-1.5 gm/day,



Iron: 30-50 mg/day.

Vitamins and minerals: Especially B, C, D, K Folic acid is important for cell division and replication. In the first few weeks, a dose of 400 has been shown to effectively reduce the risk of neural tube defects. Salt restriction



- **Effect of severe Malnutrition on the mother:**

1. Loss of weight, anaemia, and lowered resistance against infection.
2. Decalcification of bones, caries of teeth.
3. Affection of lactation.

- **Effect of Malnutrition on the fetus:**

1. Low birth weight infants.
2. Higher incidence of rickets and anaemia, in severe cases.



- **Vaccination (immunization) in pregnancy:**

The vaccines for the following diseases may be given if needed, preferably after the 1st trimester:

- Tetanus, poliomyelitis, rabies, influenza, cholera and typhoid.
- Passive immunization against hepatitis A and B may be given.

- **pregnancy according to FDA classification:**
- **Group: A** Safe during pregnancy. Examples are supplementary vitamins & levothyroxine
- **Group: B** Risky in animal, no enough data on humans. Examples are penicillin & cephalosporin
- **Group: C** Risk in human cannot be ruled out. Examples are corticosteroids & NSAID.
- **Group: D** Risky in human pregnancy, but the benefits may outweigh the risks, Tetracycline & streptomycin

HIGH RISK PREGNANCY

is a pregnancy complicated by a diseases or disorder that may a endanger the life or affect the health of the mother, fetus, or the newborn.

High risk pregnancy may be associated with:

- a. Personal factors:** elderly primigravida, pregnancy above 40 years and grand multipara
- b. Severe medical conditions** affecting the mother; such as uncontrolled diabetes mellitus, cardiac disease , etc...
- c. Obstetrical complications** that require specialized care such as: severe pre-eclampsia (PE) or eclampsia, HELLP syndrome, severe intrauterine fetal growth restriction (IUGR), and multiple high risk factors.
- d. Conditions that may require invasive procedures for** fetal diagnosis or therapy as immune and non-immune hydrops fetalis, and congenital anomalies or genetic disorders.

The elderly primigravida

- . **Definition:** Primigravida whose age is 35 years or more during pregnancy or delivery
- . **Maternal and foetal risks:** increased liability to the following;
 1. Medical disorders during pregnancy; as PE, Eclampsia, and GDM
 2. Intrauterine growth restriction placental insufficiency, and preterm labour
 3. Increased incidence of chromosomal abnormalities and congenital malformations
 4. Dysfunctional labour, abnormal uterine action, rigid perineum.
 5. Increased rate of CS to reassure fetal safety

The grand multipara

Definition: women who had five or more previous deliveries

Maternal and fetal risks: increased liability to the following;

1. Medical disorders as; Anaemia, chronic Hypertension, PE, GDM, etc..
2. Placenta previa, and placenta accreta in cases of repeat CS
3. Malpresentations and malpositions; due to weak abdominal muscles
4. Uterine inertia and more liability to atonic post partum haemorrhage
5. Obstructed labour and rupture of the uterus
6. Increased operative delivery (CS, Forceps, and Ventouse) due to all previous factors

GIT and Hepatic disorders with pregnancy

Dr. Hend Mahmoud Abdelghany
OB/GYN Department

- Learning objectives
 1. Know different GIT and hepatic disorders with pregnancy
 2. Approach to diagnosis
 3. Different treatment modalities



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GIT disorders during pregnancy

Nausea and vomiting of pregnancy (NVP)

- Incidence: 50- 80% during first trimester (7-11 weeks), usually on morning
- Mostly self limited after 16 weeks
- Management:
 1. Reassurance
 2. Diet control (high protein, low sugar, low fat)
 3. Avoid triggering agents (odors, type of food)
 4. Pharmaceutical : pyridoxine (vit B6), antiemetics (Doxylamine)





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Hyperemesis Gravidarum

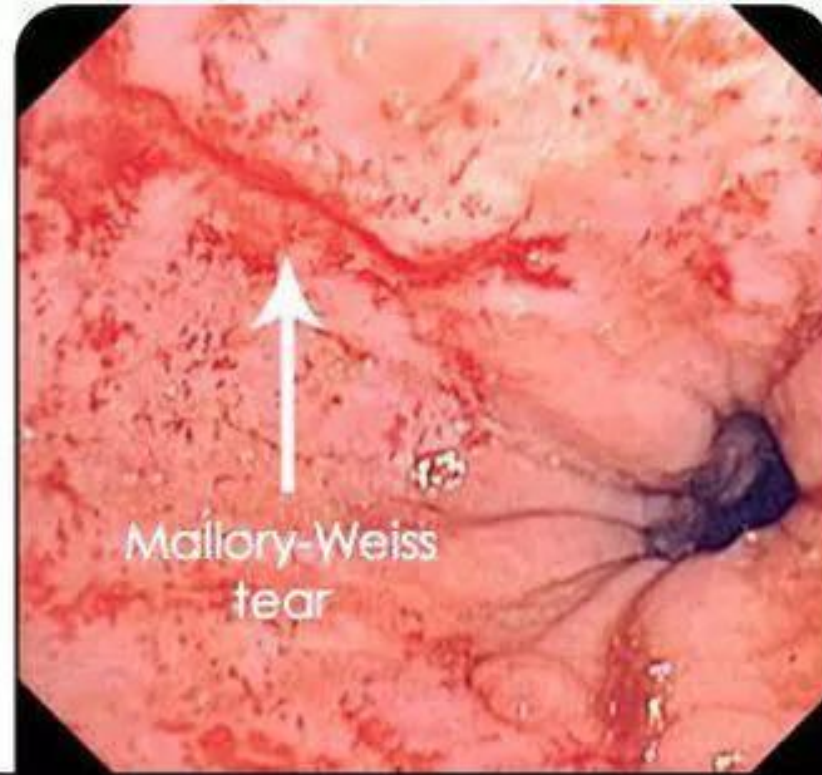
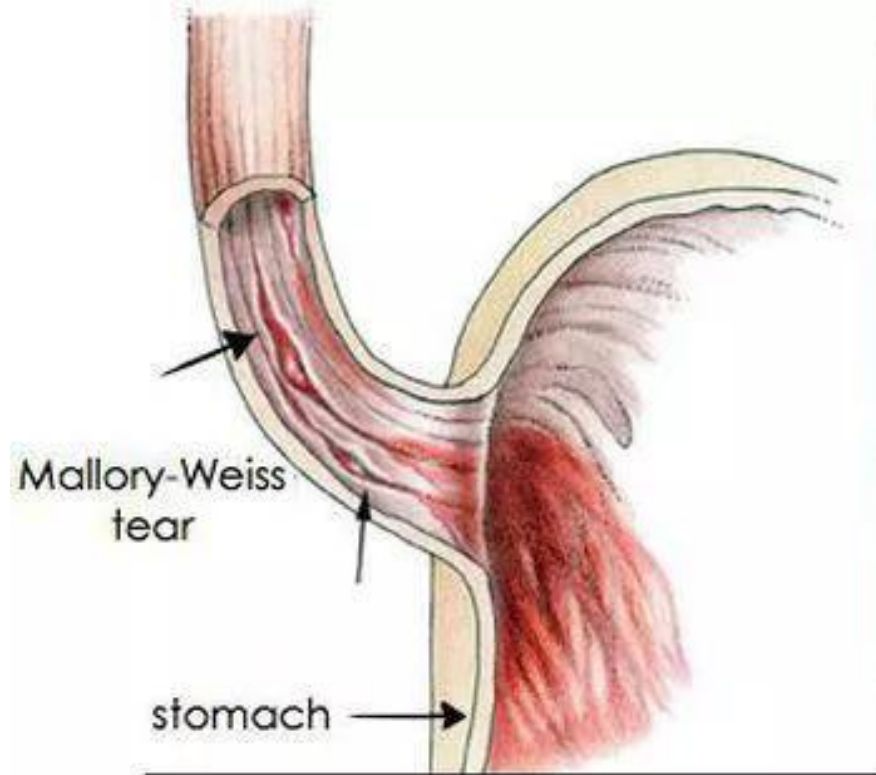


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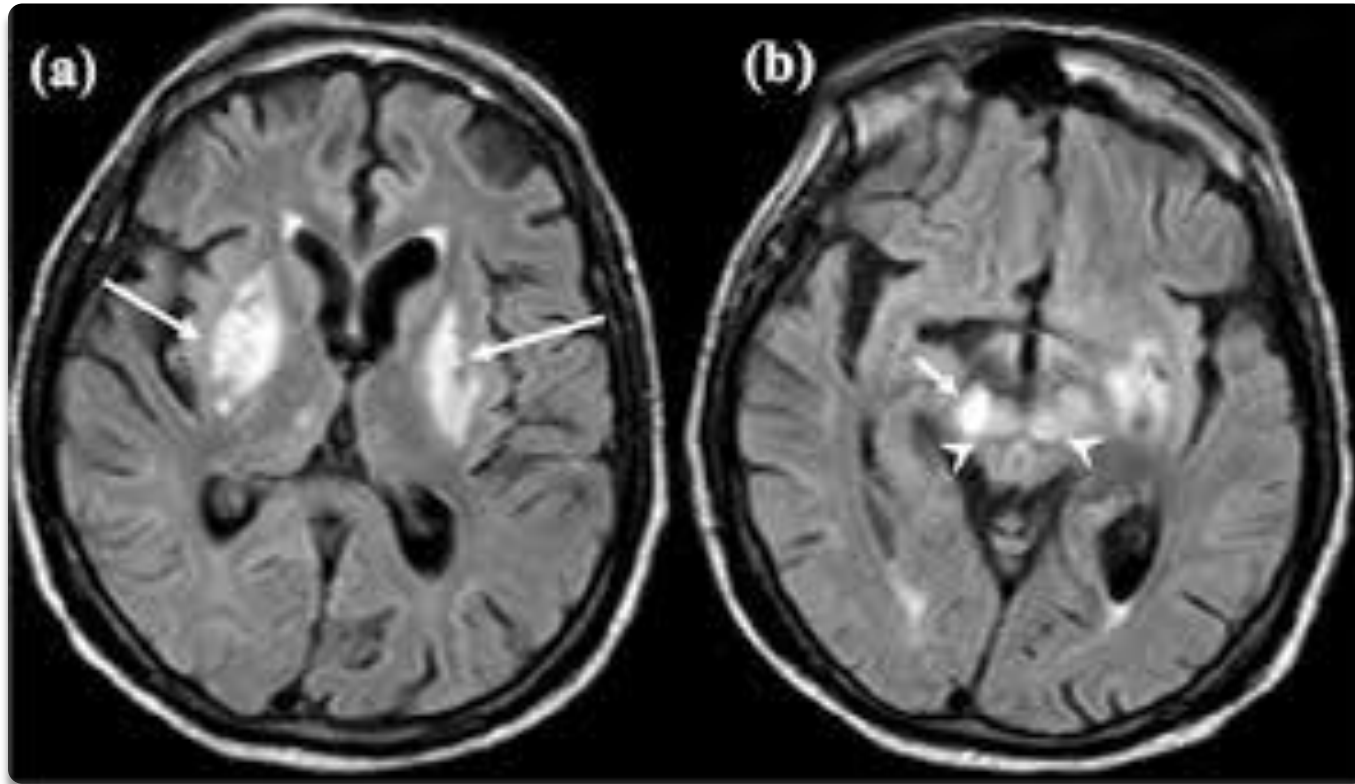
- Severe persistent vomiting before 16 weeks
weight loss, dehydration, ketosis, alkalosis, hypokalemia
- Incidence : 1-2 % of all pregnancies
- Etiology: unknown (High B-HCG, psychological, gastric dysrhythmia)
- Risk group: Hyperthyroidism, Molar pregnancy, DM, Multifetal pregnancy, underlying gastric disease, Primigravida

- Biochemical changes:
 1. Electrolyte imbalance.....Hyponatremia, Hypokalemia
 2. Ketonuria
 3. Haemoconcentration
 4. Slightly elevated liver enzymes, decreased liver glycogen
 5. Transient elevation in TSH

- Complications:
 1. Hypotension, pre-renal failure, liver failure
 2. Mallory-Weiss syndrome (esophageal tear)
 3. Wernick's encephalopathy (Brain degenerative changed due to thiamine deficiency)MRI shows petechial hemorrhage



Mallory-Weiss Syndrome



Wernick's
encephalopathy

Treatment

- Mild cases: as NVP + strong antiemetics
- Severe cases:
 - Hospitalization
 - NPO + IV fluids + Na⁺, K⁺ correction
 - Anti-emetics (Doxylamine+ pyridoxine), Meclizine , ondansetron
 - Thiamine
 - Liver support
 - Exclude Underlying disease
 - Anticoagulant is considered
- Severe resistant cases: termination of pregnancy

Gastro-Esophageal reflux disease (GERD)

Prevalence: 70 % of all pregnant women with different degrees

- Usually symptomatize with heart burn especially with advancing gestation.
- Usually after meals and in recumbent position

Coping with **PREGNANCY HEARTBURN**



- **Eat little and often**
- **Sleep on incline**
- **Stay hydrated**
- **Identify trigger foods**
- **Take walks**
- **Use home remedies**

Treatment

➤ Reassurance and instructions

- Avoid big and late meals
- avoid recumbent position
- antacids

➤ Pharmaceutical

- H₂ receptor blocker (cimetidine, Ranitidine)
- Proton pump inhibitor (omeprazole)

Hepatic disorders with pregnancy

Hepatic disorders

- Related to pregnancy
- Coincidental with pregnancy
- Dating before pregnancy



- Related to pregnancy

Intrahepatic cholestasis

- Incidence : 1-4 % of all pregnancies
- Etiology : unknown (Decreased canalicular transport of bile acid)
- Symptoms :
 - Pururitis (late pregnancy) >20 weeks especially soles and palms
 - Jaundice



- Usually has **benign** course with no matern hepatic sequalae but has tendency to **recur** next pregnancies
- Complications:
 - Preterm labour
 - fetal distress, meconium
 - perinatal death



- Investigations

- Bile acid: high levels

- liver enzymes and bilirubin:

- Normal or slightly high

- Treatment

- I. Maternal

- anti-histaminics, cold baths, emollients

- ursodeoxycholic acid

- Postpartum surveillance: rapid improvement

- I. Fetal surveillance:

- close follow up

- Delivery at 36-37 weeks

Acute fatty liver of pregnancy

- Incidence: 1:14.000 pregnancies (very rare), However it is the most common cause of liver failure during pregnancy
- Etiology: Unknown, Inborn error of metabolism
Fatty infiltration that leads to liver cell failure

- Symptoms:
 - Vomiting (late in pregnancy)
 - Abdominal pain
 - Jaundice
 - Hypertension, sometimes pre-eclampsia
 - Hematemesis
- complications:
 - Maternal Mortality
 - Peri-natal mortality

thir

- Investigations:
 - Liver enzymes moderately elevated
 - Hyperbilirubinemia
 - Hypoglycemia
 - Hypoprothrombinemia
 - creatinine elevated

Liver biopsy is diagnostic but not done

Treatment

- Supportive measures
ICU admission
- Immediate Delivery
- reversal of pathology one week after delivery

Pre-eclampsia – HELLP syndrome

Clinical syndrome related to Hypertension

- Abdominal pain, usually in the right upper side (specific to HELLP syndrome).
- Headache.
- Blurred vision.
- Nausea and vomiting.
- Fatigue.
- Edema (swelling) and quick weight gain.
- Pain when taking deep breaths.

- Investigations:
 1. CBC: Hemolysis, Hemoconcentration, Low platelet
 2. Liver enzymes: Slightly elevated
 3. kidney function tests: Creatinine and Urea may be elevated
 4. Urine : Proteinuria

- Management:

1. Termination of pregnancy
2. Mg_2SO_4 IV infusion
3. Control of blood pressure

- Dating before pregnancy

Chronic viral hepatitis

1. HBV

Pathogen: DNA virus

Mode of transmission:

Blood, Sex, perinatal

- Maternal infection:
 - Asymptomatic
 - Usually doesn't cause morbidity or mortalities
- Fetal infection
 - vertical transmission

- Investigations:
 - HbSAg
 - viral load
- Prevention of vertical transmission
 - Mothers:
 - receive ttt to lower viral load
 - vaginal delivery allowed
 - Newborn:
 - Immuno-prophylaxis + HB vaccine
 - breast-feeding allowed

- 2. HCV

Pathogen: RNA virus

Fetal complications:

Low birth weight , preterm labour, NICU admission

vaginal delivery and breast feeding allowed



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Disorder	Onset in Pregnancy	Clinical Findings	Hepatic		Renal	Hematological and Coagulation					
			AST (U/L)	Bili (mg/dL)	Cr (mg/dL)	Hct	Plat	Fib	DD	PT	Hemolysis
Hyperemesis	Early	Severe N&V	NL-300	NL-4	↑	↑↑	NL	NL	NL	NL	No
Cholestasis	Late	Pruritus, jaundice	NL-200	1-5	NL	NL	NL	NL	NL	NL	No
Fatty liver	Late	Moderate N&V, ± HTN, liver failure	200-800	4-10	↑↑↑	↑↑↑	↓↓	↓↓↓	↑	↑↑	↑↑↑
Preeclampsia	Mid to late	HA, HTN	NL-300	1-4	↑	↑	↓↓	NL	↑	NL	↑-↑↑
Hepatitis	Variable	Jaundice	2000+	5-20	NL	↑	↓	NL	NL	↑	No

↑ = increased levels; ↓ = decreased levels; AST = aspartate aminotransferase; Bili = bilirubin; Cr = creatinine; DD = D-dimers; Fib = fibrinogen; HA = headache; Hct = hematocrit; HTN = hypertension; N&V = nausea and vomiting; NL = normal; Plat = platelets; PT = prothrombin time.

Jaundice with pregnancy

- Cholestasis
- Acute fatty liver of pregnancy
- Hyperemesis Gravidarum
- Pre-eclampsia- HELLP syndrome
- Acute viral hepatitis



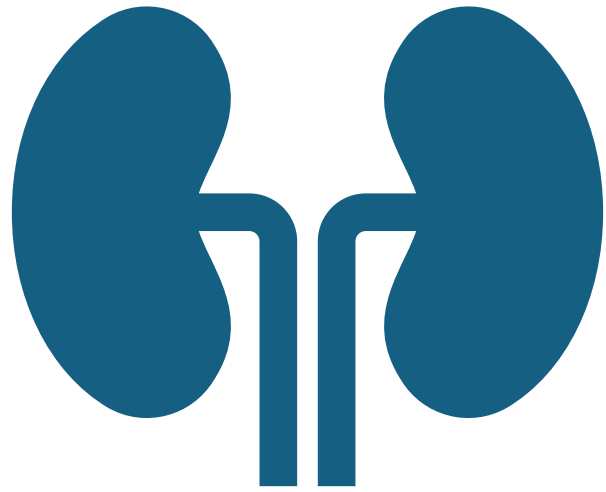
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- Obstructive jaundice
- Hemolytic jaundice
- Infections
- Drug toxicity
- Incompatible blood transfusion



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Urinary tract Infections in pregnancy

Bacteriuria of Pregnancy

Significant asymptomatic bacteriuria:

- Is defined as the presence of 100,000 or more bacteria per ml of urine in 2 freshly voided midstream specimens of urine.

INCIDENCE AND RISK

- It occurs in about 2-10% of all pregnant women (average 6%).
- Pyelonephritis will develop in 25-30% of cases with significant bacteriuria if untreated but in less than 1% of the patients without bacteriuria. Therefore, many cases of pyelonephritis in pregnancy can be prevented by finding and treating asymptomatic bacteriuria.

Bacteriuria of Pregnancy

INVESTIGATIONS :

- Urine examination for presence of pus cells
- Culture and sensitivity for identifying pathogenic organisms.

Test Name	Result	Unit	Reference Range	Method
Urine Examination				
Physical				
Colour	Lemon yellow			
Transparency	Clear			
Microscopy				
Cast	Nil			
Crystals	Nil			
Pus	0 - 2/hpf			
Erythrocytes	Nil			
Squamous Epithelial Cell	0 - 2/hpf			
Yeast	Nil			
Chemical				
pH	6		5 - 8	
Albumin	Nil			
Sugar	Nil			

Table 1. Urinalysis Results

Test	Usual Range	Indicators of Infection	Accuracy
Bacteria	Absent	Any amount	Low sensitivity, ^a high specificity ^b
Leukocyte esterase	Absent	Positive = pyuria, presence of WBCs in urine	High sensitivity, low specificity
WBC	<5	Pyuria: WBC >10	High sensitivity, low specificity
Nitrite	Absent	Positive = presence of bacteria that reduce nitrate	Low sensitivity, high specificity
RBC	<5	Hematuria common in infection	Low sensitivity, high specificity
Epithelial cells	<5	<5 = good urine sample	High epithelial cells indicate contamination with skin flora
pH	4.5-8	pH ↑ if urea-splitting organism (e.g., <i>Proteus mirabilis</i>) is present	Low specificity (there are many other causes of alkaline urine)

^a Sensitivity = likelihood of positive test when disease is present.

^b Specificity = likelihood of negative test when disease is not present.

Source: Reference 1.

Bacteriuria of Pregnancy

Treatment

Ampicillin antibiotic:

- The infecting organism is most commonly E coli. About 75% of cases are cured by a single 10 days course of ampicillin (500 mg 8-hourly).

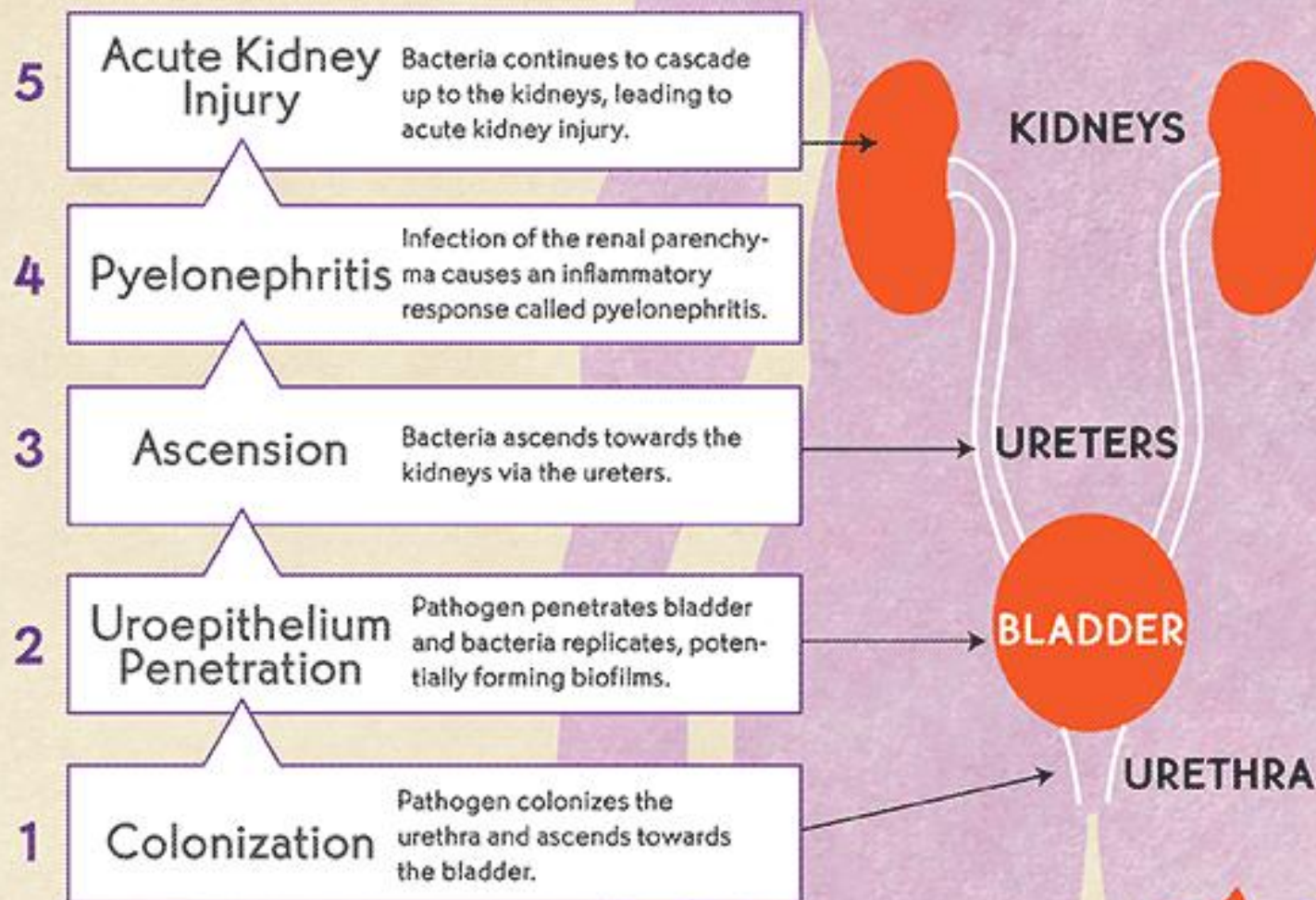
Recurrence:

- In 1/3 of the cases, bacteriuria will recur and treatment should be repeated.

Resistant and recurrent cases:

- Antibiotic should be given according to culture and sensitivity.
- After delivery resistant and recurrent cases should be offered a full urologic evaluation including an intravenous pyelography (IVP)

Stages of a Urinary Tract Infection



PYELONEPHRITIS DURING PREGNANCY

INCIDENCE:

- About 1-2% of pregnancies may be complicated by pyelonephritis.

AETIOLOGY:

• Predisposing Factors

- *Asymptomatic bacteruria* during pregnancy.
- *Urinary stasis* due to dilatation of the ureter caused by:
 - Atony of the ureter associated with outer compression by the enlarging gravid uterus.
 - Hypertrophy of the wall of the lower end of the ureter. As the uterus is usually slightly inclined to the right side, more compression on the right ureter occurs, that is why pyelonephritis is more common on the right kidney.

PYELONEPHRITIS DURING PREGNANCY

Infecting organisms:

- E. coli is the commonest organism.
- Other organisms include Streptococcus faecalis, Staphylococci, Bacillus proteus and Klebsiella Enterobacter.

Routes of infection:

- Blood borne (most common).
- Ascending infection along the periureteric lymphatics or along the lumen.
- Spread by lymphatics from the colon.

PYELONEPHRITIS DURING PREGNANCY

CLINICAL PICTURE AND DIAGNOSIS (usually occurs after 16 weeks):

- **Acute onset:** *Pain in the loin, fever, rigors and may be vomiting.*
- **Tenderness:** *Renal angle region.*
- **Urine examination:**
 - The urine is acidic, diminished in amount severe cases and contains albumin.
 - The urine contains pus cells and micro-organisms.
 - Culture of the urine is always positive



Laboquick

Urinalysis Reagent Strips

10 Parameter

Glucose

Bilirubin

Ketone

Specific Gravity

Blood

pH

Protein

Urobilinogen

Nitrite

- For in vitro diagnostic use only.
- Read instructions for use before starting.
- Do not use after the expiration date.
- Keep device out of the reach of the children.
- Dispose used materials as biological waste.
- Keep between +2...+30°C. Do not freeze.
- Keep away from radiation, moisture, odour.
- Only take required amount of strips that will be used.
- The strip should remain in the closed canister until use.
- Do not touch the reagent areas of the strip.

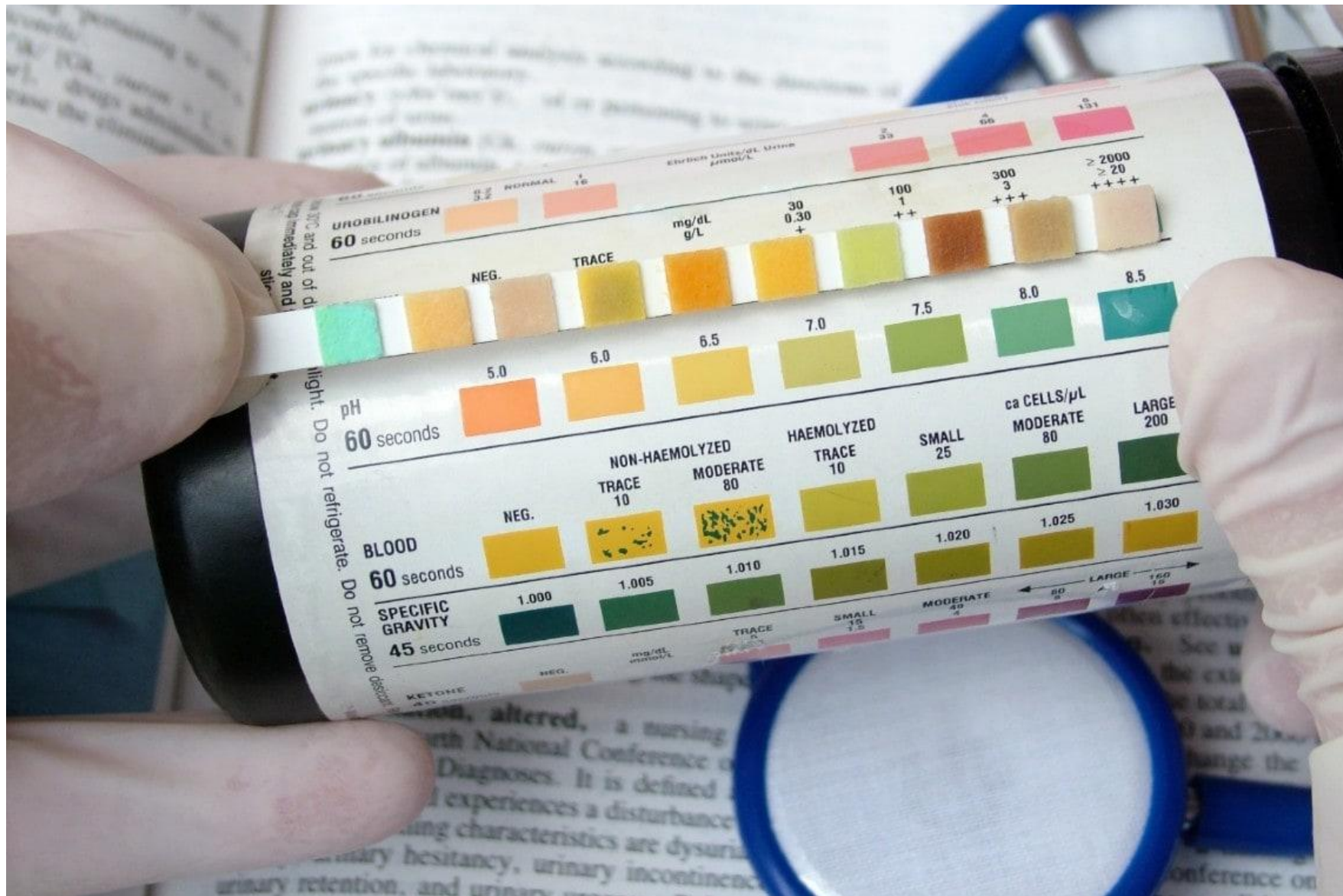


LEU 120 s						
	-	15 ±	70+	125++	500+++	Leu/μL
NIT 60 s						
	-	+	+	+	+	
URO 60 s						
	02(3.5)	1(17)	2(35)	4(70)	8(140)	12(200)
PRO 60 s						
	-	15(0.15)	30(0.3)+	100(1.0)++	300(3.0)+++	2000(20)+++
pH 60 s						
	5.0	6.0	6.5	7.0	7.5	8.0
BLO 60 s						
	-	±	+	++	+++	5-10
SG 45 s						
	1.000	1.005	1.010	1.015	1.020	1.025
KET 40 s						
	-	5(0.5)	15(1.5)	40(4.0)	80(8.0)	160(16)
BIL 30 s						
	-	1(17)+	2(35)++	4(70)+++		
GLU 30 s						
	-	100(5)	250(15)	500(30)	1000(60)	≥2000(110)

TESTING AND READING TIME

Rev. 08/2010

Leukocytes 120s	Neg.			Trace 16	Small 16	Moderate 125	Large 500	cells/ μ l
Nitrite 60s	Neg.					Positive Any degree of uniform pink color		
Urobilinogen 60s	2.2	Normal	16		32	64	128	μ mol/l
Protein 60s	Neg.		Trace 1	1.3	1.7	2.1	2.7	g/l
pH 60s	5.0	6.0	6.5	7.0	7.5	8.0	8.5	
Blood 60s	Neg.	10 Trace	10 Trace	25 Small	50 Moderate	200 Large		cells/ μ l
Specific Gravity 45s	1.000	1.005	1.010	1.015	1.020	1.025	1.030	
Ascorbate 40s	0			0.6	1.4	2.8	5.0	mmol/l
Ketone 40s	Neg.		Trace 0.5	1.5	3.0	6.0	Large 16	mmol/l
Bilirubin 30s	Neg.				Small 1.7	Moderate 30	Large 100	μ mol/l
Glucose 30s	Neg.		5 Trace	10	25	100	150	mmol/l



Result Summary for URINALYSIS WITH MICROSCOPIC

Result Information

Abnormal	Status Final result (9/18/2015 7:17 PM)	Provider Status Ordered
-----------------	--	----------------------------

Component Results

Component	Value	Ref Range & Units
UA SPECIMEN SOURCE	URINE (PRESERVED) UNSPECIFIED	
COLOR	YELLOW	YELLOW
APPEARANCE, URINE	HAZY (A)	CLEAR
PH, URINE	6.0	5.0 - 8.0
SPEC. GRAVITY, URINE	1.030 (H)	1.005 - 1.029
PROTEIN, URINE	1+ (A)	NEGATIVE
GLUCOSE, URINE	NEGATIVE	NEGATIVE
KETONE, URINE	3+ (A)	NEGATIVE
BILIRUBIN, URINE	NEGATIVE	NEGATIVE
BLOOD, URINE	NEGATIVE	NEGATIVE
UROBILINOGEN, URINE	<2.0	<2.0 mg/dL
NITRITE, URINE	NEGATIVE	NEGATIVE
LEUKOCYTE ESTERASE, URINE	2+ (A)	NEGATIVE
RBC, URINE	0	0 - 5 /HPF
WBC, URINE	3	0 - 3 /HPF
SQUAMOUS EPI CELLS	17	/HPF
BACTERIA	PRESENT	/HPF
MUCUS	PRESENT	

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^a Sensitivity = likelihood of positive test when disease is present.

^b Specificity = likelihood of negative test when disease is not present.

Source: Reference 1.





Name of the antimicrobial agents	Sensitivity
Amoxycillin	R
Nitofurantoin	S
Cephoxime	I
Enrofloxacin	R
Ceftriaxone	S
Streptomycin	S
Amikacin	I
Tetramycin	R
Chloramphenicol	S
Ciprofloxacin	R
Penicillin	R
Nalidixic Acid	R
Ampicillin	R

Surgical intervention

Bitch was premedicated with DNS @ 20ml/kg BW IV and

Antibiotics	Pattern			Antibiotics	Pattern		
	A	B	C		A	B	C
Amoxycillin				Amikacin	S		
Ampicillin				Amoxyclav	S		
Aztreonam	R			Azithromycin			
Cefepime	R			Carbimicillin			
Cefoxitin				Cefixime	R		
Ceftriaxone	R			Ceftazidime	S		
Cephalexin				Cefuroxime			
Chloramphenicol				Cephadrine			
Clarithromycin				Ciprofloxacin	S		
Cloxacillin				Clindamycin			
Cotrimoxazole	S			Colistin	R		
Erythromycin				Doxycycline			
Gatifloxacin	S			Fusidic acid			
Imipenem				Gentamicin	S		
Linezolid				Levofloxacin	S		
Meropenem	S			Mecillinum			
Nalidixic Acid				Metronidazole			
Netilmicin	S			Neomycin			
Novobiocin				Nitrofurantoin	S		
Oxacillin				Ofloxacin			
Penicillin				Pefloxacin			
Rocephin				Rifampicin			
				Spectinomycin			

PYELONEPHRITIS DURING PREGNANCY

Differential diagnosis

- Appendicitis (In acute pyelonephritis the temperature is often higher, the renal-angle is tender, the urine containing pus cells and leucocytosis is more marked).
- Acute abdomen, twisted ovarian cyst.
- Other causes of fever and vomiting during pregnancy.

PYELONEPHRITIS DURING PREGNANCY

COMPLICATIONS

Chronic

Sometimes the condition becomes chronic with recurrence. In these cases, plain X-ray and intravenous pyelography are done after delivery to exclude ureteric stone.

Obstetric complications

in severe cases: especially if proper treatment is not given early, abortion or intrauterine fetal death may occur.

Fetal complications

With less severe infection and even in cases of asymptomatic bacteruria, if proper treatment is not given, fetal growth retardation or premature labor may occur.

PYELONEPHRITIS DURING PREGNANCY

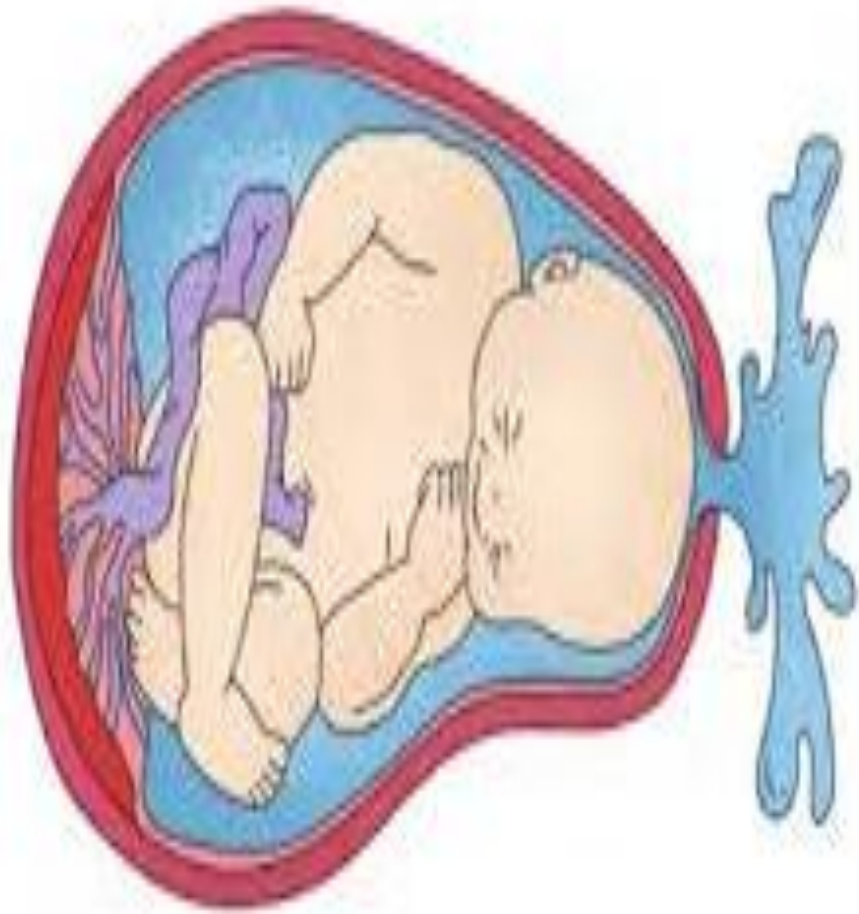
TREATMENT

General measures:

Rest, light diet, increased fluid intake, alkalinization of the urine, if it is acidic and intravenous fluids if there is excessive vomiting.

Antibiotics:

Start with ampicillin 500 mg 6-hourly, until results of urine culture are available, then continue according to the results of the culture and sensitivity.



**Amniotic fluid disorders .
Rupture of membranes.**

Amniotic fluid

- SOURCES OF AMNIOTIC FLUID-
- Fetal: urine production, amniotic membrane, and transudation from fetal circulation.
- Maternal: placental chorionic membrane and transudation from maternal circulation.

AMNIOTIC FLUID VOLUME AND COMPOSITION

- The AFL is composed of 99% water, and is characterized by having a clear aspect, pale colour, and slightly alkaline PH. It contains carbohydrates, lipids, proteins, enzymes, and mediators as cytokines. It also contains fetal excretory products as; uric acid, urea, creatinine, and sodium chloride, together with vernix caseosa and lanugo hair.

FUNCTIONS OF THE AMNIOTIC FLUID

- 1. Protection of the fetus from external trauma.
- 2. Maintains constant surrounding temperature suitable for fetal growth
- .3. Provides a suitable medium for fetal movement for muscular development and neuromuscular coordination.
- 4. Provides a source for fetal excretion and a source of fetal nutrients.
- 5. Allows for fetal lung distension and development.
- 6. Prevents occurrence of intra-amniotic bands, adhesions, and fetal compression deformities.

EVALUATION OF AMNIOTIC FLUID VOLUME

- Clinical evaluation of AFL is only subjective especially in presence of maternal obesity, twins and MFP, PI PRV, and uterine leiomyomas.
- US is the gold standard in evaluation of AFV and diagnosis of volume disorders as oligohydramnios and polyhydramnios.
- AFL pockets measurement longitudinally in cm length.. AFL Index (AFI) in which the vertical diameter of largest pocket is measured in each of the four abdominal quadrants. The sum of 4 quadrants gives an estimate of AF• Normal liquor (AFI: 10-15), oligohydramnios (AFI < 5.0), and polyhydramnios (AFI > 25).

Amniotic fluid disorders

OLIGOHYDRAMNIOS

- Oligohydramnios is defined as diminished AFV.
- By US it is diagnosed when measurement of the deepest AFL pocket is < 2 cm, or an AFI < 5 .
- INCIDENCE:- Oligohydramnios affects almost 3-4% of all pregnancies.
- - Incidence is significantly increased in association with IUGR and placental insufficiency.

ETIOLOGY

- 1. Undiagnosed ROM
- 2. Placental insufficiency: as in IUGR, Preeclampsia (PE), and post term pregnancy (PTP).
- 3. Fetal causes: Fetal congenital anomalies (FCA) as renal agenesis and renal hypoplasia
- 4. Drugs; Indomethacin Is known to decrease AFV due to decreased renal perfusion

SYMPTOMS AND SIGNS:

- 1. History of leaking amniotic fluid (ROM).
- 2. Inappropriate growth of maternal abdominal girth especially in the 3rd trimester
- 3. Inspection: reveals smaller than expected abdominal contour.
- 4. Abdominal examination: lower than expected fundal level for gestational age.

RISKS AND COMPLICATIONS:

- A) **During pregnancy**:- Complications related to IUGR and placental insufficiency .
- Pulmonary hypoplasia, compression limb deformities, and amniotic adhesions.
- B) **During labor**- Cord compression with fetal hypoxia and fetal distress
- .- Meconium passage and fetal meconium aspiration.

MANAGEMENT

- Termination of pregnancy: in cases that reached near term especially if associated with placental insufficiency or lethal anomalies or if the fetus is compromised as per the assessment of fetal well being measures.

Polyhydramnios

- Polyhydramnios is defined as excessive AFV, measured By US when the deepest vertical AFL pocket is > 8 cm, or an AFI > 25 . INCIDENCE: it may be associated with 0.5-1.5% of pregnancies.

ETIOLOGY

- 1. Idiopathic: most cases are possibly due to an imbalance between AFL production and absorption of an unknown etiology.
- 2. Fetal causes:- Twins and MFP: due to large surfaces producing AFL (placenta and membranes).- Placental tumours: as chorioangioma.- FCA: as anencephaly, open NTDs, esophageal atresia, and duodenal atresia.- Hydrops fetalis: due to large placenta & excessive transudation from fetal circulation.
- 3. Maternal causes:- Gestational diabetes mellitus (GDM): most common cause.- Rarely with PE: in association with placental edema.

SYMPTOMS AND SIGNS

- 1. Progressive rapid enlargement of abdominal girth especially in 3^o trimester.
- 2. Respiratory embarrassment and pressure symptoms only in severe cases.
- 3. Inspection: shows larger abdominal contour than expected for gestational age.
- 4. Fundal level: is higher than expected for gestational age.
- 5. Fetal parts are not easily palpated due to excessive surrounding AFL

RISKS AND COMPLICATIONS

- 1. PTL due to uterine overdistension, with fetal prematurity complications.
- 2. Dysfunctional labor due to malpresentations, macrosomia, PROM, and associated MFP.
- 3. Placental abruption due to suddenly decreased intraamniotic pressure after PROM.
- 4. Risks of associated fetal anomalies or maternal GDM

MANAGEMENT

- **1. Mild and Moderate Polyhydramnios:**
- Mild and moderate cases are best managed conservatively till spontaneous labor pains start.
- During this period the patient will need, reassurance, follow up, and exclusion of MFP, FCA, and GDM by US.2.
- **2. Severe Polyhydramnios:-** Term pregnancy > 36 weeks: termination of pregnancy by either induction of labor or CS according to fetal presentation, pelvic adequacy, and bishop score- Preterm pregnancy < 36 weeks: Repeated amniocentesis removing 1-1.5 litres every time to prolong pregnancy until lung maturity is established

Rupture of Membranes “ROM”

Premature ROM:

- spontaneous **ROM** before the onset of labour (10% of all pregnancies).

If PROM occurs < 37 wks

- **preterm PROM** (Associated with significant maternal ± fetal morbidity).

AETIOLOGY :

- **Idiopathic.**
- **Infection** (**most common** risk factor - decreasing the tensile strength of fetal membranes – usually by **GBS & Trichomonas Vaginalis**).
- **Cervical Incompetence.**
- **Polyhydramnios and Multifetal pregnancy** (**increased intra-amniotic pressure**)
- **Local membrane defects.**
- **Cigarette smoking.**

HAZARDS AND COMPLICATIONS

A. PRETERM LABOUR:

Labour usually start within **few hours after ROM**.

If this occurs **< 37 wks** → PTL.

B. PLACENTAL ABRUPTION:

ROM in cases of **polyhydramnios** and **MFP** →
sudden decrease in intra-amniotic pressure →
Placental separation → Accidental hemorrhage.

C. OLIGOHYDRAMNIOS.

D. INFECTION (CHORIOAMNIONITIS):

Prolonged ROM may be easily complicated by
ascending **LGT** infection in at least **20%** of cases.

MATERNAL CHORIOAMNIOTIS?



Neonatal Sepsis

 Signs of neonatal sepsis?

 Maternal Chorioamnionitis?

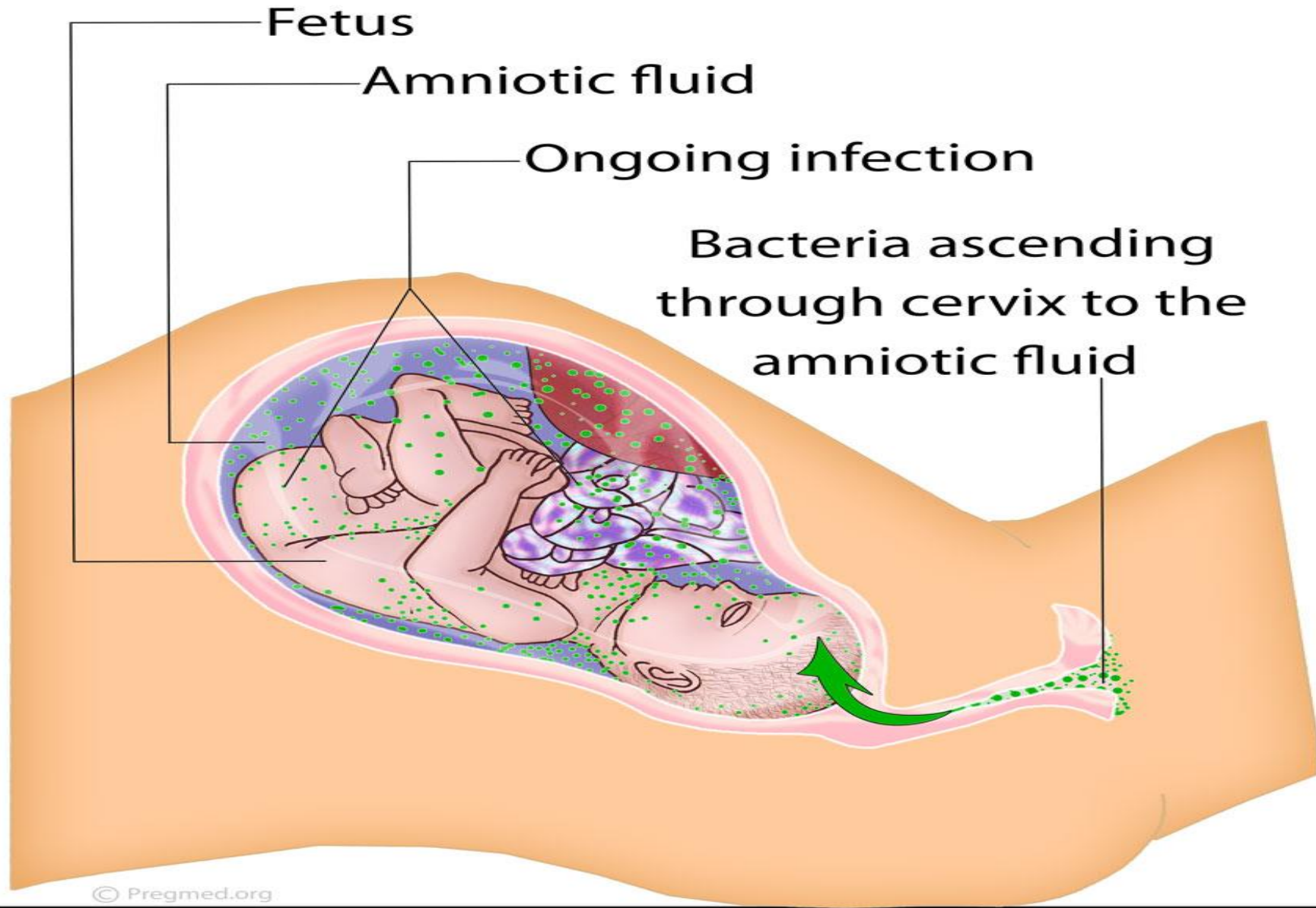
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■ Definition:

■ Maternal fever of $\geq 38^{\circ}\text{C}$ (100.4°F) and at least two of the ff criteria:

- maternal leukocytosis
($\geq 15,000$ cells/ mm^3)
- maternal tachycardia
(≥ 100 beats/minute)
- fetal tachycardia
(≥ 160 beats/minute)
- uterine tenderness, and/or
- foul odor of the amniotic fluid

Chorioamnionitis



HAEMATOLOGICAL TESTS

Test Name	Result	Unit	Ref. Range
Complete Blood Count			
Haemoglobin	8.3	gm/dl	12 - 16
Haematocrit -HCT	24.8	%	37 - 47
RBCs Count	3.02	Millions/Cmm	4.2 - 5.4
MCV	82.12	fl	78 - 95
MCH	27.48	pg	27 - 31
MCHC	33.47	%	32 - 36
RDW-CV	13.9	%	13 - 18
Platelets Count	428000	Cmm/	150000 - 440000
WBCs Count	18260	Cmm/	4000 - 11000
Neutrophils - Staff%	0	%	2 - 6
Neutrophils - Segmented %	85	%	40 - 75
Lymphocytes%	9	%	20 - 45
Monocytes%	4	%	2 - 10
Eosinophils%	2	%	0 - 6
Basophils%	0	%	0 - 1
Blast%	0	%	0 - 0
Promyelocytes%	0	%	0 - 0
Myelocytes%	0	%	0 - 0
Metamyelocytes%	0	%	0 - 0
Normoblast	0	/100 leucocyte	0 - 0
Others	--	.	0 - 0

Comment : RBCs show moderate normocytic normochromic anemia with normal platelets. WBCs show mild leucocytosis with relative neutrophilia & lymphopenia.

CLINICAL PRESENTATION

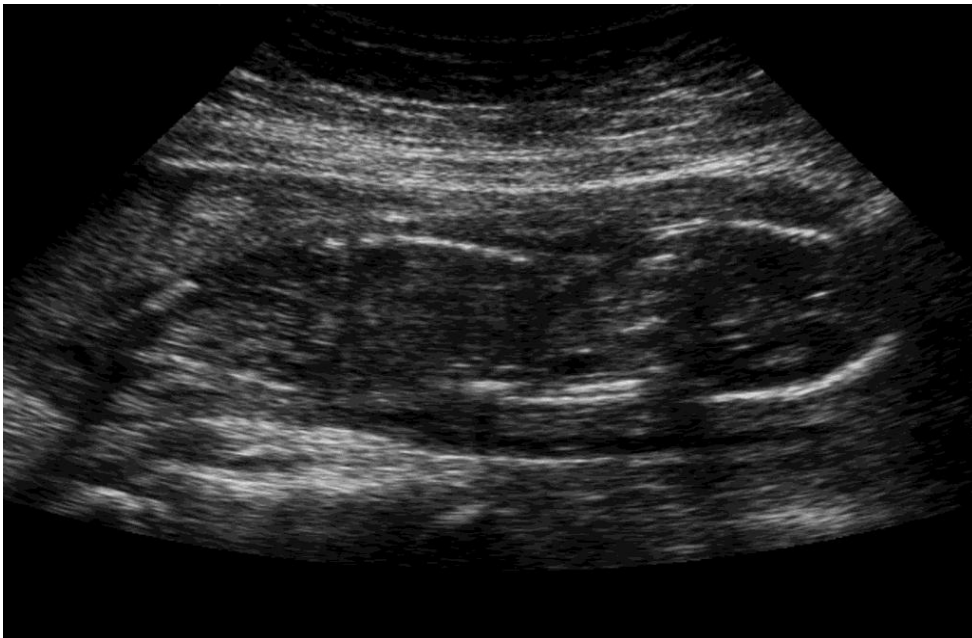
Sudden gush of a **significant amount** of **colourless fluid** through the **vagina**.

Inspection: reveals **clear, colourless, odourless fluid** flowing out of the vagina.

PV examination: **should be avoided** (to minimize risk of infection).

Ultrasound

- Decreased amount of AF around the fetus (OLIGOHYDRAMNIOS). Amniotic fluid pockets <2 cm.
- AFI < 5 cm.



CONFIRMING DIAGNOSIS OF PROM

POOLING TEST



Nitrazine test

Intact membranes

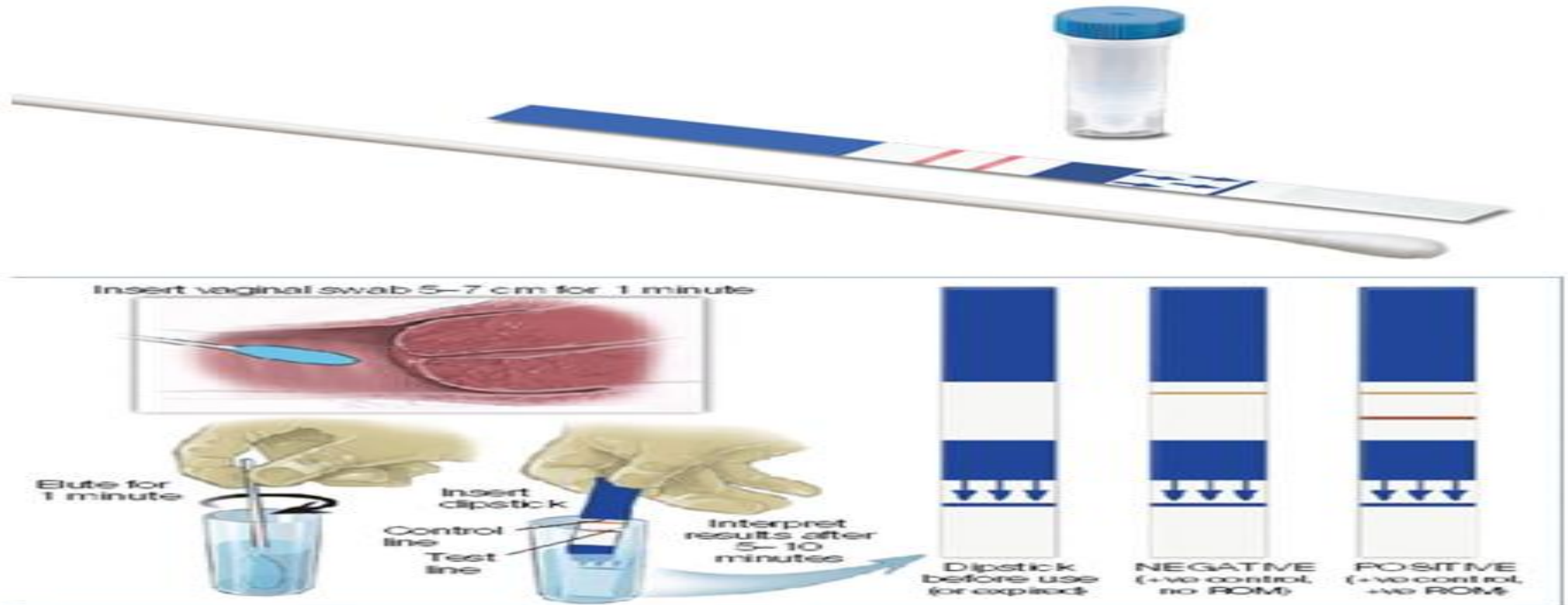


Ruptured membranes



Amnisure

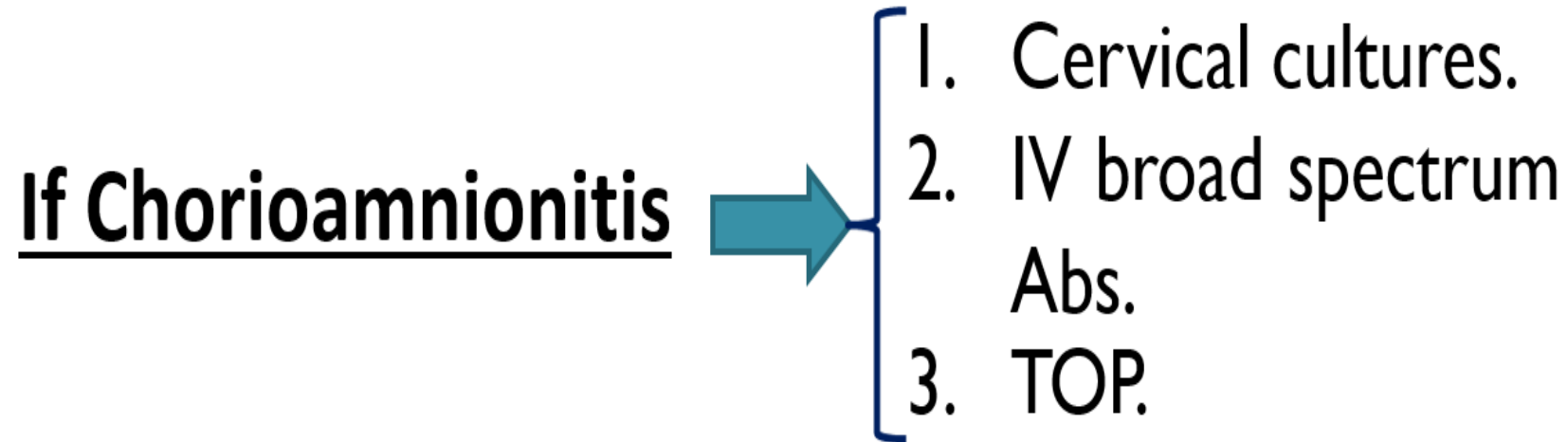
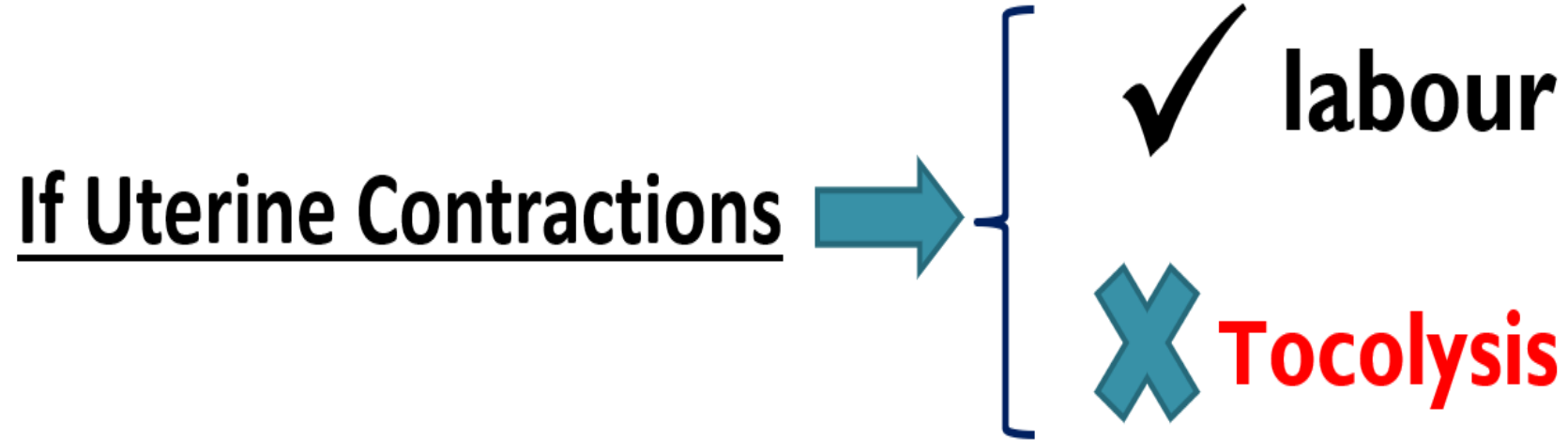
measures placental alpha macroglobulin

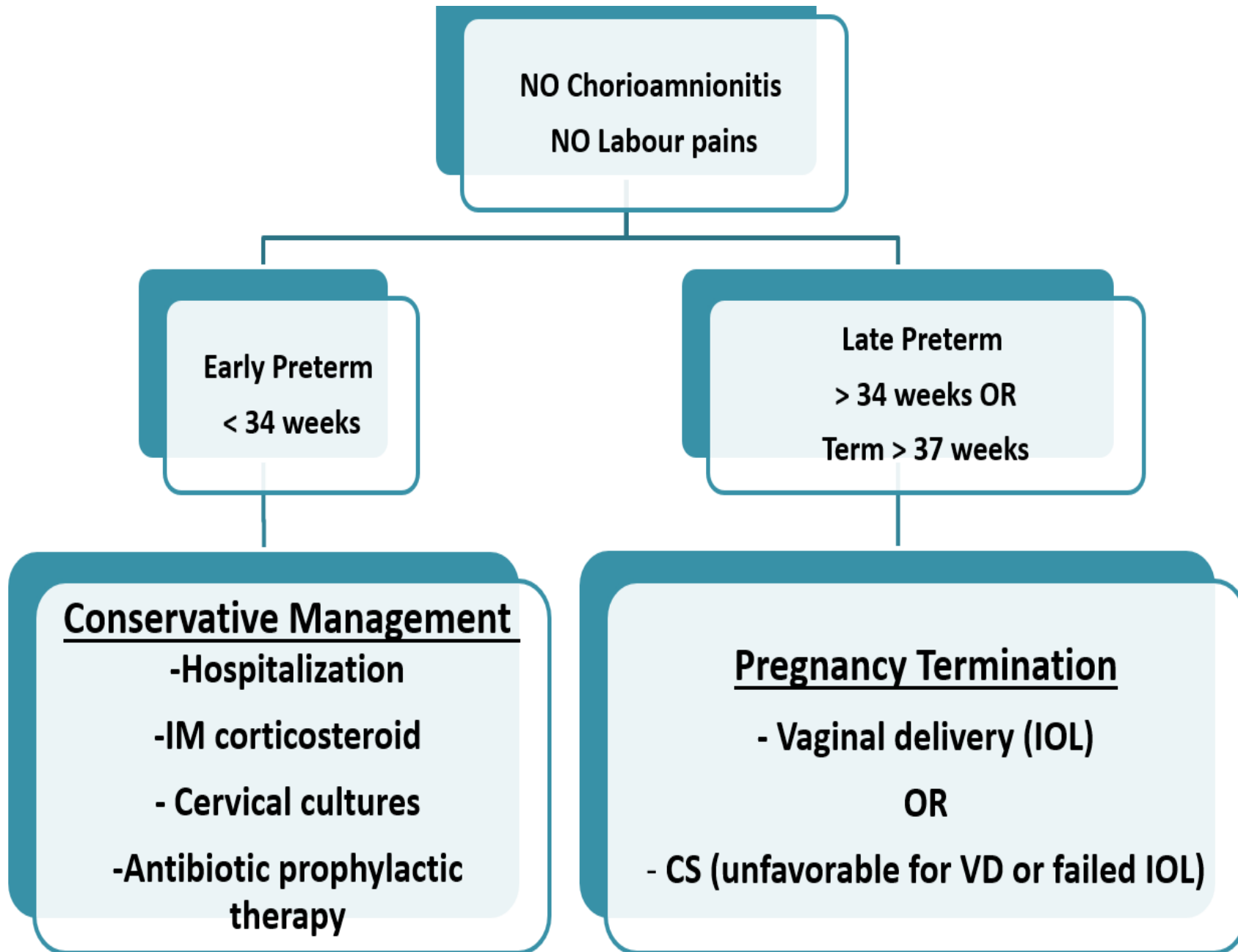


MANAGEMENT OF PROM

Depends on:

1. GA
2. Presence of uterine contractions
3. Infection







Title: **FETAL SURVIELENCE**
DR: - **Mohamed Shabaan**
OB/GYN Department



Learning objectives

1. Early detection and, possibly prevention, of complications specific to fetal distress
2. Detection and management of any disorder complicating fetus during labour
3. Finally classification of patients into normal or high risk to put the plan of proper management.

ANTEPARTUM FETAL SURVETLLANCE

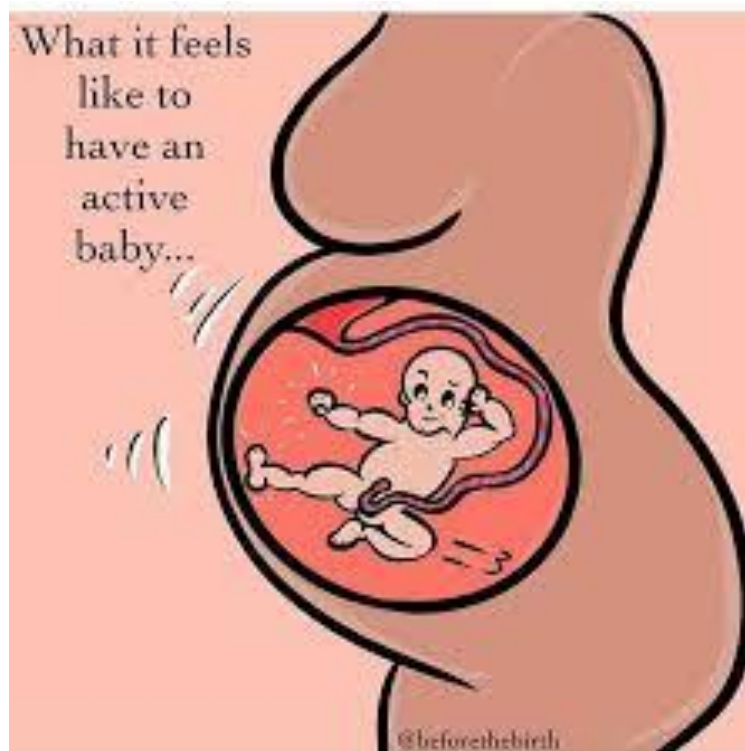
- Antepartum Fetal Surveillance (APFS) will allow identification of pregnancies at high risk for placental insufficiency, and evaluation of fetal wellbeing to avoid fetal loss.
- Fetal surveillance is performed during routine PNC in all pregnancies but is done more frequently in pregnancies at a higher risk for placental insufficiency.

CASES AT HIGH RISK FOR PTACENTAL INSUFFICIENCY

- a) Postdate pregnancies
- b) Oligohydramnios
- c) Intrauterine growth restriction (IUGR)
- d) Medical disorders as preeclampsia, hypertensive disorders, and gestational diabetes mellitus
- e) Recurrent APH associated with placental abruption or placenta praevia

Daily Fetal Movement Count (DFMC)

- Recording the number of fetal kicks over a period of 1 hour after 30 weeks gestation may help in selecting cases at risk for placental insufficiency, for further evaluation.
- If More than 10-12 gross movements within 1 hour indicate good fetal wellbeing.
- If Less than 10-12 gross movements, need extension of the test to another 1 hour if decreased DFMC calls for further tests for evaluation of wellbeing



THE NON-STRESS TEST (NST):

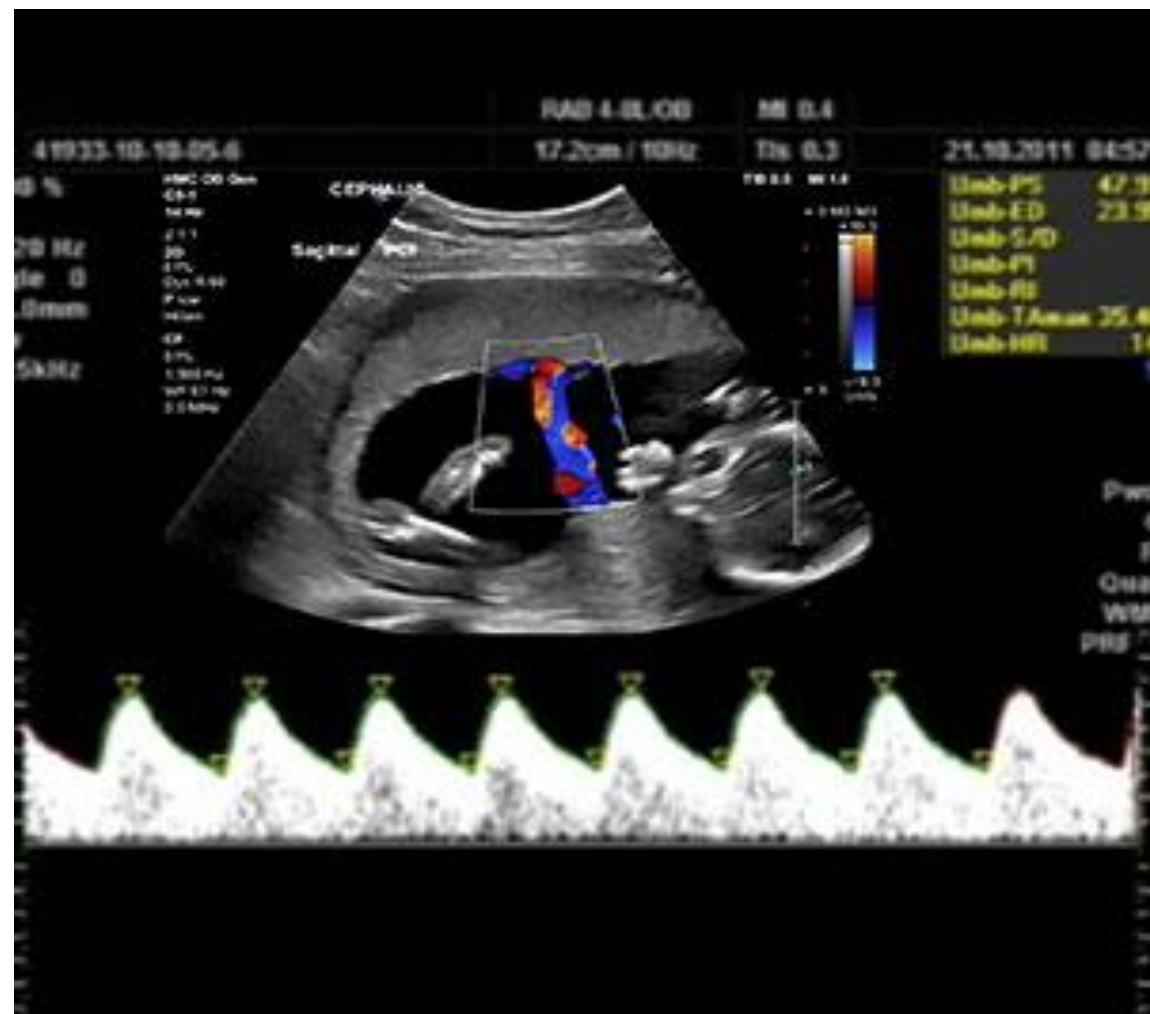
- Electronic monitoring of FHR changes in relation to fetal movements, known as the NST, allows evaluation of fetal wellbeing and selecting pregnancies at high risk for acute placental insufficiency during labor.
- Normal Baseline FHR lies between 120-140 bpm
- Normal reactive fetus when FHR accelerations of > 15 bpm, lasting > 15 seconds occur with each fetal movement for 15-20 minutes test duration.
- Non-Reassuring NST: when movements are minimal or accelerations are not consistent with each movement. Extend the period of the test to another 20 minutes, immediate Biophysical profile score and Doppler tests

THE BTOPHYSTCAL PROFTLE SCORE

- is a simple test based on ultrasound evaluation of fetal tone (flexion / extension), fetal movements (limbs and body), fetal breathing activities (expansion movements of the abdomen), amniotic fluid volume (using AFI), and placental maturity grading (Grannum score).
- Each tested parameter is given a score (either 0 or 2, with a complete score of 10 to 0, in a duration of 30 minutes US examination).
- The BPPS is performed any time in the 3'd trimester and can efficiently pick up those fetuses at risk of hypoxia, and when normal can be repeated every 2 weeks for follow up.

COLOR DOPPLER STUDIES OF FOETAL BLOOD FLOW

- Measurement Of The Fetal Blood Flow Indices Of: -
 - The Fetal Umbilical Artery
 - Middle Cerebral Artery
 - Ductus Venosus
 - Yield Excellent Information About The State Of Fetal Organ Perfusion
- In Addition To US Findings Of: -
 - SGA Fetus
 - Asymmetric IUGR
 - Oligohydramnios.
- Color Doppler Indices Suggestive Of Placental Insufficiency:
 - A) High Resistance Bed For Umbilical Artery Blood Flow (Increased Resistance And Pulsatility Index)
 - B) Low Resistance Bed For Middle Cerebral Artery, Denoting The Vascular Shift Phenomenon
 - C) Absent Or Reversed Umbilical Artery End Diastolic Flow.
 - D) Reversed Flow In The Fetal Ductus Venosus.



INTRAPARTUM FETAL SURVEILLANCE

- Childbirth is a normal physiologic process that will usually end safely both for the mother and her fetus, however 20-30% of pregnancies are considered high risk, with 50% of perinatal morbidity and mortality occurring in this group.
- High risk cases will require a close FHR monitoring during labor to ensure a good fetal outcome other than traditional FHR auscultation by a Sonicaid.

METHODS OF MONITORING FHR DURING LABOUR

A) Auscultation Of The FHR

- For a low-risk pregnancy FHR should be listened to every 30 minutes in the 1st stage and at least every 15 minutes in the 2nd stage of labor.
- For a high-risk pregnancy FHR should be listened to every 15 minutes in the 1st stage and every 5 minutes in the 2nd stage of labor.
- Persistent bradycardia < 100 bpm is an ominous sign for hypoxia and fetal distress



B)continuous electronic FHR monitoring

- An external tocodynamometer is placed on the abdomen to record the **frequency and intensity** of uterine contractions, and a duplex US probe fixed on the abdomen will continuously record FHR with the results plotted on a screen and a special paper that will show basal FHR together with FHR changes in relation to uterine contractions



- **Normal Baseline FHR:** 100-160 b/m, Tachycardia: > 160 b/m and Bradycardia: < 100 b/m
- Normal Beat-to-beat variability: fluctuations of +/- 5 to 10 b/m periodically
- **Early decelerations (head compression):** slowing of FHR with onset of contractions with recovery by its end. Early decelerations are not usually associated with fetal distress,
- **Late decelerations (uteroplacental insufficiency):** slowing of FHR at the peak of the uterine contraction, and recovery short after it ends. Severe repetitive late decelerations usually indicate fetal metabolic acidosis and low arterial PH.
- **Variable decelerations (cord compression):** slowing of FHR which is not related to uterine contractions

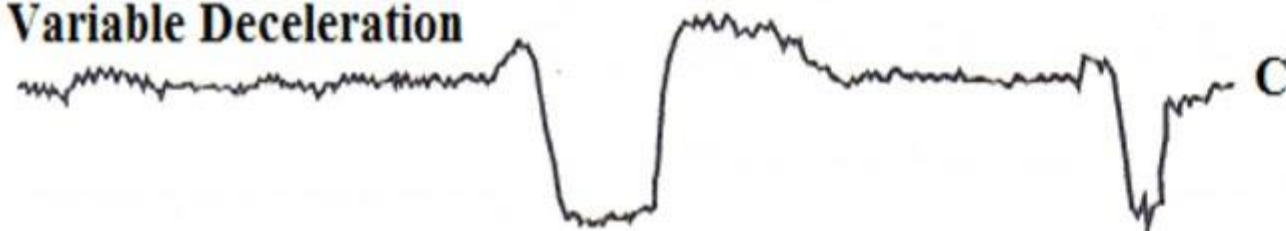
**Fetal Heart Rate
Acceleration**



Early Deceleration



Variable Deceleration



Late Deceleration



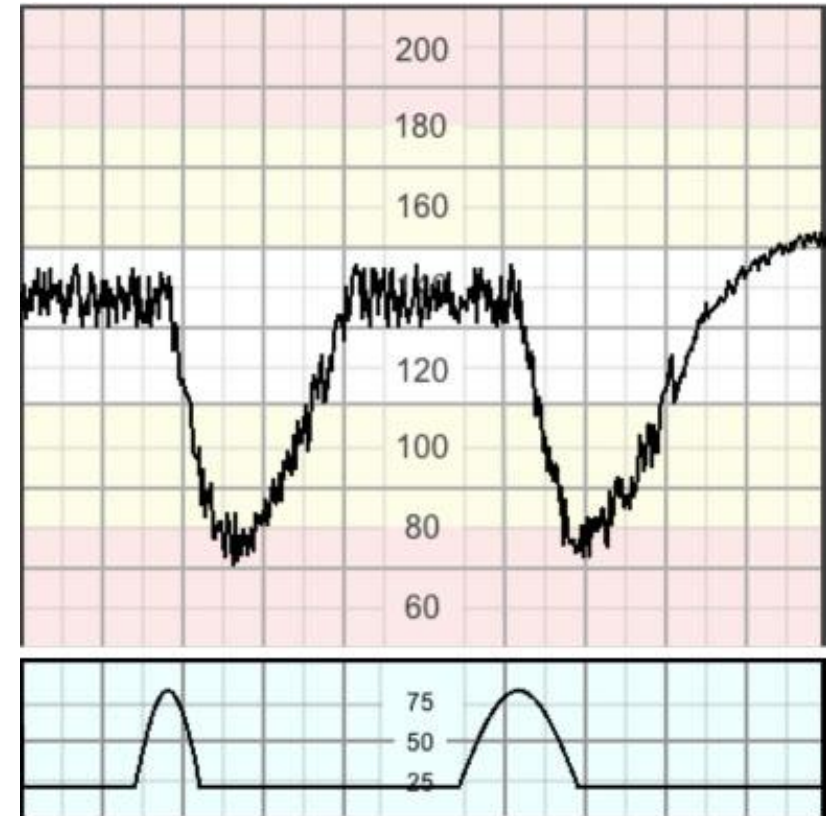
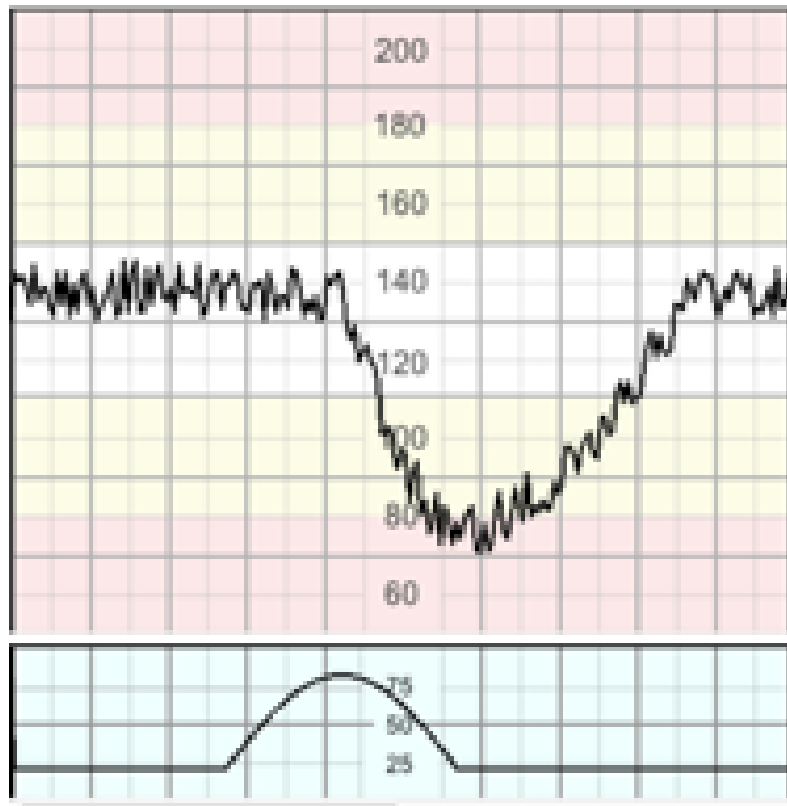
**Uterine
Contraction**



HEAD COMPRESSION

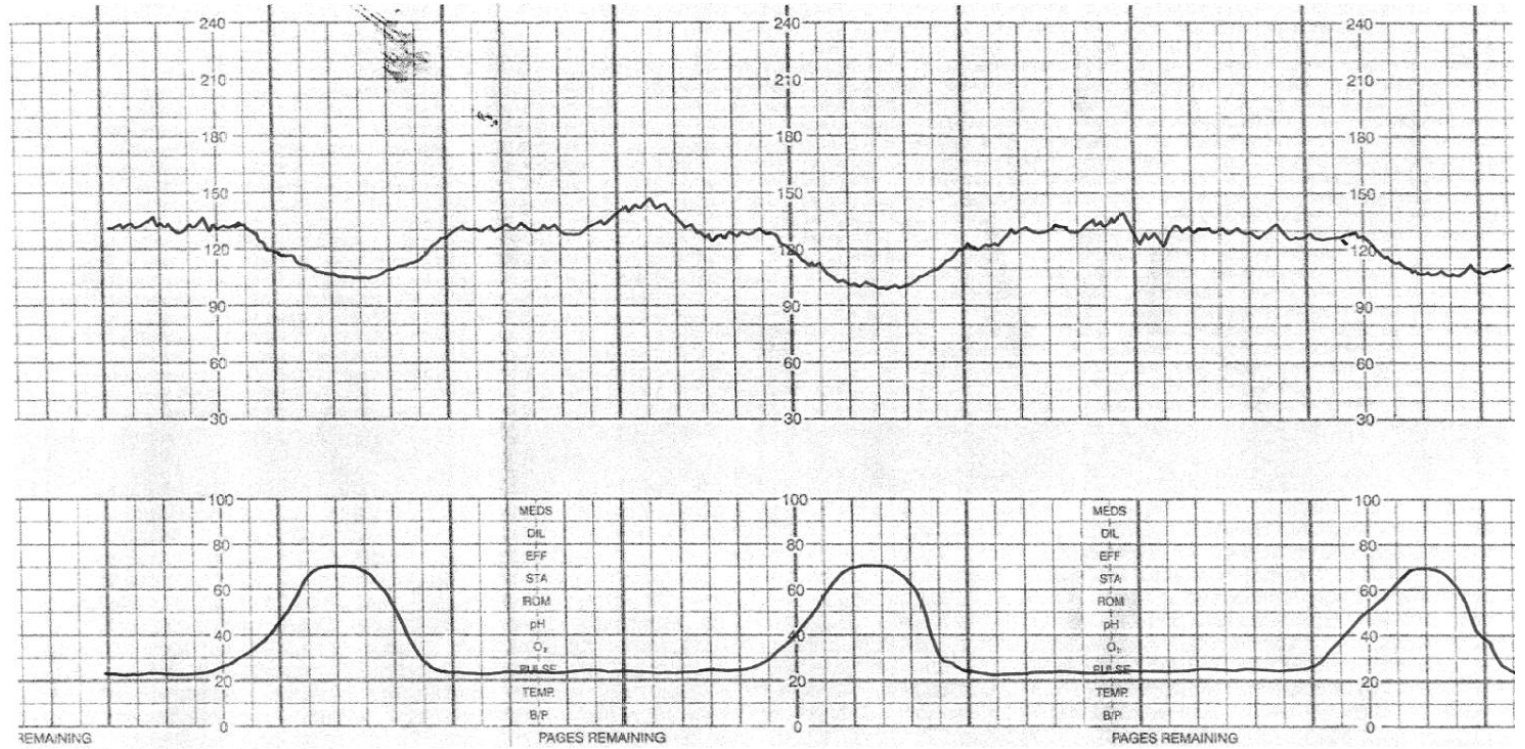
CORD COMPRESSION

UTEROPLACENTAL
INSUFFICIENCY



Late deceleration

Early deceleration



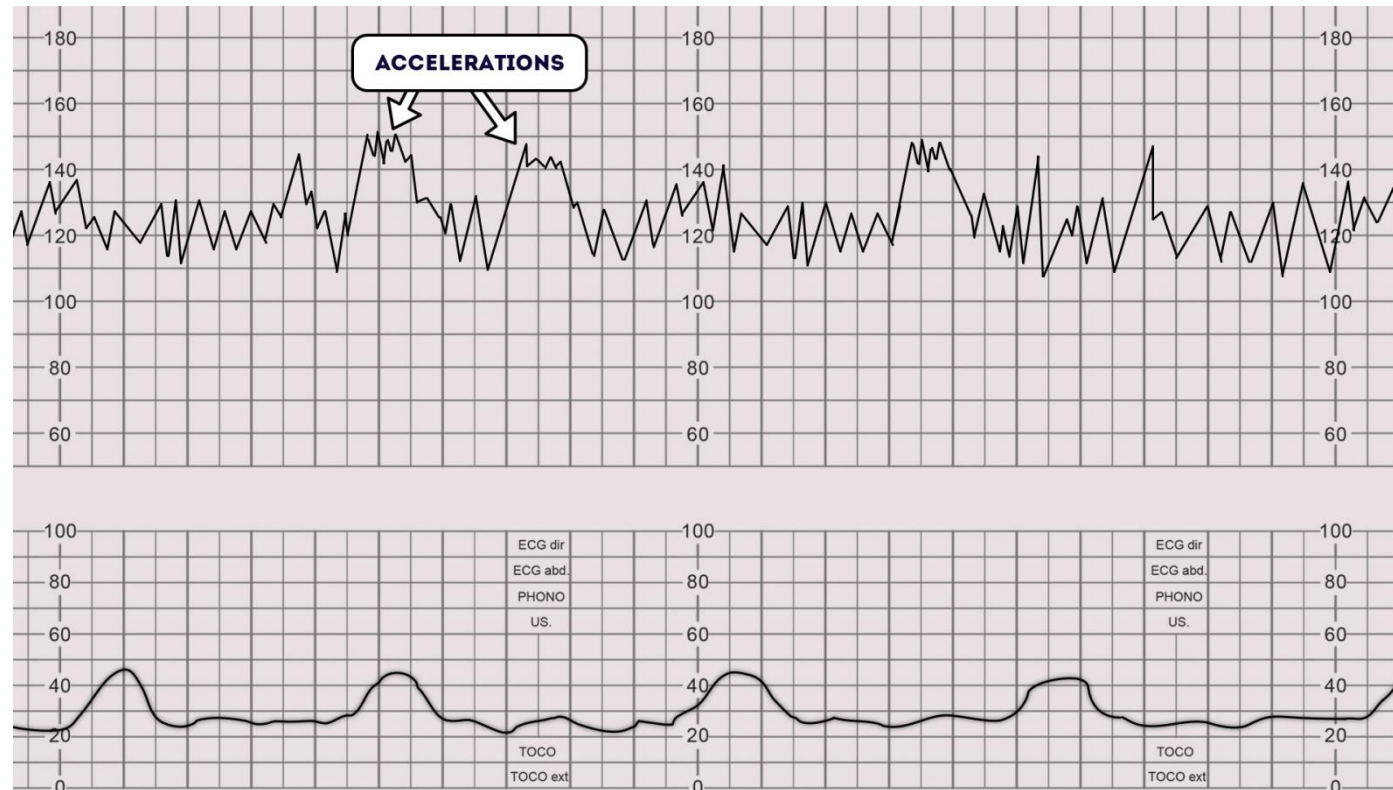


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Faculty of Medicine



Other signs of fetal distress during labour

- A) Fetal tachycardia.
- B) Meconium in amniotic fluid

- **MANAGEMENT OF FETAT DISTRESS DURING LABOUR**

f CTG records abnormal FHR patterns; the following should be done:

- a) Stop oxytocin, if it has been infused
- b) Change position of the mother left lateral position (allows increased blood supply).
- c) O2 mask to the mother (to avoid maternal hypoxia).
- d) I.V. Fluids (to avoid maternal dehydration)



Remember

1. APFS detects fetuses at risk secondary to utero-placental insufficiency but cannot predict sudden events.
2. All pregnant women at high risk for placental insufficiency should do a DFMC in the 3rd trimester.
3. The NST should be offered to cases with decreased DFMC or to high-risk cases in general.
4. Abnormal NST calls for performing a BPPS and Doppler velocimetry studies.
5. Normal NST may be repeated on weekly or biweekly intervals.
6. Abnormal NST confirmed by BPPS and Doppler studies necessitate termination of pregnancy.



- **Remember**

7. Cases at high risk for fetal hypoxia should be selected and carefully monitored throughout labor.
8. Cases with normal FHR patterns are allowed to continue their trial for a safe vaginal delivery
9. Abnormal FHR patterns are managed according to their severity and impact on fetal outcome



Fetal asphyxia, fetal growth disorders, fetal birth injuries

DR: Nadia Helmy
OB/GYN Department

Learning objectives

1. Identify different types of fetal growth disorders, how to diagnose and how to manage.
2. Understanding the causes of fetal asphyxia and management
3. Identify important commonly avoidable causes of neonatal morbidity.

Fetal growth disorders

- ❑ **Appropriate for gestational age (AGA):** fetuses between 10th - 90th percentile for their GA
- ❑ **Small for gestational age (SGA)** fetuses are <10th percentile weight for gestational age
- ❑ **Large for gestational age (LGA)** fetuses are >90th percentile weight for gestational age.

I- INTRAUTERINE GROWTH RESTRICTION (IUGR)

Fetuses with birth weights <10th percentile for their GA.

Types:

A) SYMMETRICAL IUGR (TYPE I):20%

Causes(intrinsic to the fetus):

- Intrauterine infections e.g. rubella.
- Chromosomal abnormalities.
- Skeletal anomalies e.g. osteogenesis imperfecta.
- Congenital malformation e.g. renal hypoplasia

3) **ASYMMETRICAL IUGR (TYPE II):80%**

Causes(extrinsic to the fetus):

1. **Maternal vascular disease:** Chronic hypertension, Preeclampsia, and Diabetic vasculopathy.
2. **Antiphospholipid syndrome:** antibodies cause vascular lesions in uterine and placental bed leading to diminished blood supply and impaired placental infusion.
3. **Chronic renal disease**
4. **Chronic Hypoxia:** heavy smoking and cyanotic heart disease
5. **Placental and cord abnormalities:**
 - ❑ Chronic focal placental abruption, extensive infarction.
 - ❑ Marginal and velamentous insertions of the cord.



Baby with IUGR (left) at term, compared to average sized baby at term

DIAGNOSIS OF IUGR :

1. Clinical Examination: inaccurate

a fundal level lower than expected (2 finger breadths lower than GA) and longitudinal symphysial fundal height (2 cm less than GA) can only suspect a SGA fetus.

2. Ultrasound Assessment:

a) Estimated fetal weight < 10th percentile for GA.

b) Foetal Biometry:

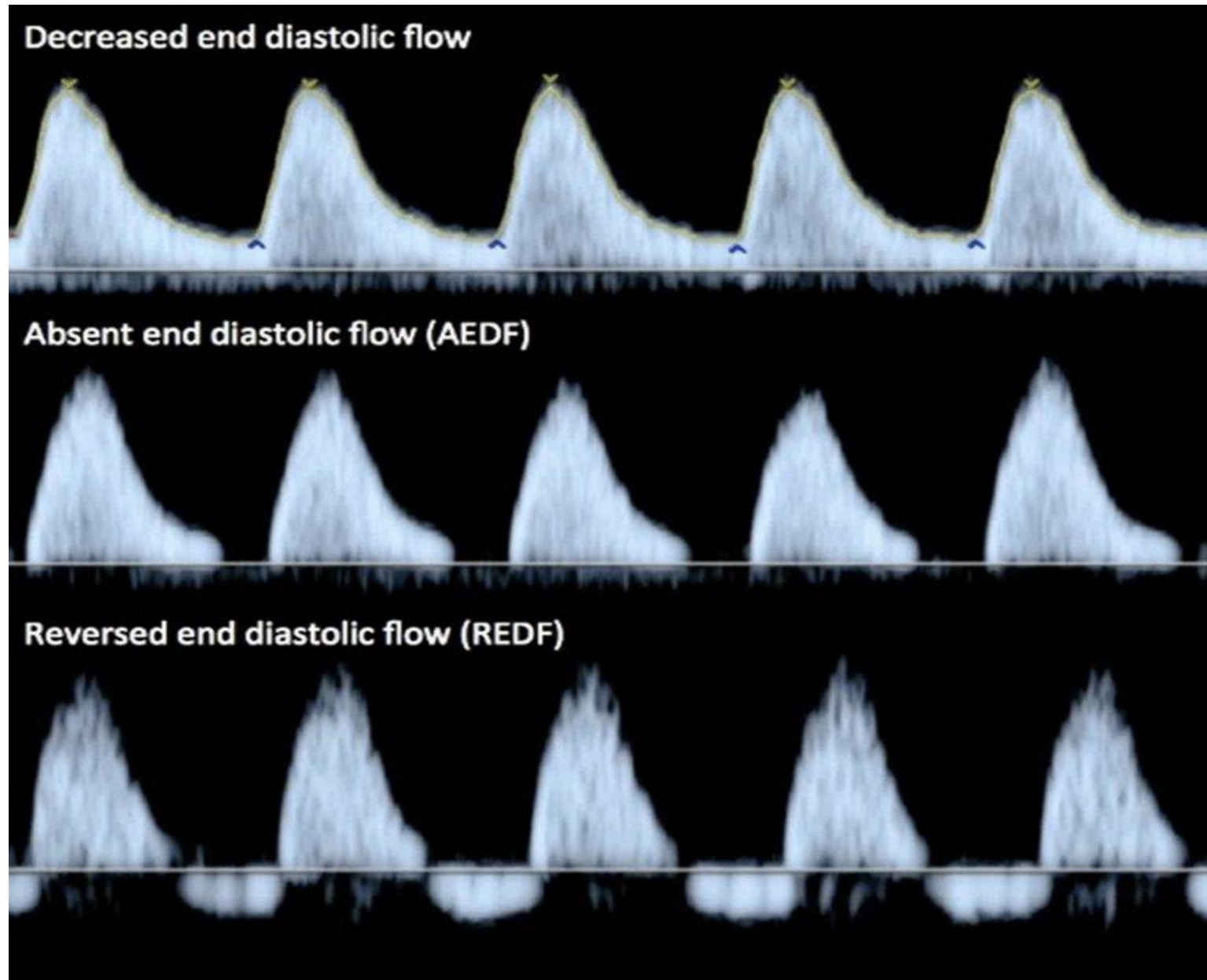
☐ Type I IUGR: Symmetrical reduction in HC and AC

☐ Type II IUGR: altered HC/AC ration due to brain sparing phenomenon.

c) Oligohydramnios is common with type II IUGR.

d) Accelerated placental aging with calcifications.

e) Abnormal umbilical and cerebral artery Doppler flow indices (decreased umbilical and preserved or increased cerebral vascular flow in asymmetrical IUGR).



4) *TERM and NEAR TERM IUGR:*

☐ Termination of pregnancy is through either induction of labor or CS according to obstetric indications, guidelines, facilities, and available team experience

B) PRETERM IUGR: conservative management

1. GENERAL MEASURES:

- Physical and mental rest: hospitalization is individualized according to each case
- Stop smoking and improve nutrition
- Low dose aspirin (LDA)

2. REGULAR FOETAL SURVEILLANCE:

- Daily fetal kick count done by the patient
- Bi-weekly Non stress test (NST); for evaluation of foetal well being
- Bi-weekly US for assessment of a biophysical profile score and amniotic fluid index
- Weekly US Doppler for umbilical artery and middle cerebral artery blood flow assessment

II. FOETAL MACROSOMIA

Fetuses with birth weights >90th percentile for their GA.

RISK FACTORS

Maternal diabetes is the most common and most important risk factor for foetal macrosomia.

Post-term pregnancy

Maternal obesity

Grand multipara

A previous LGA baby.

DIAGNOSIS

1. Clinical estimates of foetal size and weight based on fundal height measurements, are often unreliable and are markedly affected by clinical experience and maternal obesity.
2. Ultrasound estimates of foetal weight are reasonably accurate with only 10-15% error range, however US estimates of foetal weight are more difficult in obese women.

PREVENTION

1. Early diagnosis of GDM and meticulous control of established cases.
2. Control of weight gain during pregnancy at regular antenatal care visits.

MANAGEMENT

1. During pregnancy: regular antenatal care, control excessive weight gain, repeated investigation to discover GDM, serial US estimates for growth curves, and anomaly scans.
2. Vaginal delivery: If attempted, anesthesia staff and neonatal resuscitation team must be available. Assisted instrumental vaginal delivery **MUST BE AVOIDED**



COMPLICATIONS

A) Fetal Complications:

- IUFD (in diabetic macrosomia or serious congenital malformations)
- Birth trauma (shoulder dystocia and brachial plexus palsy)
- Hypoglycemia, polycythemia, hypocalcemia and jaundice.

B) Maternal:

- Higher incidence of C.S. deliveries (due to prolonged, or obstructed labor).
- Traumatic injuries of birth canal: (rupture uterus, cervical and vaginal lacerations)
- Postpartum hemorrhage (PPH), whether atonic (due to uterine overdistension), or traumatic (due to lacerations).

Fetal asphyxia

Intrauterine foetal asphyxia (IUFA) and neonatal asphyxia (NNA) are serious situations, which might put the foetus (infant) at risk for loss of life or permanent CNS disability.

IUFA is a state of inadequate oxygenation and inadequate elimination of CO₂, which if allowed to be continued, will result in metabolic acidemia (umbilical arterial blood pH < 7.2).

CAUSES OF IUFA

I. Maternal causes: (conditions leading to imperfect oxygenation of maternal blood): Severe anemia, Hemorrhage & shock, Respiratory failure, and heart failure.

II. Placental causes:

- ❑ Placental compression: interfering with its circulation as in tonically contracted uterus
- ❑ Placental separation as in accidental haemorrhage.
- ❑ Placental insufficiency: (extensive degeneration, multiple infarcts & abnormally small placenta).

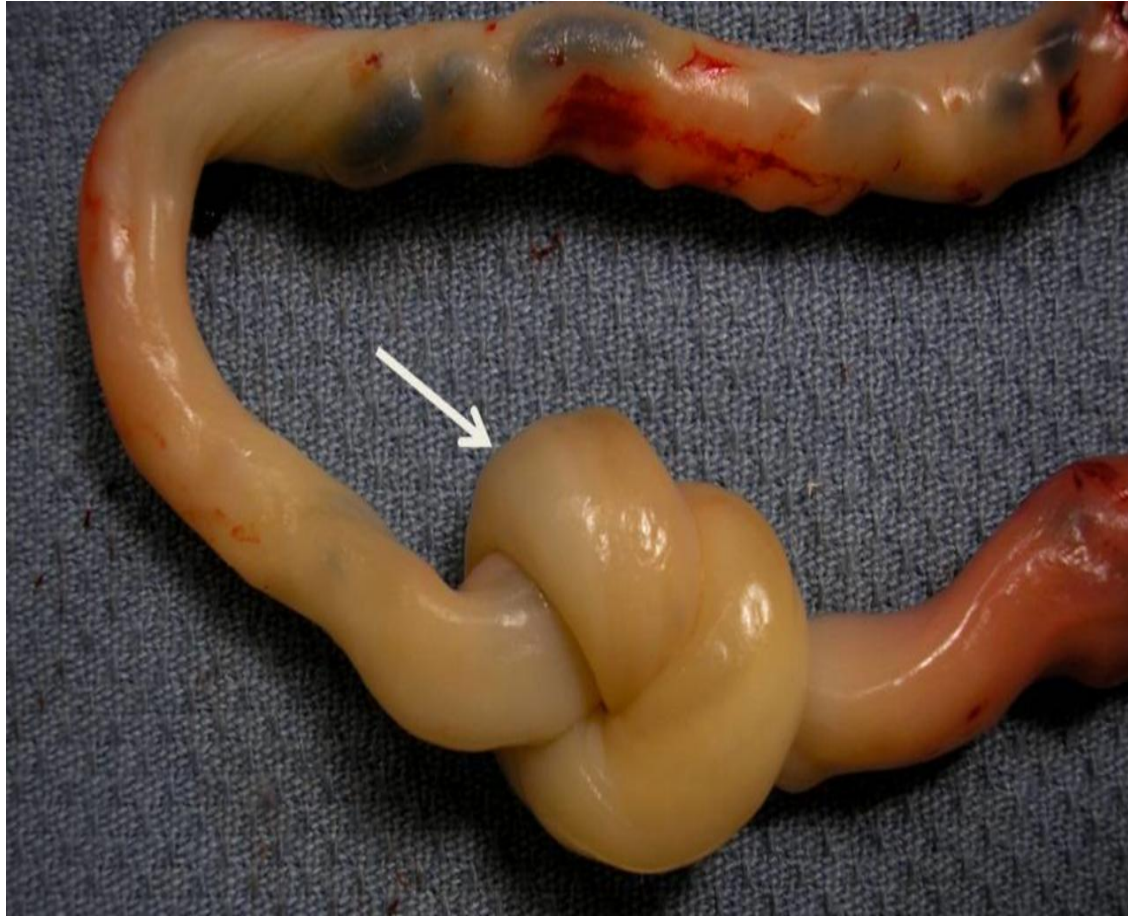
I. Causes in the umbilical cord: Obstruction of the circulation, which may be due to:

1. Tight coils around the neck.
2. True knots of the cord.
3. Prolapsed cord.
4. Compression of the vessels by hematoma of the cord or by blades of the forceps.

IV. Prolonged fetal head compression :

This will cause oedema and ischemia, which interfere with the blood supply of the medulla leading to depression of the respiratory center. Prolonged compression may be due to:

1. Contracted pelvis (C/P disproportion).
2. Rigid perineum.
3. Intracranial hemorrhage.
4. Forceps application for a long time.



True knot



Cord around the neck

CLINICAL FEATURES OF IUFA (FOETAL DISTRESS FD):

1. Abnormal Fetal heart rate (FHR)
 - Bradycardia (FHR < 100 bpm); most dangerous sign
 - Tachycardia (F.H.R. >160 bpm)
2. Delay of return of the FHR to their normal rate after uterine contraction.
3. If continuous electronic FHR monitoring and CTG would suggest fetal distress:
 - a. Late deceleration
 - b. Variable deceleration.
 - c. Loss of beat-to-beat variation in FHR
4. Passage of meconium in cephalic presentations, due to relaxation of the anal sphincter due to anoxia and increased intestinal peristalsis.
5. Foetal acidosis: Detected by taking blood samples from the scalp of the foetus during labour; pH below 7.2 indicates foetal asphyxia (N. 7.25- 7.35).

Fetal Heart Rate

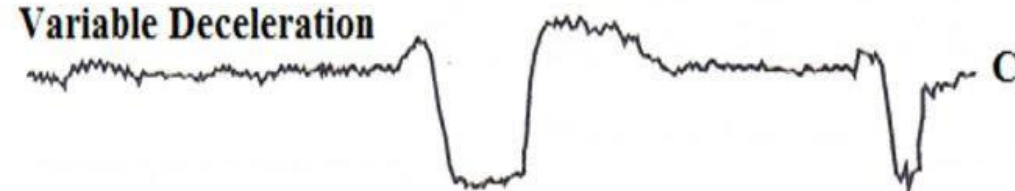
Acceleration



Early Deceleration



Variable Deceleration



Late Deceleration



Uterine Contraction



HEAD COMPRESSION

CORD COMPRESSION

UTEROPLACENTAL
INSUFFICIENCY

Description	Feature		
	Baseline (beats/minute)	Baseline variability (beats/ minute)	Decelerations
Normal/ reassuring	100–160	5 or more	None or early
Non- reassuring	161–180	less than 5 for 30–90 minutes	Variable decelerations: • dropping from baseline by 60 beats/minute or less and taking 60 seconds or less to recover, • present for over 90 minutes • occurring with over 50% of contractions OR Variable decelerations: • dropping from baseline by more than 60 beats/minute or taking over 60 seconds to recover • present for up to 30 minutes • occurring with over 50% of contractions OR Late decelerations: • present for up to 30 minutes • occurring with over 50% of contractions
Abnormal	Above 180 or below 100	Less than 5 for over 90 minutes	Non-reassuring variable decelerations (see row above): • still observed 30 minutes after starting conservative measures • occurring with over 50% of contractions OR Late decelerations • present for over 30 minutes • do not improve with conservative measures • occurring with over 50% of contractions OR Bradycardia or a single prolonged deceleration lasting 3 minutes or more

MANAGEMENT OF INTRAUTERINE ASPHYXIA (FOETAL DISTRESS)

1. Elimination of the cause of asphyxia if possible:

- a. The patient should be turned onto her side: left lateral position.
- b. Oxytocin infusion, if started, should be discontinued
- c. Any hypotension should be corrected

2. Oxygen (100%) should be administered to the mother by facemask.

3. Atropine given to the mother may be beneficial in some cases of fetal bradycardia.

4. If FD is relieved: careful observation by EFHR monitoring till delivery is accomplished.

5. If foetal distress is not relieved within several minutes, immediate delivery is indicated;.

☐ If the cervix is not fully dilated: C.S. is immediately performed regardless presentation

☐ If the cervix is fully dilated; delivered by:

- Forceps : in cephalic presentation, engaged head, CPD excluded
- Breech extraction: in breech presentation.

Fetal birth injuries

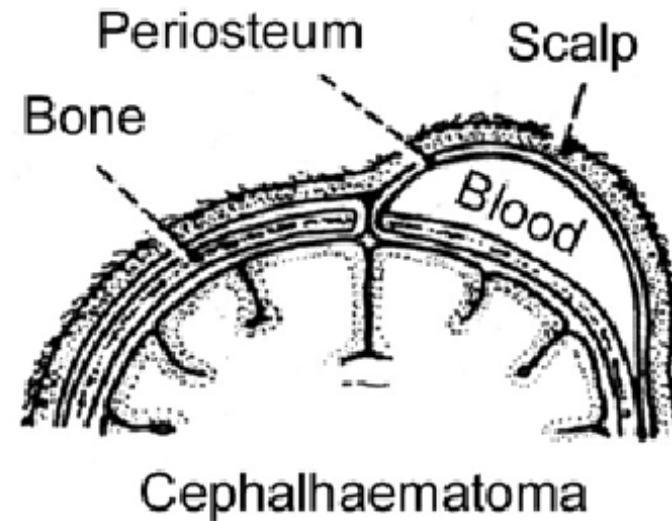
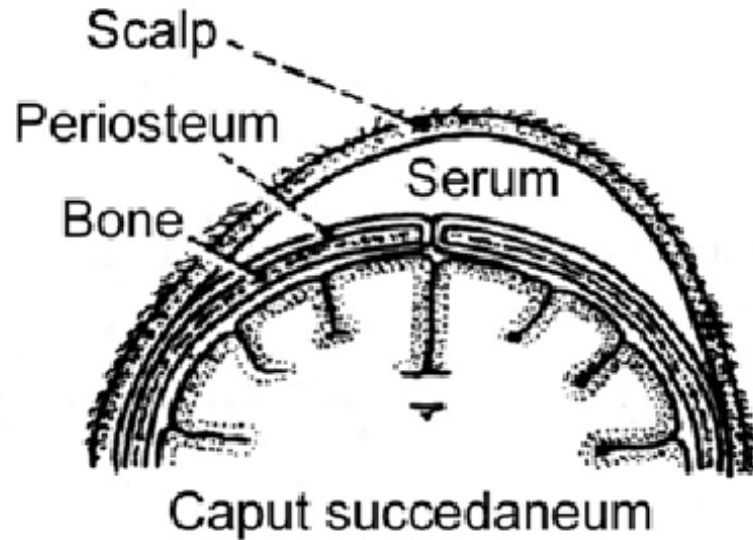
Fetal birth injuries represent an important commonly avoidable cause of neonatal morbidity and mortality. They vary from minor skin abrasions to severe intracranial hemorrhage.

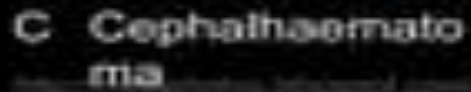
CEPHALHEMATOMA (Sub-periosteal hematoma)

		Caput succedaneum	Cephalhaematoma
Cause		<i>Obstructed labour</i>	<i>Forceps or ventouse</i>
		<i>Ventouse</i>	<i>Difficult delivery through contracted pelvis</i>
Appearance		<i>At birth</i>	<i>Few hours after birth</i>
Character	<i>Edges</i>	<i>ILL defined</i>	<i>Well defined</i>
	<i>Skin</i>	<i>May be ecchymotic</i>	<i>Normal</i>
	<i>Sutures</i>	<i>Overlap sutures and cover more than 1 bone</i>	<i>No overlap & limited to one bone</i>
Complications		<i>No</i>	<i>Calcification, infection and hyper-bilirubinaemia</i>
Treatment		<i>No treatment (disappears after 1 -2 days)</i>	<i>Expectant treatment (disappears within few weeks)</i>

? **Treatment:**

- Expectant as spontaneous resolution mostly occurs
- Vit K 1 mg IM could be given (to guard against increase in size due to fragile blood vessels)
- Antibiotics are sometimes given.





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Caput succedaneum



Cephalhaemato
ma

11138 *Chickadee*, G. Clayton, Illustrated
Newspaper of Ornithology 1911



INTRACRANIAL HAEMORRHAGE

AETIOLOGY:

1. Prematurity, due to:

☐ Fragile blood vessels. ☐ Hypoprothrombinemia. ☐ Increased susceptibility to birth trauma.

2. Breech delivery: due to sudden compression and decompression of cranial bones.

3. Excessive compression, due to:

☐ Excessive moulding, in cases of cephalopelvic disproportion. ☐ Excessive compression by forceps (oblique application or persistent locking).

4. Asphyxia: leads to hypoxia of the walls of blood vessels with subsequent leakage.

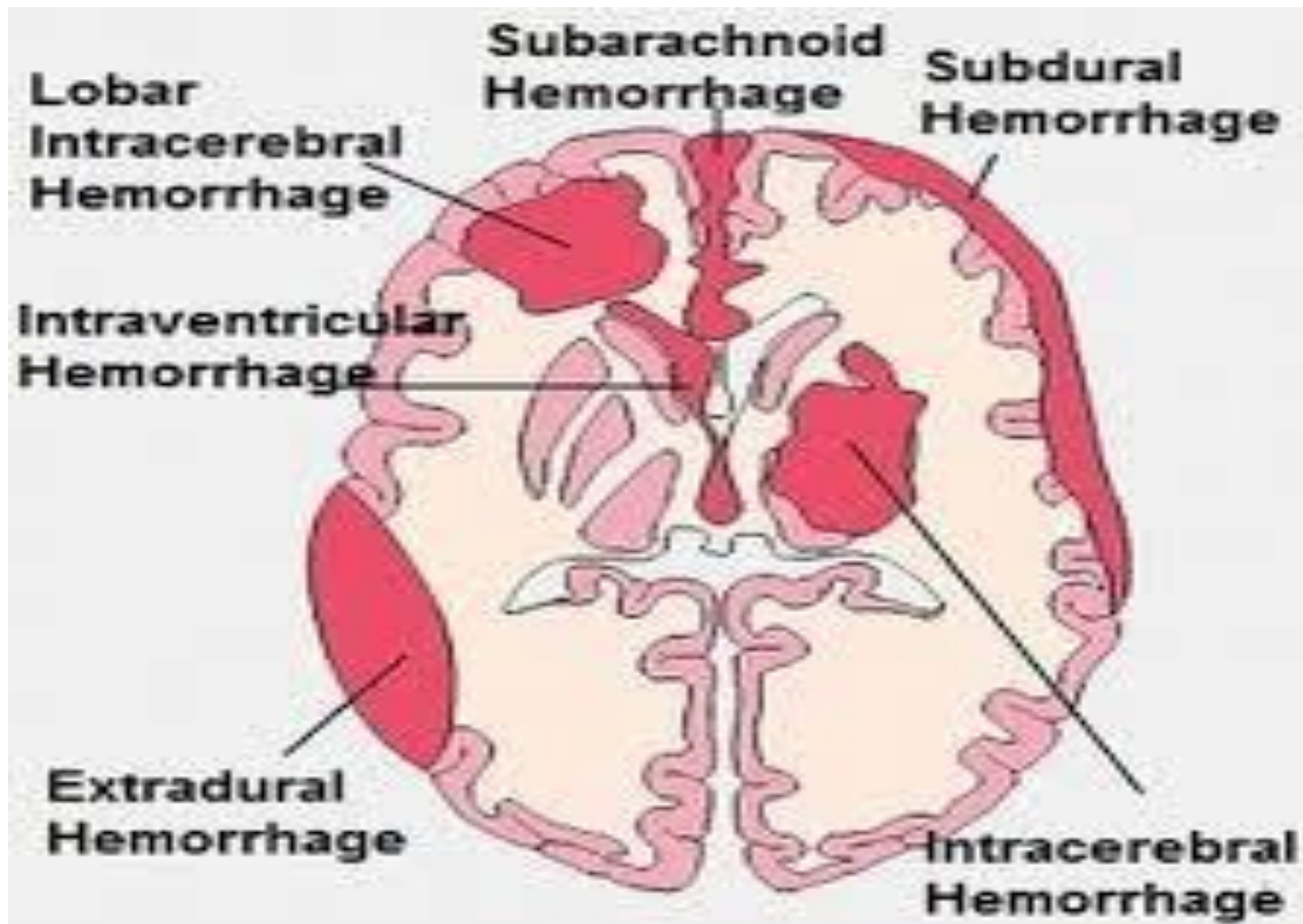
5. Haemorrhagic disease of the newborn.

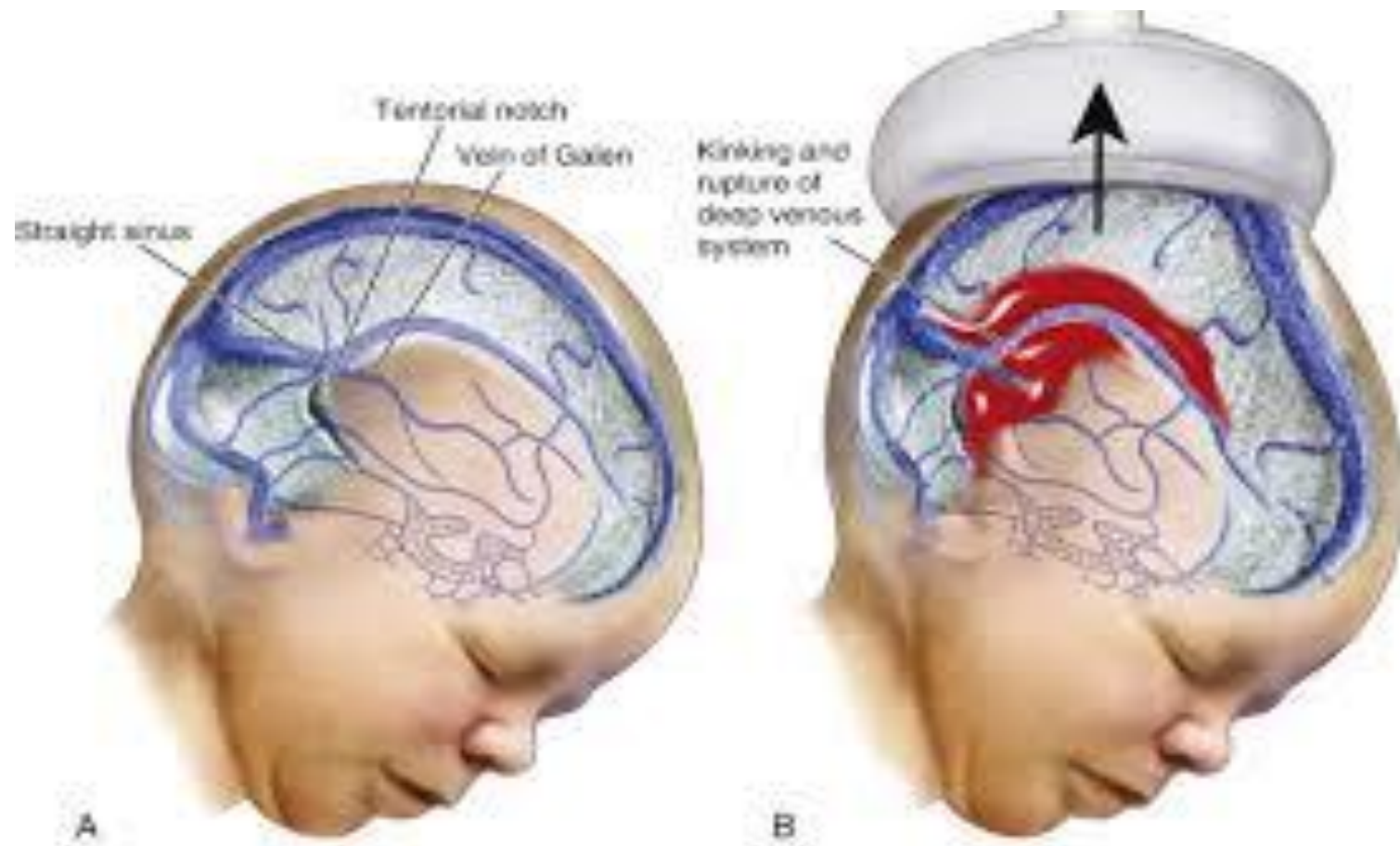
• Sites of Haemorrhage:

1. Intra-ventricular hemorrhage.
2. Intra-cerebral hemorrhage.
3. Subdural hemorrhage.
4. Subarachnoid hemorrhage,

N.B.: Subdural and subarachnoid hemorrhages usually develop with traumatic delivery.

- The vein of Galen is torn due to tear in dura at junction of falx cerebri with tentorium cerebelli (This usually results from excessive moulding or forceps delivery)





• **CLINICAL PICTURE:**

- 1. Stillbirth or neonatal asphyxia with very difficult resuscitation.
- 2. Drowsy, refuse suckling with irregular breathing and sudden sharp cry.
- 3. Neurological manifestations as Convulsions, rigidity and paralysis are common.
- 4. Tense bulging anterior fontanel.
- 5. Vomiting.

• **DIFFERENTIAL DIAGNOSIS:**

1. Asphyxia neonatorum.
2. Neonatal convulsions.

INVESTIGATIONS: Brain CT scan.

• Treatment:

- Prophylactic Treatment

1. Avoid the cause as traumatic delivery, prolonged labor and forcible delivery of premature babies.
2. Vitamin K for the mother (10 mg IM with preterm labor)

- Active Treatment

- 1. sedative for convulsions and combat brain edema by NaCl or MgSo4 per rectum.
- 2. Vitamin K for the fetus (1 mg IM).
- prophylactic antibiotic.

NERVE INJURIES

I. Facial Nerve Palsy:

Cause: Compression of the nerve by blade of forceps on stylomastoid foramen results in oedema and hematoma around the nerve.

Clinical picture: unilateral and temporary

- ❑ Absent nasolabial fold.
- ❑ Angle of the mouth is deviated to the healthy side.
- ❑ Absent blinking on the affected side.

The power of suckling is not usually affected.

Treatment:

Conservative management. May need corticosteroids.

Facial Palsy





facial palsy

NERVE INJURIES

II. Brachial Plexus Injury:

Cause: Forcible lateral flexion of the head during delivery causes damage of the roots of brachial plexus (oedema and haematoma around the nerves) especially in cases of shoulder dystocia.

Clinical picture:

A) Upper injury (Erb's palsy):

❑ Injury to C5 and C6.

❑ Characters: Policeman tip position:

1. The affected limb is adducted to the body and internally rotated.
2. Elbow is extended.
3. Wrist is flexed.

Erb's palsy



B) Lower injury (Klumpke's palsy):

❑ Injury to C7, C8 and T1.

❑ Characters:

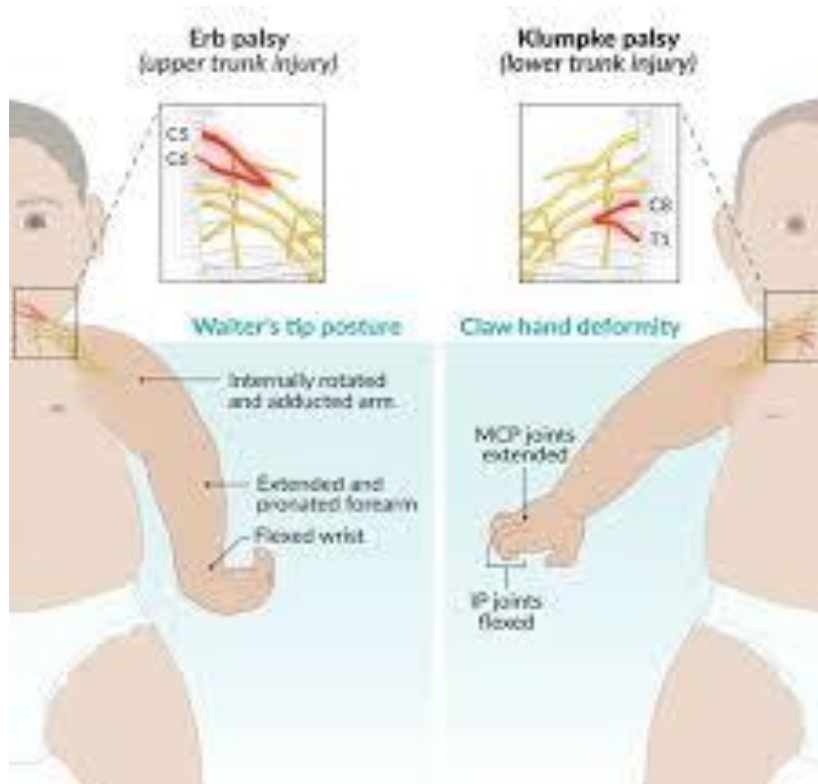
1. Wrist drop (Ape hand).
2. Absent grasp reflex.
3. Paralysis of small muscles of hands (atrophy).
4. Weakness of long flexors of wrist and fingers

Treatment:

Injuries vary from nerve contusion to actual nerve cut.

❑ Physiotherapy. ❑ Rarely, it may need plastic correction.

Klumpke's palsy



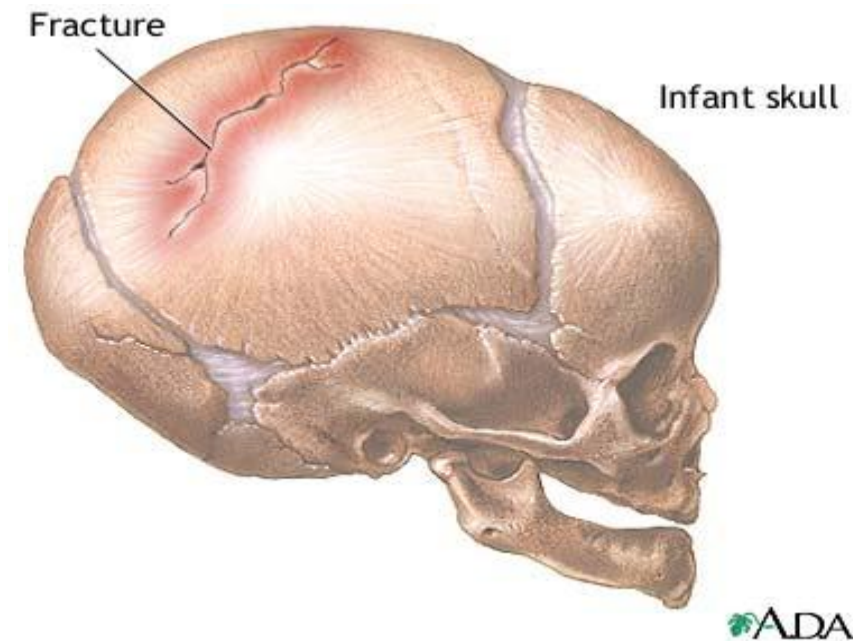
Other INJURIES

Fracture of the skull

Cause: may occur after difficult forceps delivery

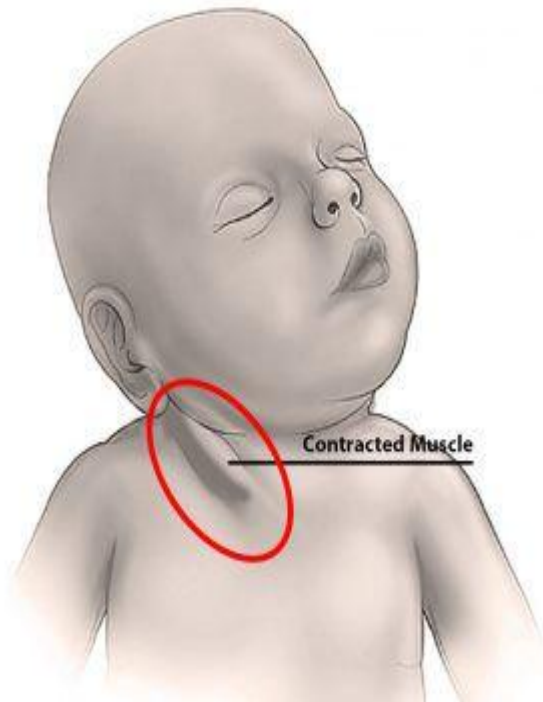
It may be linear or depressed fracture.

The latter is usually associated with intracranial hemorrhage.



Other INJURIES

Injury to vertebral column, abdominal organs or sternomastoid muscle may occur in difficult breech delivery.





Title: -Prenatal Diagnosis Of Foetal Chromosomal And Congenital Abnormalities

DR: - Marwa Abdalla
OB/GYN Department



- Learning objectives

1. To know how to understand prenatal diagnosis.
2. To know how to diagnose pregnancy complicated with Foetal Chromosomal And Congenital Abnormalities.

- **Prenatal diagnosis** entails in utero diagnosis of fetal **congenital** or **chromosomal** abnormalities, to facilitate dealing with such problems appropriately during pregnancy or immediately after delivery. Prenatal diagnosis can be achieved in early pregnancy through;
 - **Screening tests:** as *maternal blood sampling* and *level 2 ultrasonography* **and/or**
 - **Diagnostic tests:** as *fetal sampling* and *level 3 ultrasonography*.



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- **FOETAL STRUCTURAL CONGENITAL ANOMALIES**

- These describe abnormalities or malformations in organ development as occurs in:
 - - *Neural tube defects (NTDS)* as anencephaly, spina-bifida, hydrocephalus, etc...
 - - *Abdominal wall defects* (exomphalos) and congenital *diaphragmatic hernia*.
 - - *Cardiac anomalies* as ASD , VSD, and transposition of great vessels
 - - *Renal anomalies* as renal agenesis, unilateral or pelvic kidney and obstructive uropathy.
 - - *Tumours* as sacro-coccygeal teratoma, congenital goitre, ...
 - - *Lymphatic channel obstruction* as cystic hygromas, and many others



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- **FOETAL CHROMOSOMAL ANOMALIES:**

- **Aneuploidy** occurs when the chromosome **number** is not an exact multiple of **23**. Most common aneuploidy are **trisomy 21** (Down syndrome), trisomy **18** (Edward syndrome) and trisomy **13** (Patau syndrome).
- **Nondisjunction** is an error in **meiosis** that results in a gamete with one chromosome too few, or too many. They increase with advancing maternal age. Fertilization of the gamete with one extra chromosome results in a conception with **47** chromosomes. Nondisjunction may involve sex chromosomes resulting in abnormalities such as 47 XXY, 47 XYY, etc...
- **N.B.:** Maternal age **above 35** years is a risk factor for a foetus with **nondisjunction** chromosomal defect, **aneuploidy**, and some structural **congenital anomalies**. The incidence increases with advancing maternal age.

- **HIGH-RISK CASES FOR PRENATAL SCREENING AND DIAGNOSIS**

1. Maternal **age** more than **35** years (increased chromosomal and structural anomalies) Cases with a history of **familial** genetic disease, past **obstetric** history of previous babies with con-genital or chromosomal anomalies, and cases with previous recurrent pregnancy loss.
2. History of exposure to **teratogenic drugs** or large doze ionizing **irradiation** in early pregnancy.
3. Exposure to certain **infections** during pre-conception or early pregnancy.

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TYPES OF PRENATAL TESTS

■ SCREENING TESTS:

Screening tests should be *cheap, safe, non-invasive, and readily available*, with a reasonable diagnostic accuracy & low false negative rates (sensitivity). The aim of screening is to select a subgroup which will benefit from more complicated expensive diagnostic procedures. Prenatal screening include;

- Maternal *serum biochemical markers* tests (double marker DMT & triple marker TMT)
- US screening for nuchal translucency (NT)
- New techniques as *cell-free foetal DNA* (cfFDNA).

• SCREENING TESTS FOR FOETAL CONGENITAL ABNORMALITIES

• 1. MATERNAL SERUM BIOCHEMICAL MARKERS

• THE DOUBLE MARKER TEST (DMT)

- Combining maternal **serum markers** (hCG & PAPP-A) together with **US** measurement of the nuchal translucency (**NT**) at the foetal neck in one test performed at **11-13** weeks gestation, as shown later, can improve the predictive accuracy for the risk for **Down syndrome** to **> 85%**, with a false positive rate of **< 5%**. This will allow for early diagnosis and early intervention when indicated.
- **Down syndrome** fetuses usually show **high** hCG, **low** PAPP-A levels, together with increased NT **> 3.0** mm at 12 weeks gestation. A high risk score calls for further diagnostic tests.
- **Risks** for **trisomies** 18 & 13 (Edwards and Patau syndromes) can be included in screening by the Double marker test (DMT).

- **SCREENING TESTS FOR FOETAL CONGENITAL ABNORMALITIES**
- **1. MATERNAL SERUM BIOCHEMICAL MARKERS**
- **THE TRIPLE MARKER TEST (TMT)**
- The combination of **MSAFP**, **hCG**, and **UE3**, in one test performed in the second trimester (15-16 weeks gestation) yields good information about the risk of trisomies, taking in consideration maternal age, weight, and gestational age.
- **Down syndrome** risk is high when the test shows **high** hCG levels with **low** MSAFP and uE_3 levels. The test is less commonly used nowadays in favor for the double marker test which is both more accurate and provides an earlier first trimester diagnosis.

SCREENING TESTS FOR FOETAL CONGENITAL ABNORMALITIES

2. ULTRASOUND SCREENING

■ US MEASUREMENT OF NUCHAL TRANSLUCENCY (NT):

Measurement of skin fold thickness of the foetal neck at 11 -13 weeks gestation, shows good correlation with the presence of foetal chromosomal abnormalities. NT thickness **> 3.0 mm** weeks should call for more invasive diagnostic work up to exclude Down syndrome as CVS and amniocentesis .



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Increased NT: examples



Normal NT thickness:
Normal fetus



Increased NT thickness:
Fetus with Down's syndrome

ultrasoundlink.net © 2012

SCREENING TESTS FOR FOETAL CONGENITAL ABNORMALITIES

3. CELL FREE FOETAL DNA (cfFDNA) for screening and diagnosis:

Fragments of **foetal DNA** are released in minute amounts by the foetus and the placenta into maternal circulation with increasing concentration from 10 weeks gestation onwards. By amplifying cfFDNA using **PCR** it is possible to check the foetal genotype, define the sex, and exclude many chromosomal abnormalities starting **10-12** weeks gestation.

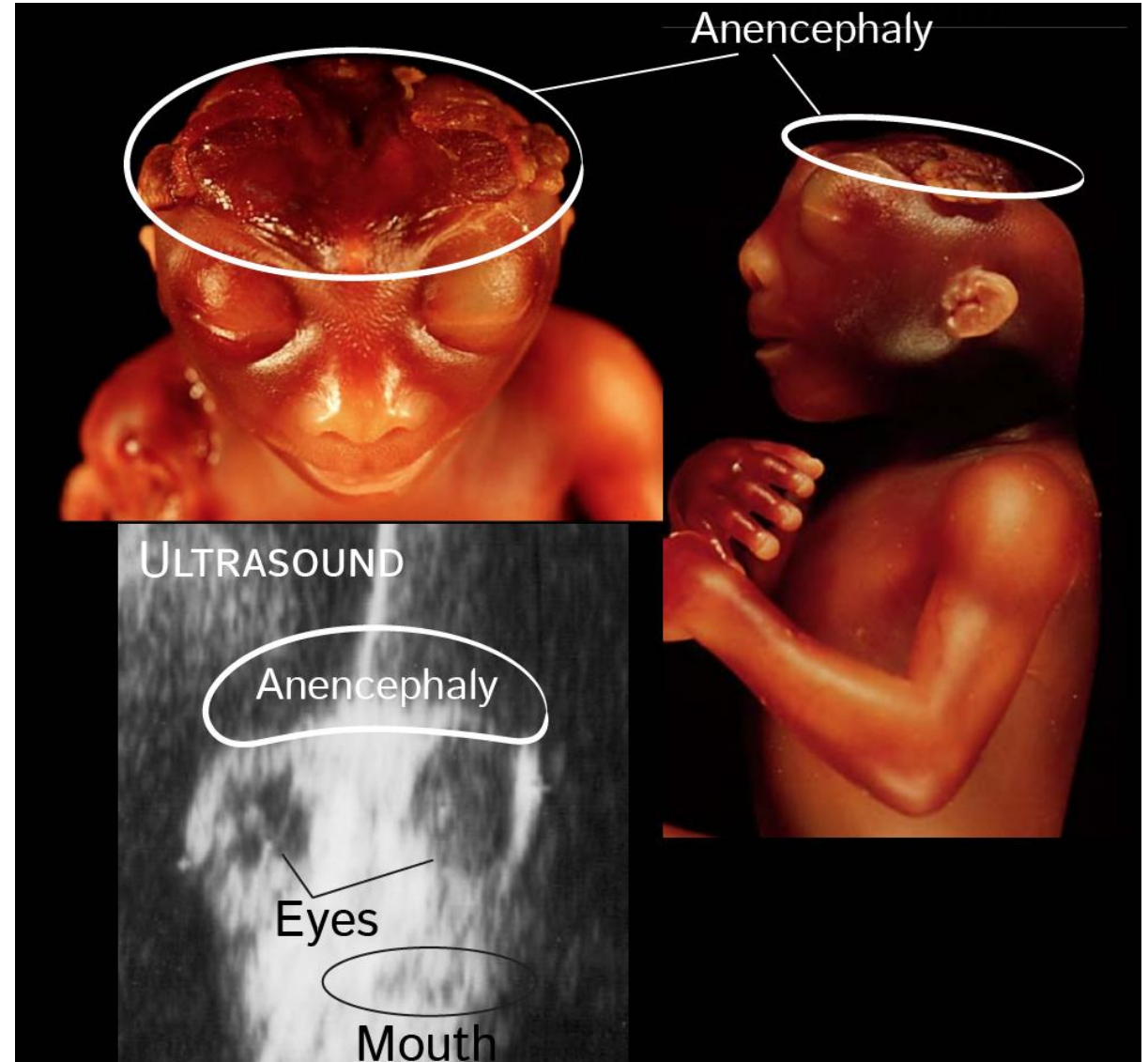
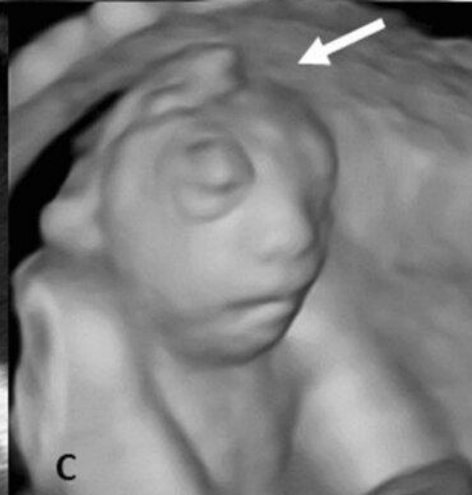
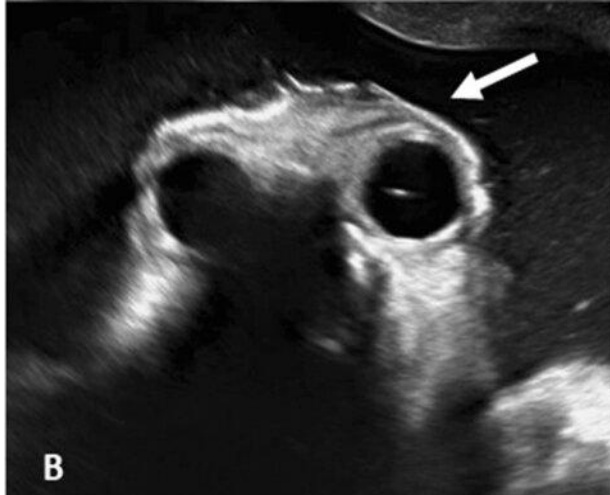
The test although being a novel promising **non-invasive diagnostic** technique with high accuracy yet its use is limited by an expensive test, not readily available in many centers, and is still of **limited** capacity in diagnosing only 5-12 chromosomes at a time compared to CVS and Amniocentesis which give a complete accurate genetic diagnosis of all 23 pairs of chromosomes in human pregnancies.

DIAGNOSTIC TESTS FOR FOETAL CONGENITAL ABNORMALITIES

1. ULTRASOUND DIAGNOSIS OF FOETAL CONGENITAL ANOMALIES

- a. **First Trimester:** by the end of the first trimester some important anomalies can be diagnosed as anencephaly, cystic hygromas, and some skeletal abnormalities
- b. **Second Trimester Anomaly scan:** in the 2nd trimester between **18-22** weeks a detailed fetal anatomy survey is usually performed for diagnosis of important major fetal congenital anomalies as; NTDs, most of the skeletal, cardiac, renal, and GIT anomalies, in addition to diaphragmatic and ventral hernias. This may allow proper decision-making for the continuation of pregnancy or planning the place and time of delivery for better foetal outcome.

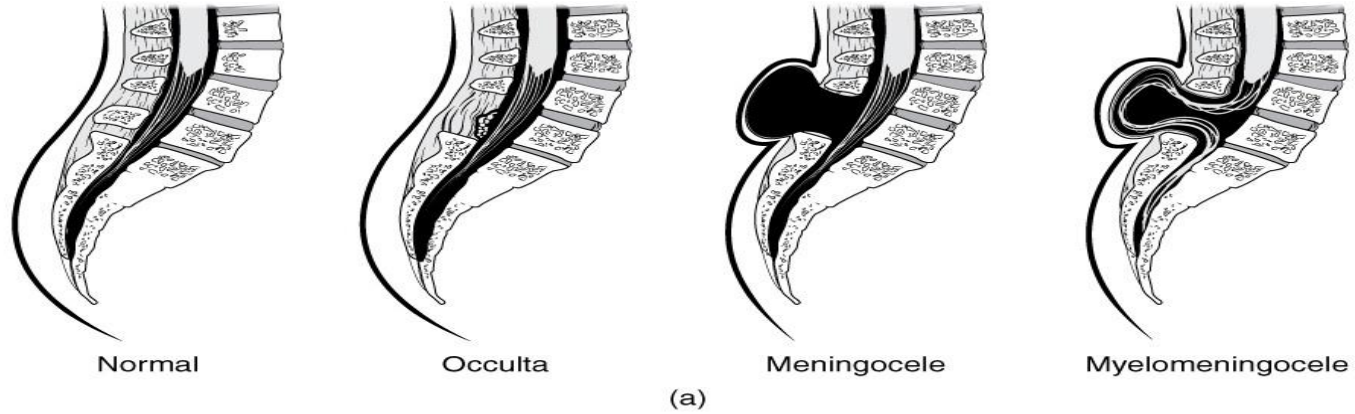
Anencephaly



cystic hygromas



NTDs



(b)



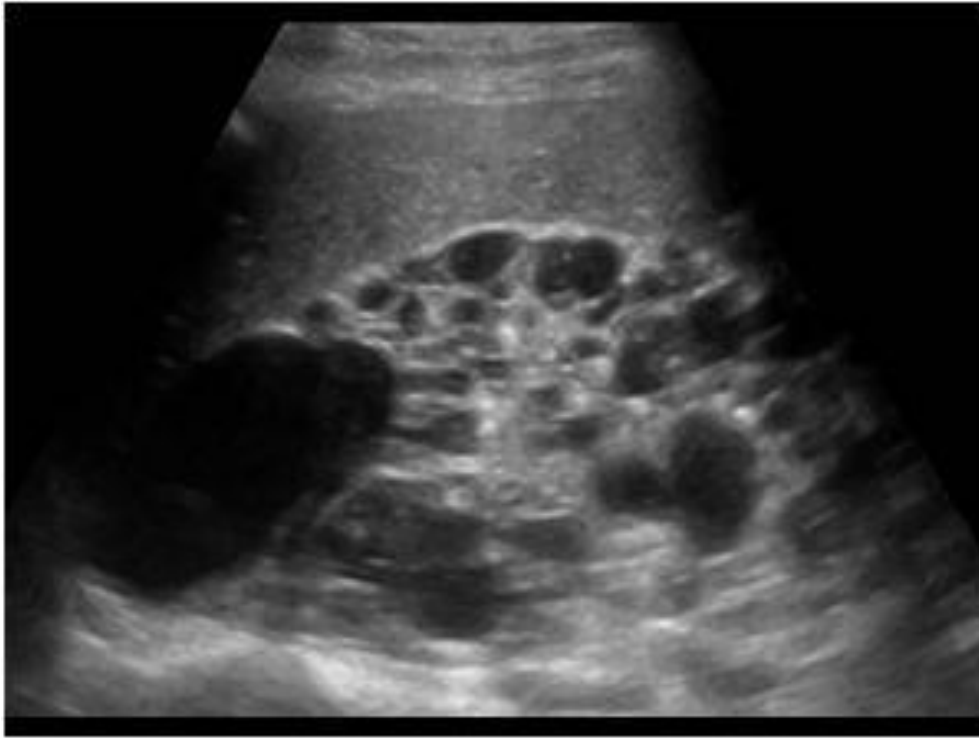
Skeletal dysplasia



VSD



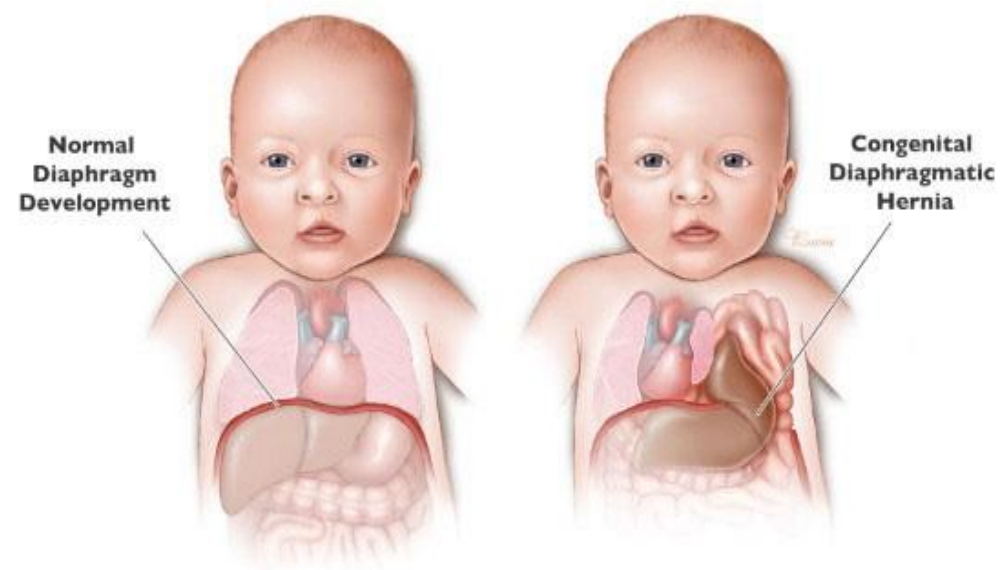
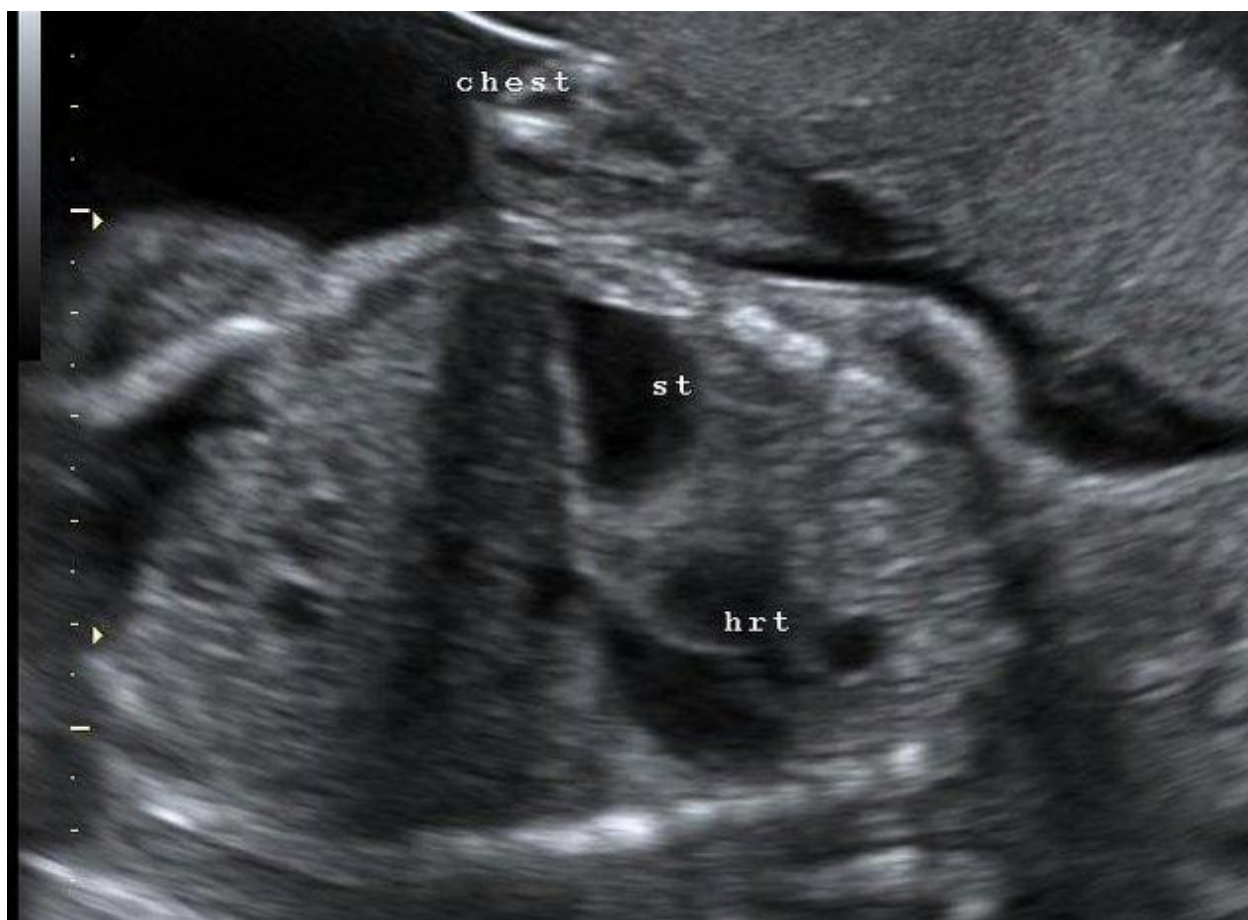
polycystic kidney



Duodenal atresia



Diaphragmatic hernia

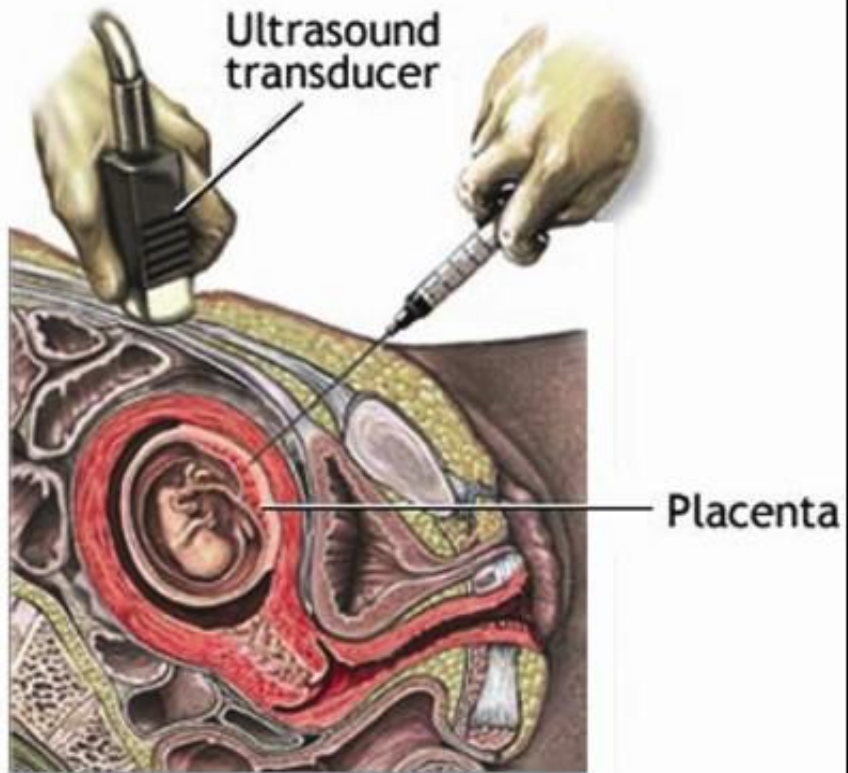


2. CHORIONIC VILLOUS SAMPLING (CVS)

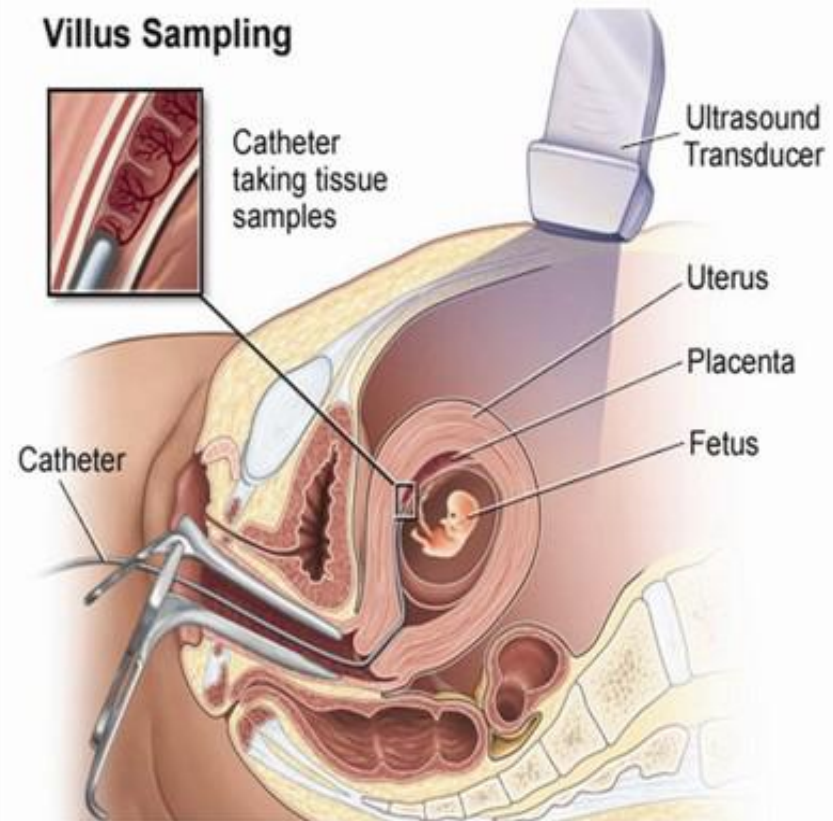
CVS procedure aims at taking a sample of the rapidly dividing **trophoblastic** cells in the placental bed during the first trimester as early as **11** weeks, performed with US guidance;

- Transabdominal: by passing a 22 gauge needle through the abdominal wall and myometrium
- Transvaginal: by passing a fine catheter through the cervix into the placenta.
- Cells obtained are subjected to **karyotyping** for genetic diagnosis of chromosomal defects
 - N.B.: the overall miscarriage risk for a CVS procedure is around **2%**

Transabdominal procedure



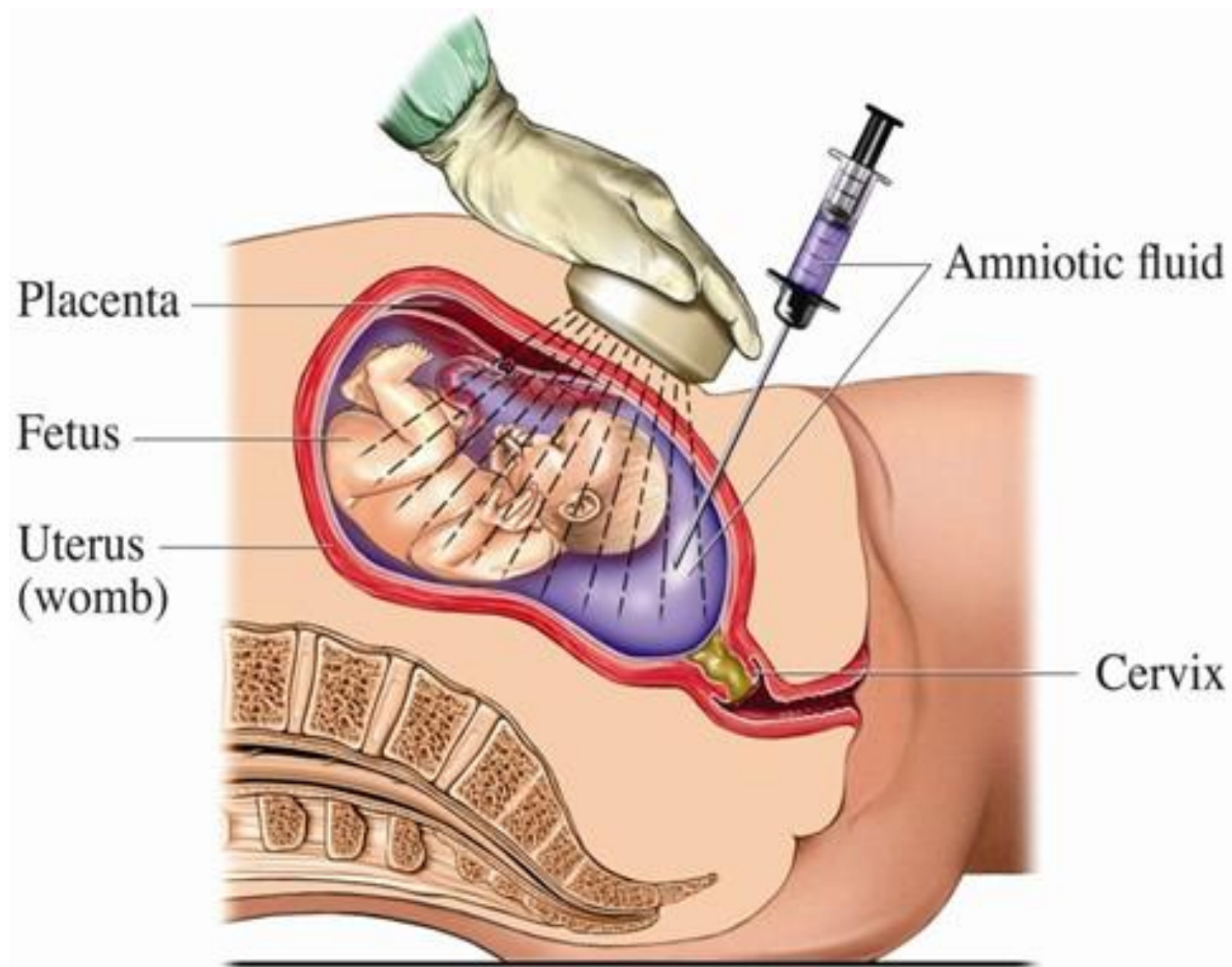
Villus Sampling



3. AMNIOCENTESIS

Amniotic fluid contains amniocytes and fibroblasts shed from foetal membranes, skin, and GIT. In amniocentesis, a sample (15-20 ml) of amniotic fluid is obtained via a **fine needle** 18-22 gauge introduced through the **abdominal wall** and myometrium under **ultrasound** guidance.

- *Timing of amniocentesis:* **14-16** weeks gestation to obtain adequate fluid sample.
- Foetal cells obtained will be subjected to **Karyotyping**, foetal sexing, **DNA** analysis using polymerase chain reaction (PCR), and enzyme assay to achieve a final accurate diagnosis of several chromosomal, biochemical, and hereditary disorders within **2-4** weeks.



4. CORDOCENTESIS

- *Cordocentesis* entails obtaining foetal blood sample or injecting foetal RBCs through the umbilical cord in utero
- *Indications:* It is performed when fetal blood is needed, or when a rapid full culture for karyo-type is needed. The most common reason is for suspected severe fetal anemia such as in cases of RH incompatibility with availability of immediate intra-uterine transfusion with packed RBCs.
- *Technique:* A needle is passed under ultrasound guidance through the abdominal wall and myometrium and, most commonly, into the umbilical cord at the point where it inserts into the placenta.
- *Timing:* Cordocentesis can be performed from about 20 weeks' gestation. The risk of miscarriage varies with indication and position of the placenta.

PREIMPLANTATION GENETIC DIAGNOSIS (PGD):

Early in human gestation, a single cell at the 8-cell stage, or a dozen cells at the blastocyst stage, can be removed without subsequent damage to the foetus. These cells provide sufficient DNA for PCR-directed molecular analysis for inherited diseases or fluorescent in situ hybridization for aneuploidy. Preimplantation biopsy for PGD involves participation in an IVF program and is currently available for a limited number of genetic conditions.

SCREENING TESTS FOR TRISOMY 21 (DOWN SYNDROME):

a. First trimester:

- *US scan*: for abnormal NT measurement and Nasal Bone hypoplasia at **11-13** weeks
- *Double Marker Test (DMT)*: US + MSAFP & PAPP-A at **11-13** weeks

b. Second trimester:

- *Triple Marker Test (TMT)*: MSAFP, hCG, & UE3 at **14-16** weeks.
- *US Foetal Anatomy Survey (FAS)*: for soft markers at **18-22** weeks as hypoplastic nasal bone, tricuspid regurge, short femur, pelvicalyceal dilatation, and cardiac echogenic focus,

DIAGNOSTIC TESTS FOR TRISOMY 21:

- **CVS:** done starting **11** weeks in cases with high-risk screening tests (US, DMT or TMT)
- **Amniocentesis:** done at **14-16** weeks in cases with high-risk screening tests refusing CVS
- **Cell free foetal DNA:** starting **10** weeks can be used as a non-invasive screening test (however expensive) or as diagnostic test whenever US, DMT, or TMT show a definite high risk result for a chromosomal defect (trisomy 21, 13, or 18).



Obstetric analgesia and types of episiotomy

DR: Mohamed Abozeid
OB/GYN Department



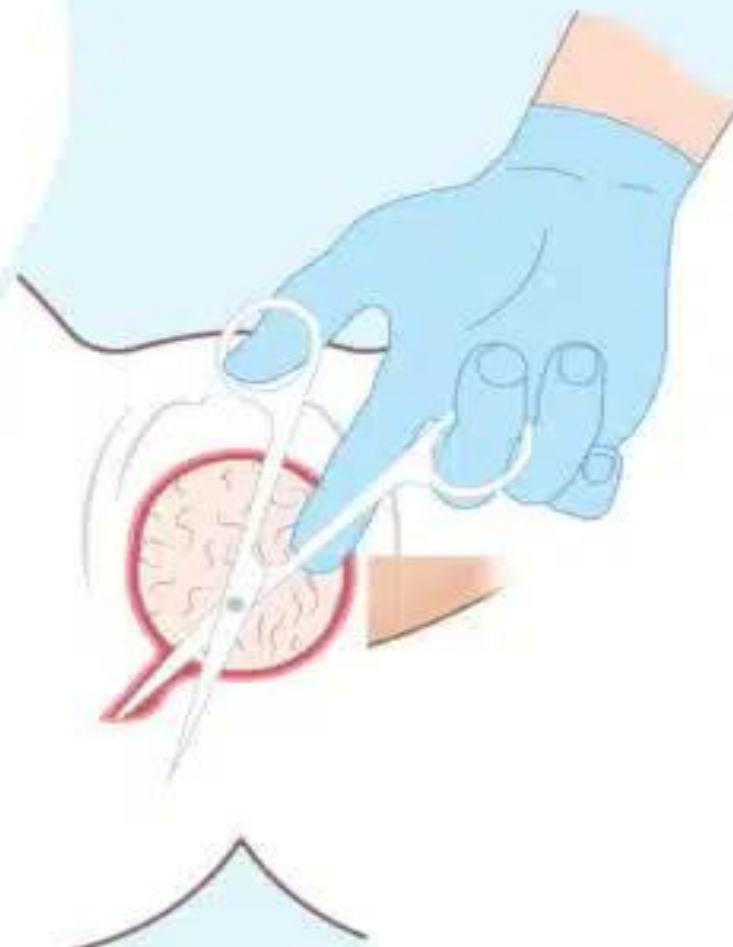
Learning objectives

1. Discriminating between different types of episiotomy and indications of each type
2. Understanding the methods of obstetric analgesia

EPISIOTOMY

MEDIOLATERAL
EPISIOTOMY

MIDLINE
EPISIOTOMY



It is an intrapartum surgically planned incision of perineum and the posterior vaginal wall during The second stage of labour .

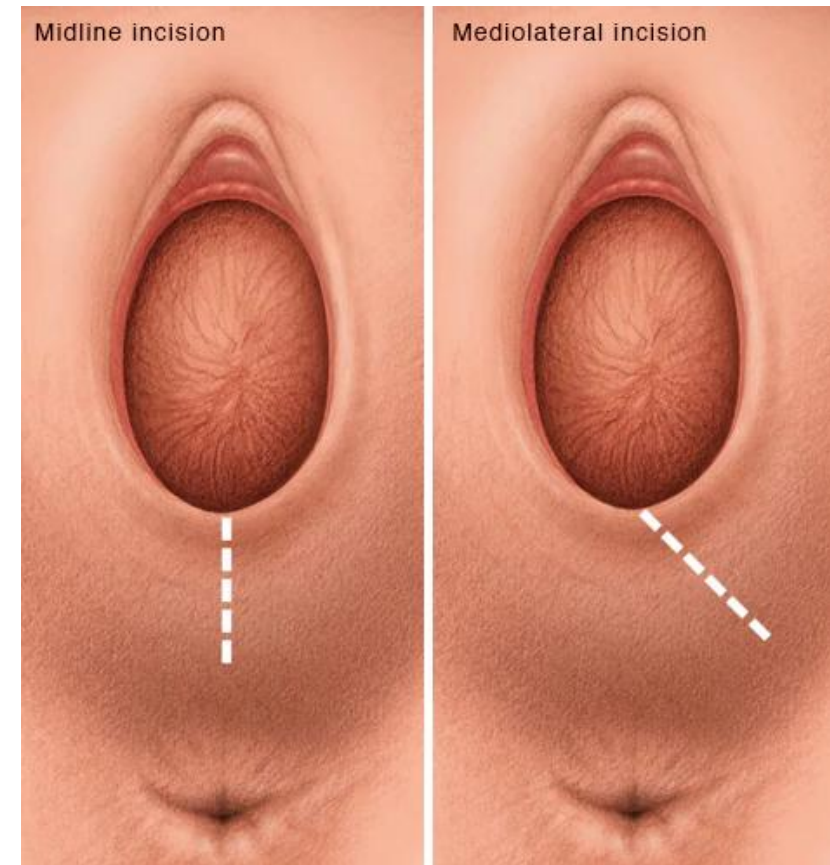
types :

Median (midline)

Mediolateral .

Lateral.

J-shaped.



Relative merits and demerits of median and medio-lateral episiotomy

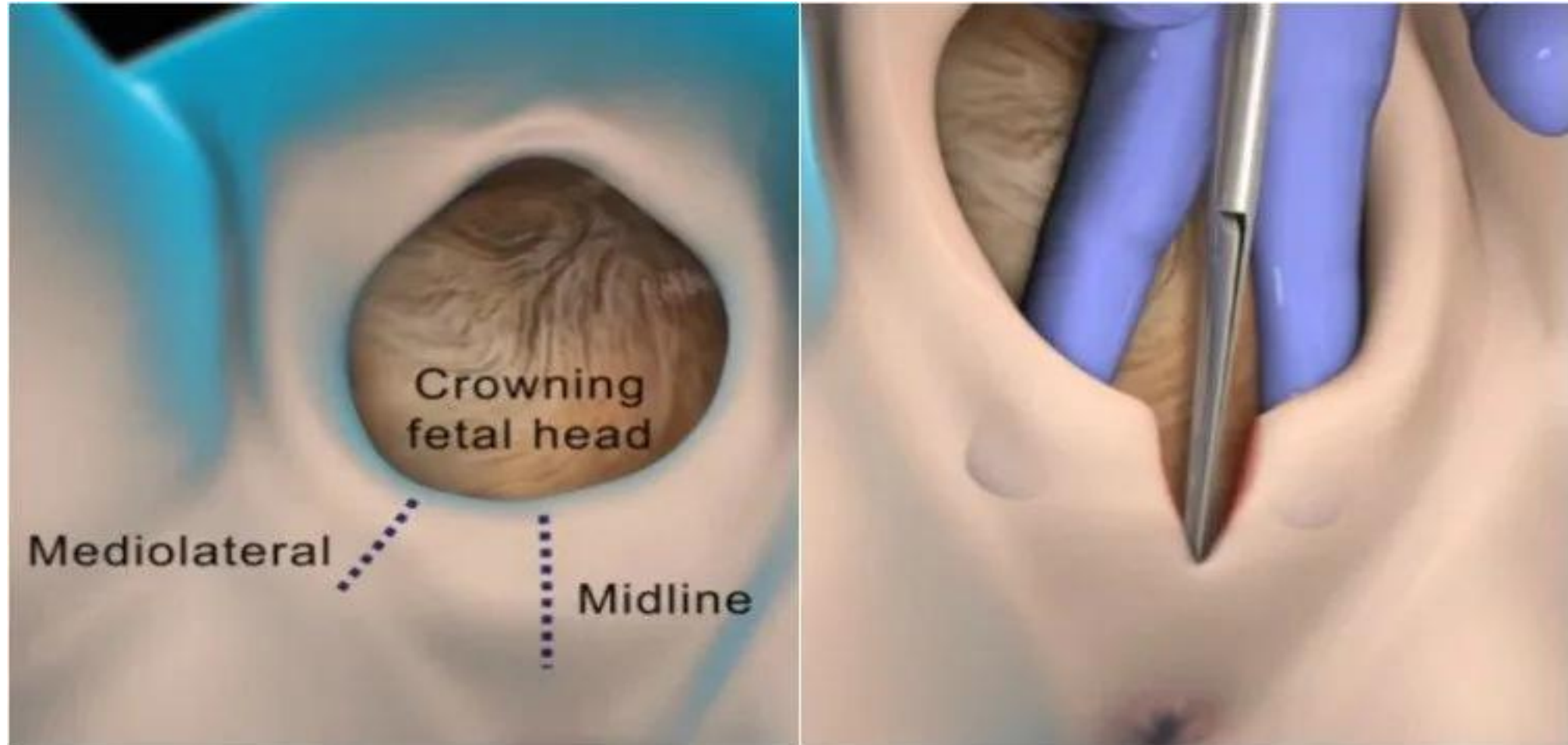
	Median	Medio-lateral
Merits	➤ The muscles are not cut ➤ Blood loss is least ➤ Repair is easy ➤ Post operative comfort is maximum ➤ Healing is superior ➤ Wound disruption is rare ➤ Dyspareunia is rare	➤ relative safety from rectal involvement from extension. ➤ if necessary, the incision can be extended.
Demerits	➤ Extension, if occurs, may involve the rectum. ➤ Not suitable for manipulative delivery or in abnormal presentation or position. As such, its use is selective.	➤ Apposition of the tissues is not so good. ➤ Blood loss is little more, ➤ Post operative discomfort is more ➤ Relative increased incidence of wound disruption ➤ Dyspareunia is comparatively more

- **Indications :-**

Indications of episiotomy:

- Fetal distress.
- Short or inelastic perineum.
- Shoulder dystocia.
- Fetal malposition, e.g. occipito-posterior.
- An instrumental or breech delivery.
- Previous pelvic floor surgery.

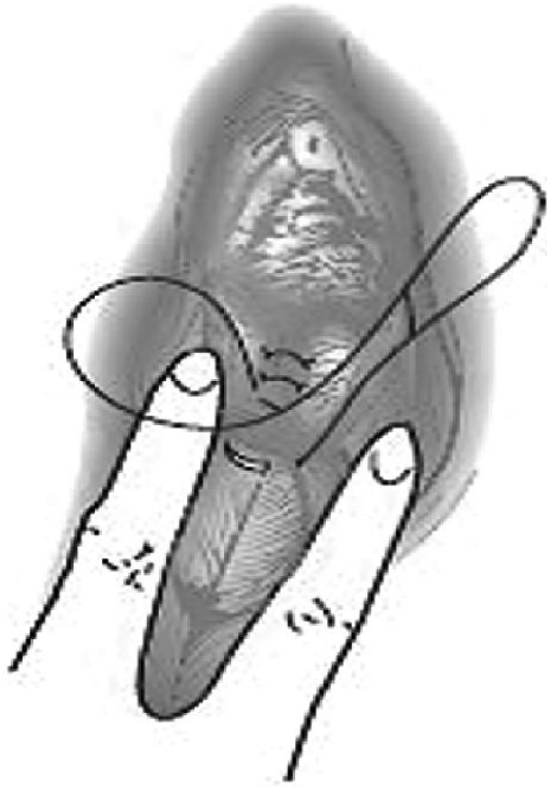
3) timing of episiotomy



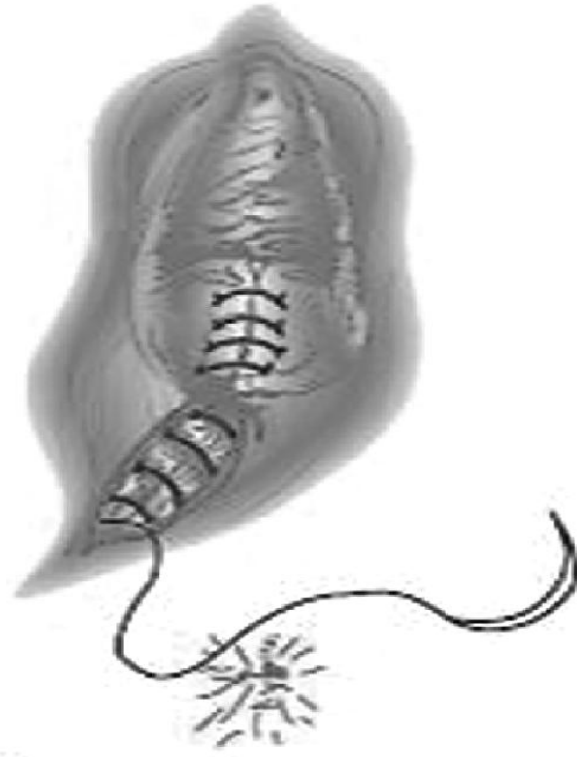
Steps of episiotomy

- Just after crowning :
- Use local infiltration with lignocaine
- Make sure there are no known allergies to the drug used
- Infiltrate beneath the vaginal mucosa , beneath the skin of the perineum and deeply into the perineal muscle .
- Pull back on the suringe plunger before infiltration to be sure that no vessel has been injured .

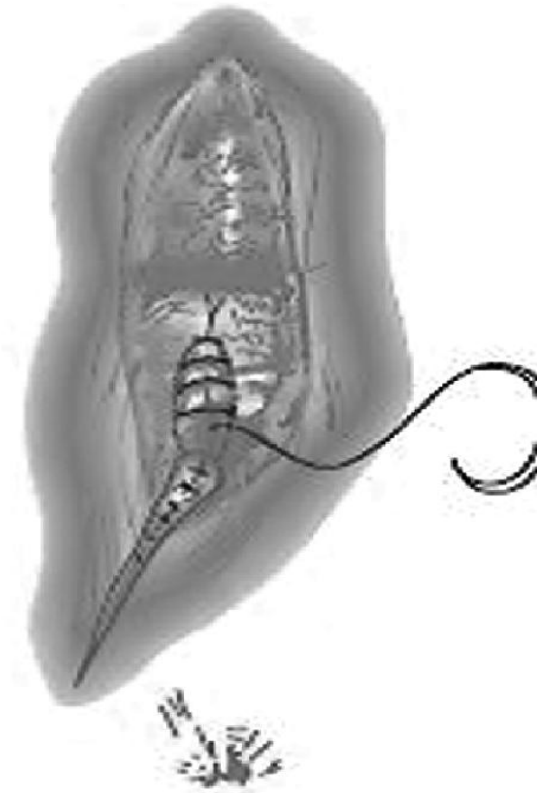
Repair of episiotomy



a. Loose, continuous, non-locking stitch to vaginal wall



b. Loose, continuous, non-locking stitch to perineal muscles



c. Closure of skin using a loose subcutaneous stitch

Complication

COMPLICATION OF EPISIOTOMY

Immediate:

- ▶ Extension of incision
- ▶ Vulval hematoma
- ▶ Infection
- ▶ Wound dehiscence
- ▶ Injury to anal sphincter causing incontinence of flatus or feces.

Remote:

- ▶ Dyspareunia
- ▶ Change of perineal laceration
- ▶ Scar endometriosis (rare)

Obstetric anesthesia



TYPES OF LABOR ANALGESIA

1. Non-pharmacological analgesia: Birthing techniques such as hydrotherapy, hypnobirthing, patterned breathing, relaxation, and visualization can increase the production of endogenous endorphins that bind to receptors in the brain for pain relief

2. Pharmacological analgesia

3. Regional analgesia

Pharmacological analgesia

- **Nitrous oxide (Entonox):**

Entonox is premixed NO and O₂ as a 50:50 mixture. It is self administered and has quick onset of action and a short half-life.

Side effects include feeling faint, nausea and vomiting.

- **Narcotic agents:**

Pethidine is administered at a dose of 50-150mg, onset of action is 15-20mins, lasts for 3-4 hrs, usually given with an antiemetic. If given within 2 hrs of delivery, can cause neonatal respiratory depression and naloxone may be needed.

Diamorphine is also used in some units at a dose of 2.5-5mg. There is controversy about the extent and timing of neonatal respiratory depression, but it may be up to 3-4 hrs after last dose.

REGIONAL ANALGESIA AND ANESTHESIA

- 1) Epidural block.
- 2) Spinal block.
- 3) Combined Spinal Epidural (CSE).
- 4) Continuous spinal analgesia.
- 5) Paracervical block.
- 6) Lumbar sympathetic block.
- 7) Pudendal block.

EPIDURAL ANALGESIA

Indications of EA

- ☐ Pain experienced by a woman in labor
- ☐ When medically beneficial to reduce the stress of labor

Advantages of EA

- It provides effective analgesia in labour.
- Reduced maternal secretion of catecholamines, which benefits the fetus.
- Can be used when topped up for an operative delivery and for any complications of the 3rd stage of labour, e.g. retained placenta or repair of perineal tears.
- Can provide effective postoperative analgesia.
- Can be used as an additional method of controlling blood pressure in pre-eclampsia.

DISADVANTAGES AND COMPLICATIONS OF EPIDURAL ANALGESIA

- Failure to site, or a patchy or incomplete block.
- Hypotension from sympathetic blockade.
- Decreased mobility.
- Tenderness over the insertion site.
- Inadvertent dural puncture
- Respiratory depression.
- Extremely rare complications resulting in neurological deficits.

CONTRAINDICATIONS TO EPIDURAL ANALGESIA

Absolute

- Maternal refusal
- Lack of personnel/facilities
- Pre-existing coagulopathy
- Local infection at insertion site.
- Raised intracranial pressure (risk of coning)
- Drug allergy.

Relative

- Haemodynamic instability.
- Anatomical abnormalities.
- Neurological disorders (medicolegal implications)
- Systemic infection.

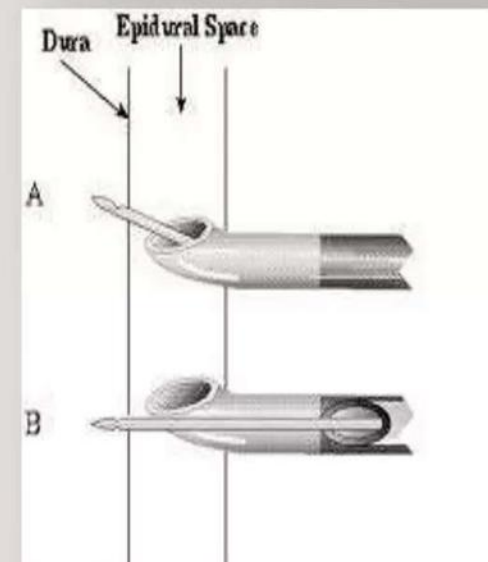
COMBINED SPINAL EPIDURAL (CSE)

Advantages of CSE for Labor Analgesia

1. Rapid onset of intense analgesia.
2. Very low failure rate.
3. Less need for supplemental boluses.
4. Minimal motor block ("walking epidural")

Maintenance of epidural analgesia can be achieved by

- I. regular top-ups
- II. an epidural infusion
- III. patient-controlled epidural analgesia (PCEA).

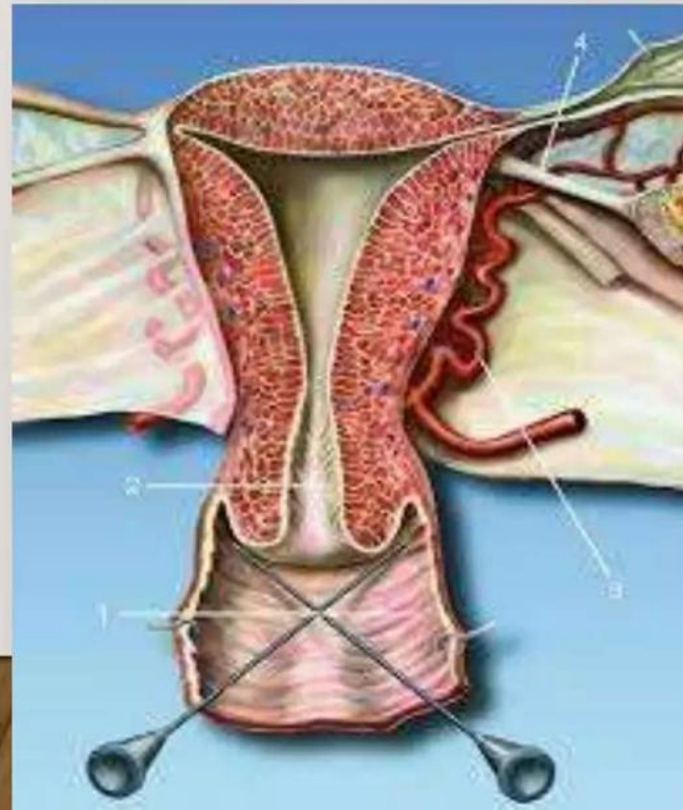


COMBINED SPINAL EPIDURAL ANALGESIA



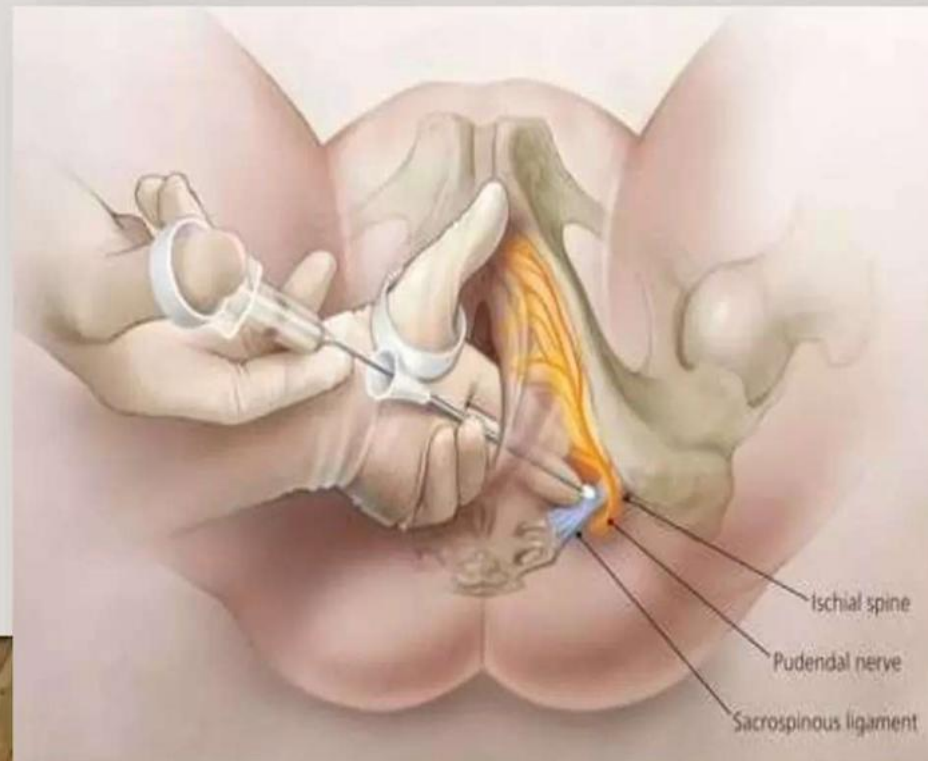
PARACERVICAL BLOCK IN LABOR

An injection of pain medicine into the tissues around the cervix is called a **paracervical block**.
A **paracervical block** is another form of local anesthesia. It reduces the pain caused by contractions and stretching of the cervix. A **paracervical block** lasts about 1 to 2 hours.



PUDENDAL BLOCKS IN LABOR

To relieve pain associated with the second (pushing) stage of **labor**, an injection called a **pudendal block** can be given through the vaginal wall and into the **pudendal** nerve in the pelvis, numbing the area between the vagina and anus (perineum). **Pudendal blocks** do not relieve the pain of contractions



COMPLICATIONS OF REGIONAL ANESTHESIA

- 1) Post Dural Puncture Headache (PDPH).
- 2) Neurological complications.
- 3) Venous puncture e.g. of dural veins
- 4) Catheter breakage
- 5) Extensive block (including unplanned blocks)
- 6) Shivering
- 7) Backache - Long-term backache is not a complication of neuraxial techniques.
- 8) Drug side effects
 - ☐ Nausea and vomiting (opiates)
 - ☐ Respiratory depression (opiates)
 - ☐ Anaphylaxis
 - ☐ Toxicity (including intravascular injection of local anaesthetics).

GENERAL ANAESTHESIA

Indications for GA includes:

- I. Maternal request.
- II. Urgent surgery.
- III. Regional anaesthesia contraindicated (coagulopathy, maternal hypovolaemia).
- IV. Failed regional anaesthesia.
- V. Additional surgery planned at the same time as cesarean section.

Problems with GA:

- Potential airway difficulties.
- Pulmonary aspiration of gastric contents.
- Awareness.

INDUCTION OF LABOUR AND ABORTION

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OB/GYN Department



IOLs

1. What is meant by induction of labour and abortion
2. Indications and contraindications
3. Complications of induction
4. Methods used



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Definition:

Termination of a pregnancy by inducing labour

Indications of induction of labour at term:

Absolute Maternal Indications:

- PE / eclampsia.
- Medical Problems:
 1. Diabetes mellitus
 2. Chronic renal disease
 3. Chronic pulmonary disease

Relative Maternal Indications:

- Chronic hypertension
- Gestational diabetes

Absolute Fetal Indications:

- Post-term pregnancy (>42 weeks).
- IUGR.
- Iso-immunization.
- Placental abruption
- chorioamnionitis

Relative Fetal Indications:

- PROM
- Fetal Macrosomia
- Previous still-birth
- Unexplained oligohydramnios

Contraindications for induction of labour:

Absolute Contraindications:

Maternal Contraindications:

Active genital herpes
Serious chronic medical condition

Fetal Contraindications:

Malpresentation.
Fetal distress.

Uteroplacental Contraindications:

Cord prolapse
Placenta praevia
Vasa praevia

Relative Contraindications:

Maternal Contraindications:

Cervical Carcinoma
Pelvic deformities

Fetal Contraindications:

Extreme fetal macrosomia

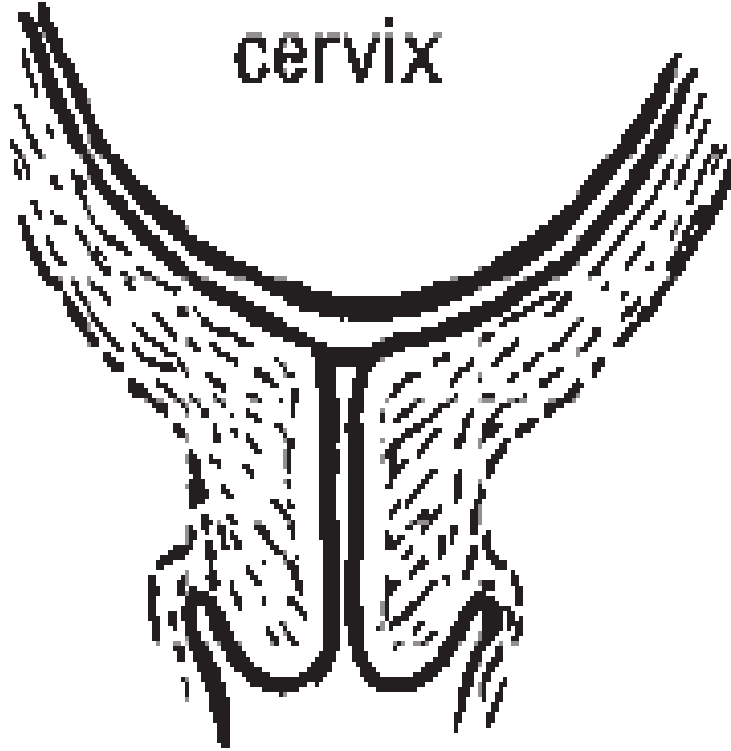
Uteroplacental Contraindications:

Cord presentation
Unexplained vaginal bleeding

Prerequisites for successful induction of labour:

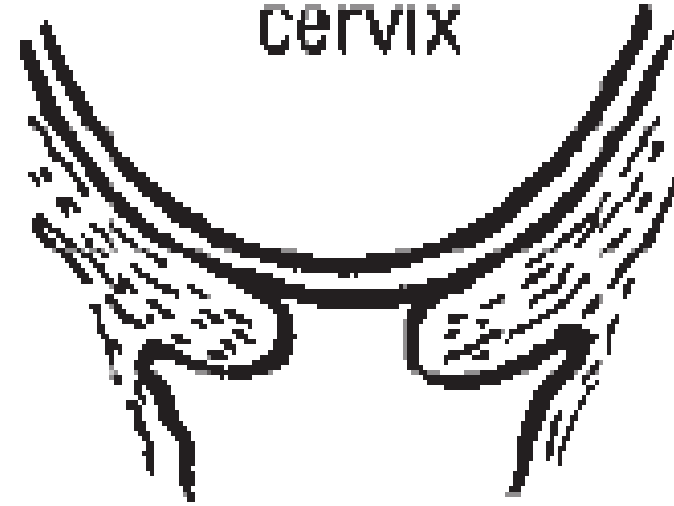
The condition of the cervix (cervical ripening) is the most important factor in successful induction and, where ripening has not occurred, there is a greater chance of a long labour, fetal hypoxia and operative delivery.

Unripe cervix



An unripe cervix is hard, long, closed and not effaced.

Ripe cervix



A ripe cervix is soft, effaced or becoming effaced and admits the finger.

The Bishop score

- It is an accepted method of recording the degree ripeness before labour.
- It takes account of the cervical length (formed / effaced), cervical dilatation (closed, 1.0, 2.0, 3.0 cm, etc), consistency of the cervix (soft / rigid), and the level of the fetal head (station -2, -1, 0, +1, +2).
- Scores of 0, 1, or 2 are given accordingly and a score of 9 or higher is favourable for safe induction of labour.

Assessment of cervical status by Bishop score

	Score			
	0	1	2	3
Position	Posterior	Middle	Anterior	--
Consistency	Firm	Medium	Soft	--
Effacement	0-30%	40-50%	60-70%	80%+
Dilation	Closed	1-2cm	3-4cm	5+cm
Station	-3	-2	-1/0	+1/+2

Additional factors: +1 point for each previous vaginal birth, -1 point for first time birth givers|

Add the score for each factor.

Scores lower than 5 suggest labour will not begin without induction.

Scores 9 and higher indicate labour will likely begin spontaneously.

Scores 3 and lower may indicate that an induction would not be successful

METHODS OF INDUCTION OF LABOUR

- **RIPENING THE CERVIX**
- **AMNIOTOMY** (Artificial Rupture of Membranes)
- **INTRAVENOUS OXYTOCIN**



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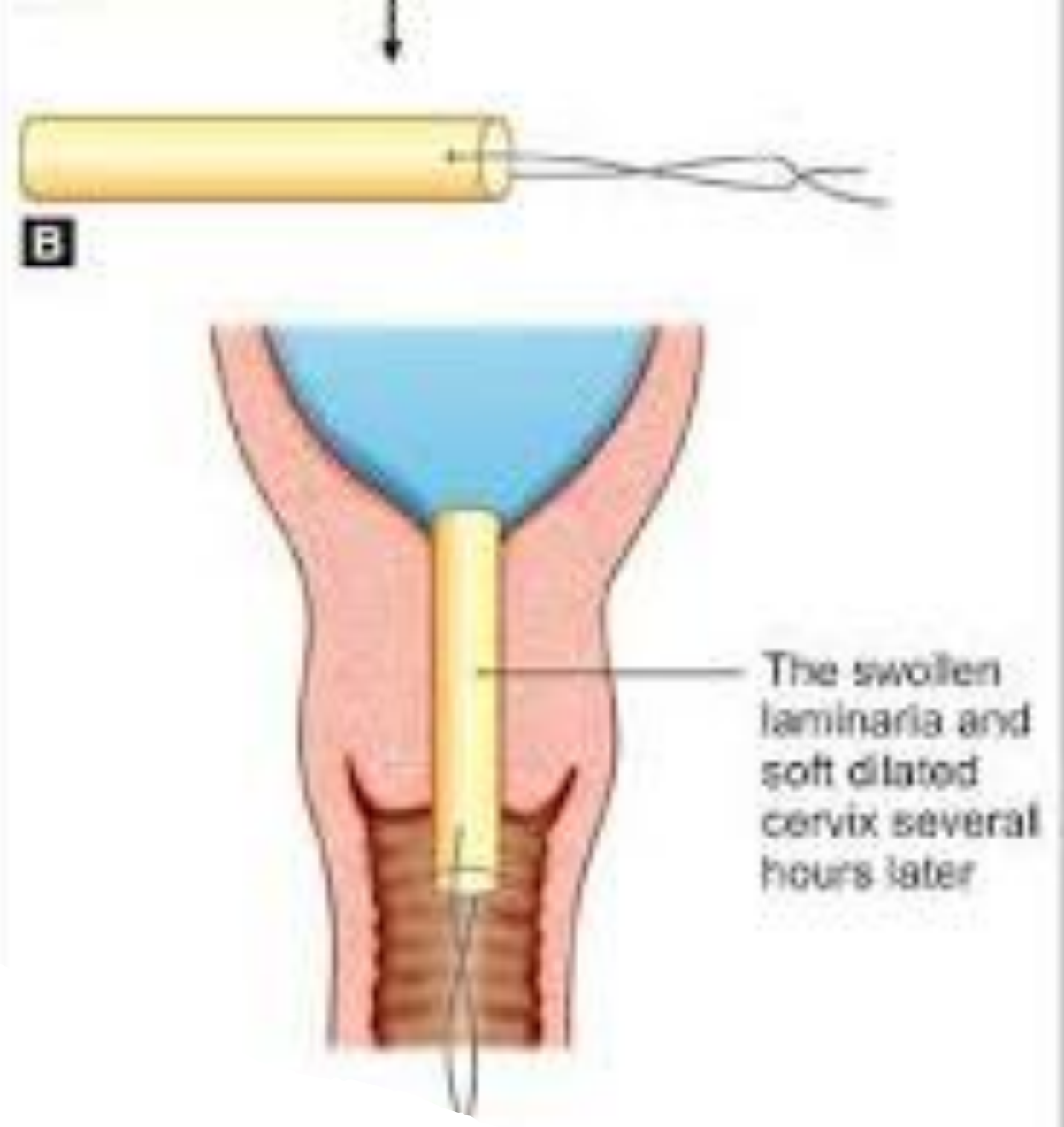
1. RIPENING THE CERVIX



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- The collagen fibers of the cervix can be much softened by local application of PGs.
- A vaginal tablet or gel containing PGE_2 Dinoprostone, or PGE_1 Misoprostol 25-50 ug, may be inserted into the posterior fornix to soften the cervix.
- Osmotic dilators (Laminaria) is a kind of seaweed that can be sterilized, rolled and dried to create a thin stick or “tent” that can be inserted into the cervix to force dilation





Laminaria tent

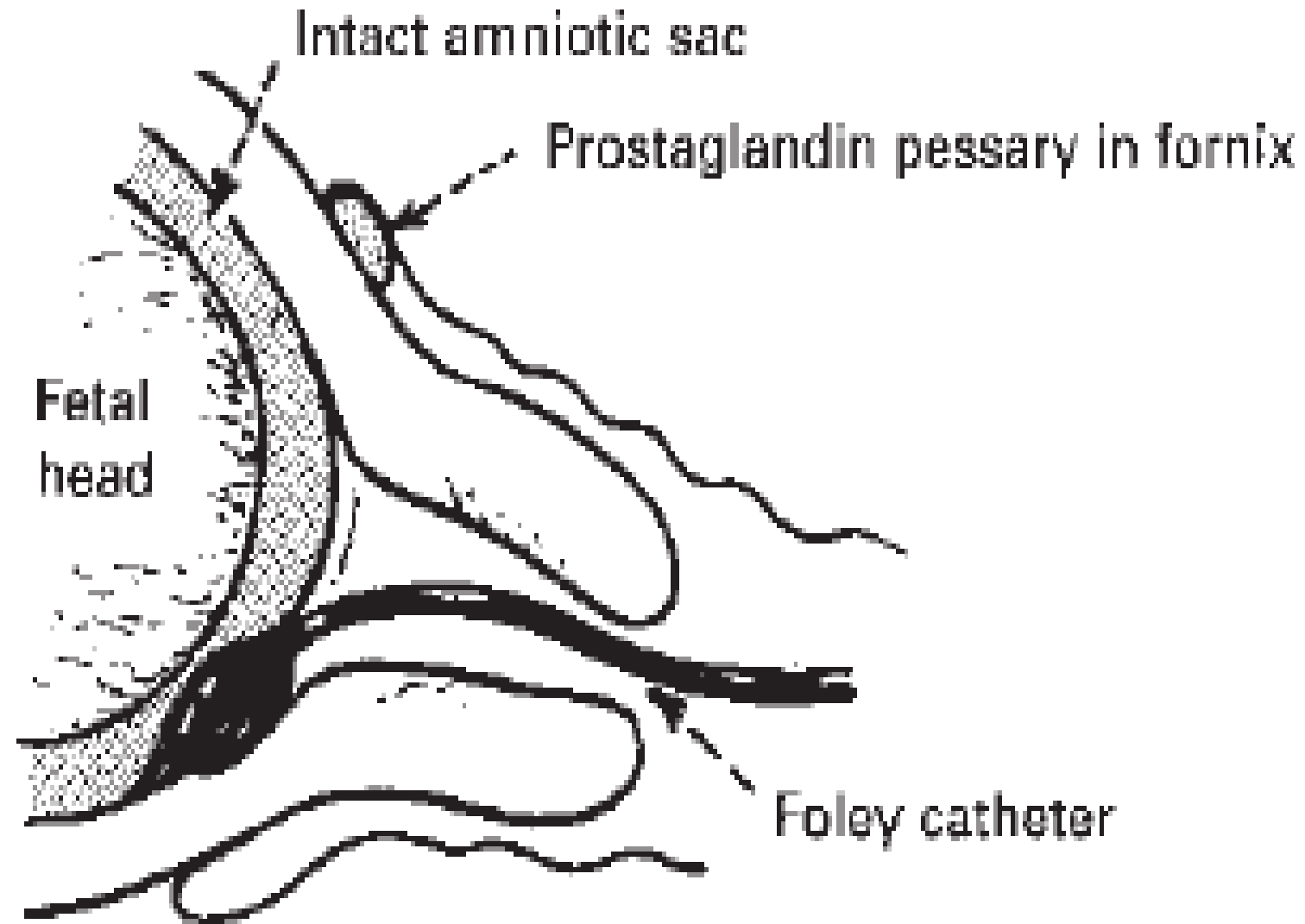


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2. AMNIOTOMY

(Artificial Rupture of Membranes)

- It is done to:
 - initiate labour (surgical induction) or
 - accelerate the labour process or
 - allow a fetal scalp blood PH estimation

Amniotomy appears to release local PGs.

- Amniotomy, using a special hook, may be used to rupture the membranes overlying the presenting part, but care must be taken not to damage cervical or fetal tissues.
- The operation may be done by passing the instrument along the fingers or by direct vision using a speculum.
- The colour and quantity of the liquor should be noted.
- Prolapse of the umbilical cord should be excluded at beginning and end of the procedure.



Amniohook

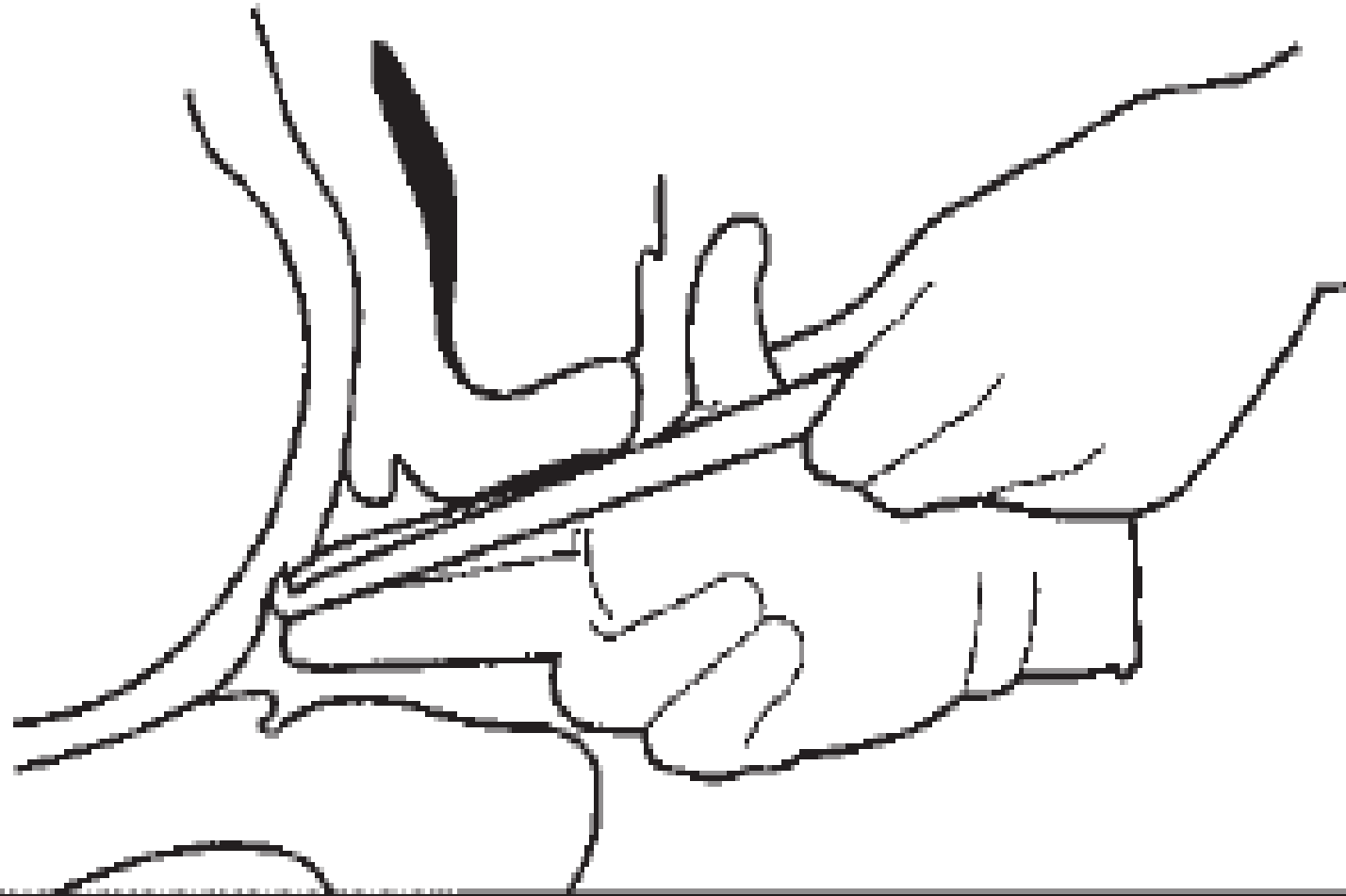


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Complications of Amniotomy

- *Failure to induce effective contractions*
- *Placental separation (Abruptio)*
- *Vaginal Bleeding*
- *Prolapse of the cord*
- *Infection*
- *Pulmonary embolism of amniotic fluid*



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3. INTRA VENOUS OXYTOCIN



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- Synthetic oxytocin by continuous I.V. is commonly used after amniotomy to stimulate uterine contractions.
- Synthetic oxytocin is a powerful drug and sometimes unpredictable, as uterine sensitivity can show a wide variation.

- The best method of administration is by a suitable semi-automated infusion system incorporating an accurate drop counter.
- A solution of 2 units of syntocinon in 500 ml of Hartmann's solution is used beginning at a dose of 1.0 mIU/min. (10-15 drops/min). This is increased by 1.0 mIU/min. every 15 minutes until satisfactory contractions are established.



Oxytocin





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Complications of Oxytocin

- *Poor uterine action*
- *Abnormal fetal heart rate patterns*
- *Uterine Hyperstimulation*
- *Rupture of the uterus*
- *Water intoxication*



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Abortion may be induced following a diagnosis of non-continuing pregnancy.

It can be induced by the following methods according to the duration of pregnancy

MEDICAL METHODS FOR INDUCTION OF ABORTION

I- First trimester:

1- oral anti-progesterone Mifepristone (RU 486), and vaginal PG.

2- oral PG (Misoprostol) 200 mcg.

3- Mifepristone 600 mg orally and 48 hours later vaginal PG
1 mg is given as a vaginal pessary, or Oral Misoprostol 400 mcg.

II- Mid trimester:

1- Vaginal prostaglandin pessaries: (into the posterior fornix 4 hourly for up to five pessaries)

2- prostaglandin Misoprostol in doses of 50 - 200 Ug (orally and/or vaginally)

3- Osmotic dilators or Extra-amniotic saline infusion: This may be employed where a patient finds difficulty in using pessaries or where there has been no response to vaginal prostaglandin.

CYTOTEC



Mifepristone & Misoprostol Dosing Chart

Recommended Regimens 2023

≤12 weeks	13-17 weeks	18-24 weeks	25-27 weeks	≥28 weeks	Postpartum Use
Induced Abortion Mifepristone 200mg PO Wait 1-2 days then, Misoprostol 800µg BU/SL/PV x1 ≥10 weeks give misoprostol 800µg BU/SL/PV every 3 hours until expulsion ¹	Induced Abortion Mifepristone 200mg PO Wait 1-2 days then, Misoprostol 400µg every 3 hours BU/SL/PV until expulsion ⁵	Induced Abortion Mifepristone 200mg PO Wait 1-2 days then, Misoprostol 400µg every 3 hours BU/SL/PV until expulsion ⁵	Induced Abortion Mifepristone 200mg PO Wait 1-2 days then, Misoprostol 200µg every 4 hours until expulsion BU/SL/PV ^{5,9}	Induced Abortion Mifepristone 200mg PO Wait 1-2 days then, Misoprostol 50-100µg every 4 hours PV OR Misoprostol 50-100µg every 2 hours PO ^{6,9}	Prophylaxis of Postpartum hemorrhage (PPH) Misoprostol 600µg SL x 1
Missed Abortion/ Anembryonic Pregnancy Mifepristone 200mg PO Wait 1-2 days then, Misoprostol 800µg BU/SL/PV x1 ≥10 weeks give misoprostol 800µg BU/SL/PV every 3 hours until expulsion ¹	Missed Abortion Mifepristone 200mg PO Wait 1-2 days then, Misoprostol 400µg every 3 hours BU/SL/PV until expulsion ⁵	Fetal Demise Mifepristone 200mg PO Wait 1-2 days then, Misoprostol 400µg every 3 hours BU/SL/PV until expulsion	Fetal Demise Mifepristone 200mg PO Wait 1-2 days then, Misoprostol 200µg every 4 hours BU/SL/PV until expulsion ⁵	Fetal Demise Mifepristone 200mg PO Wait 1-2 days then, Misoprostol 25-50µg every 4 hours PV OR Misoprostol 50-100µg every 2 hours PO ⁶	Treatment of Postpartum hemorrhage (PPH) Misoprostol 800µg SL x 1
Incomplete Abortion 400µg misoprostol SLx1 600µg misoprostol PO x1 800µg misoprostol BU x 1 dose ⁵	Incomplete Abortion Misoprostol 400µg every 3 hours BU/SL until expulsion	Incomplete Abortion Misoprostol 400µg every 3 hours BU/SL until expulsion	Induction of Labor Misoprostol 25-50µg every 4 hours PV ^{7,8} OR Misoprostol 50-100µg every 2 hours PO ^{6,7,8}	Induction of Labor Misoprostol 25-50µg every 4 hours PV ⁷ OR Misoprostol 50-100µg every 2 hours PO ^{6,7}	
Cervical Preparation Before Aspiration Not required ²	Cervical Preparation Before Aspiration Misoprostol 400µg 1-3hrs BU/SL/PV before the procedure ³	Cervical Preparation Before D&E (Use of multiple modalities is recommended) Mifepristone 200mg PO & Osmotic Dilators 1-2 days before. ⁴	LEGEND: Buccal(BU) Sublingual (SL) Per Vagina (PV) Per Oral (PO)		

Complications of prostaglandin administration

GIT symptoms: Nausea, vomiting or diarrhea.

Cardiac symptoms: Palpitations and mild pyrexia.

Reports on uterine rupture and cervical damage

Prostaglandins should **NEVER be used** to stimulate uterine contractions during labour.

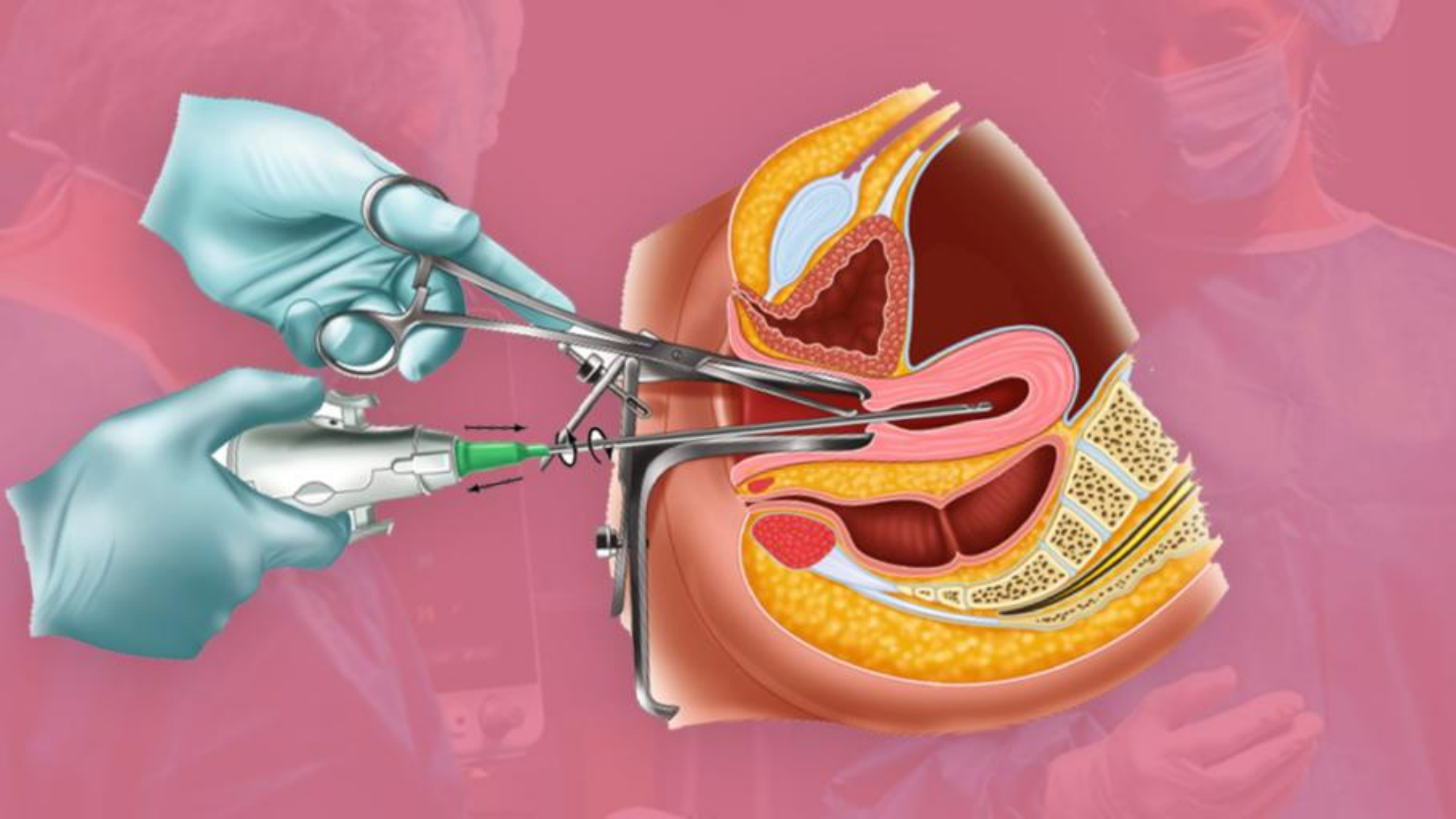
It **might be used** to induce labour or to accomplish cervical ripening in term pregnancies.

SURGICAL METHODS FOR INDUCTION OF ABORTION

· ***Vacuum aspiration:***

- commonest method of uterine evacuation in pregnancies up to 12 weeks' gestation.
- It is usually preceded by cervical ripening (by vaginal PGs) to facilitate dilatation of the cervix and thus avoid damage.
- After dilatation, the products of conception are removed by suction using a plastic curette.
- ***Complications:*** cervical trauma, perforation of the uterus and incomplete evacuation leading to sepsis.





. Abdominal hysterotomy

-Emptying of the uterus by the abdominal route through a small suprapubic incision, before fetal viability, is rarely needed.

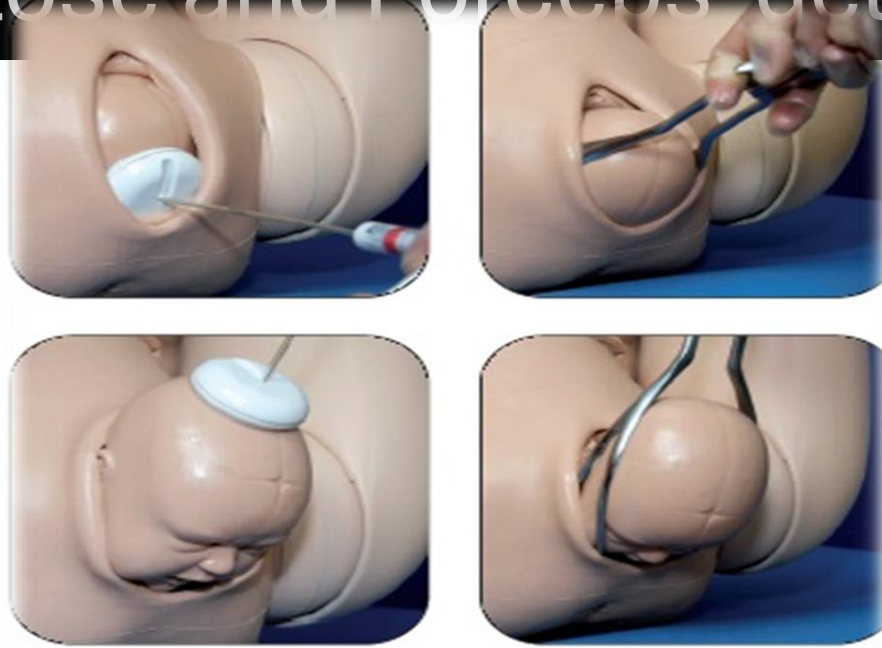
-The main disadvantages of the procedure are the risks of any abdominal operation and a slight risk of rupture in subsequent pregnancy.

-After removal of the foetus and placenta the cavity should be confirmed to be empty.

-Spillage of the decidua should be avoided as there is a well-recognized risk of implantation endometriosis in the abdominal scar.

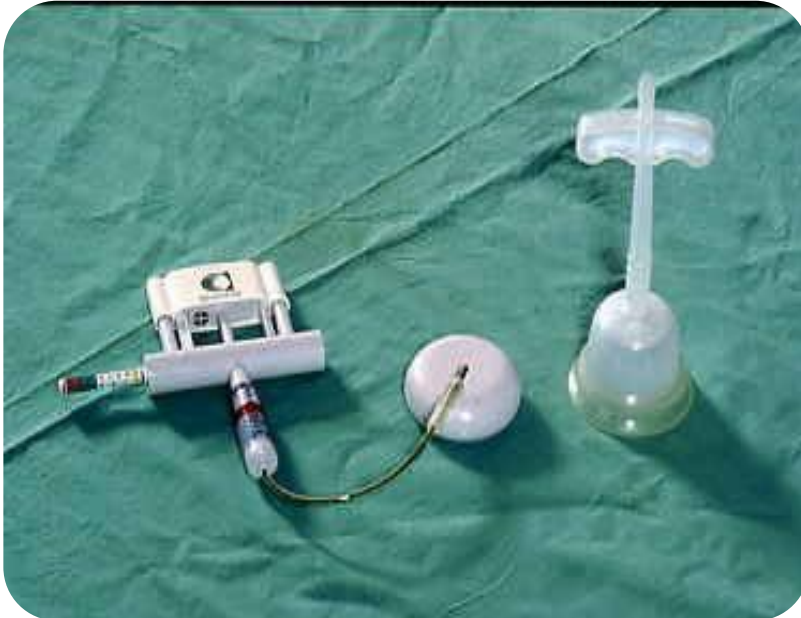


Ventose and Forcens delivery



By Dr.Mohamed Ali
Under supervision of
Prof . Ahmed Hussien

Vacuum /ventouse



Indications

MATERNAL

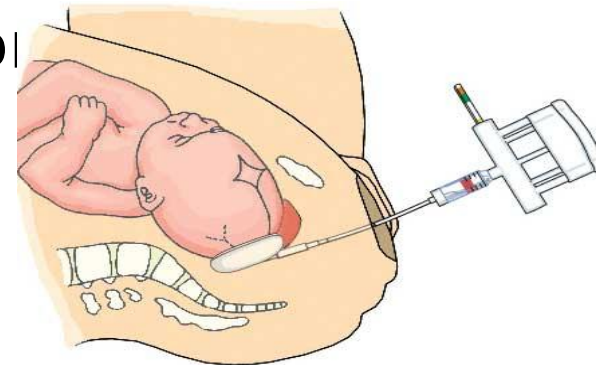
- Exhaustion
- Prolonged second stage
- Cardiac / pulmonary disease

FETAL

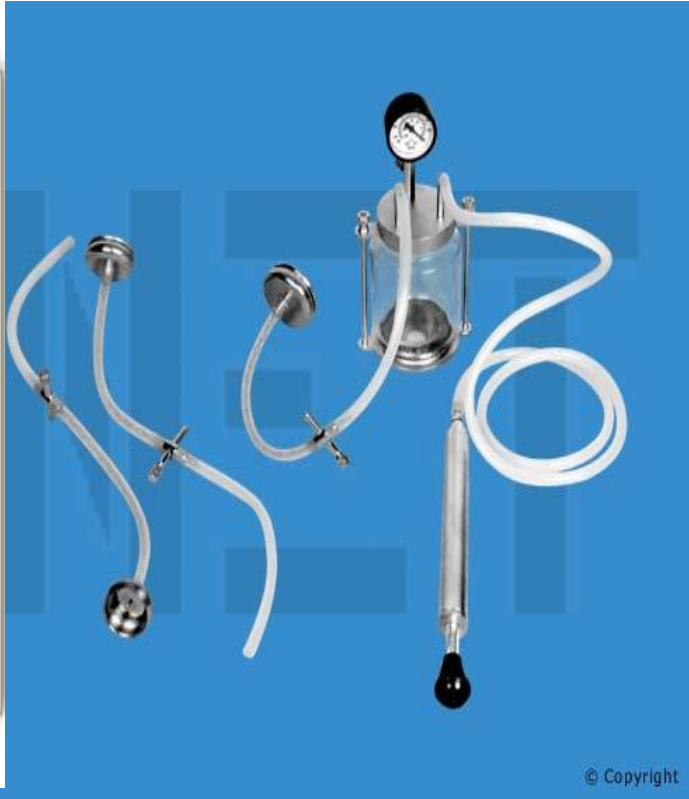
- Failure of the fetal head to rotate
- Fetal distress
- Should not be used for preterm, face presentation or breech

MNEMONIC

- **A – Anesthesia** ⇒ adequate
⇒ appropriate positioning & access
- **B – Bladder** ⇒ catheterization
- **C – Cervix** ⇒ fully dilated / membranes ruptured
- **D – Determine** ⇒ position
adequacy



- **E – Equipment** ⇒ inspect vacuum cup, pump, tubing,
⇒ check pressure



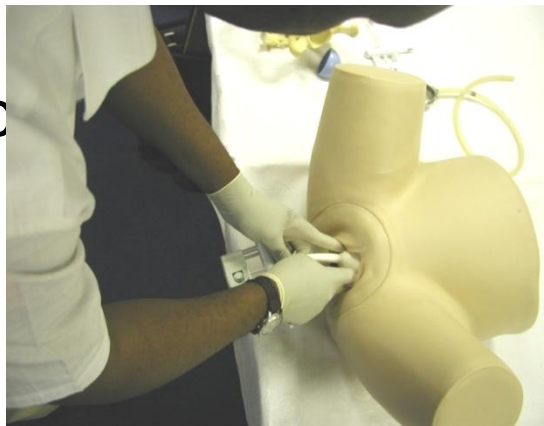
- **F – Fontanelle** ⇒ position the cup over the posterior fontan



⇒ -ve pressure ↑ 10 cm H₂O initially & between cont

⇒ sweep finger around cup to clear maternal tissue

cc 60 cm H

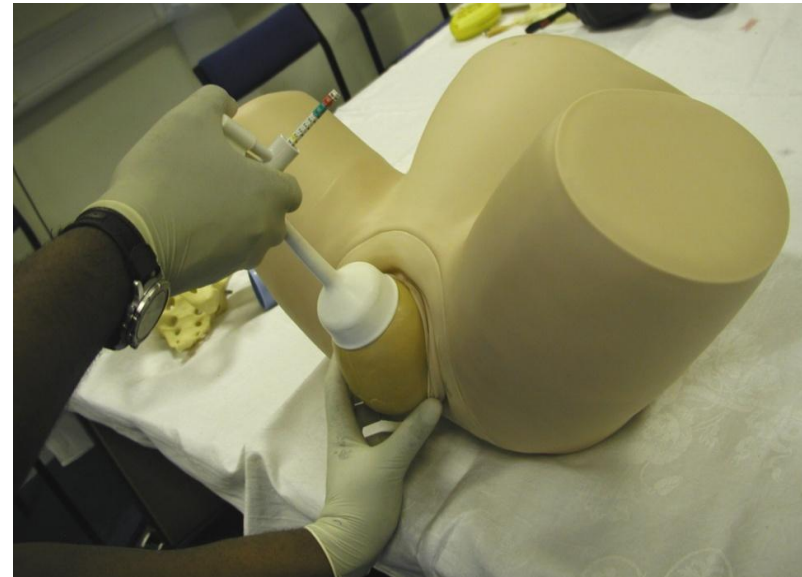
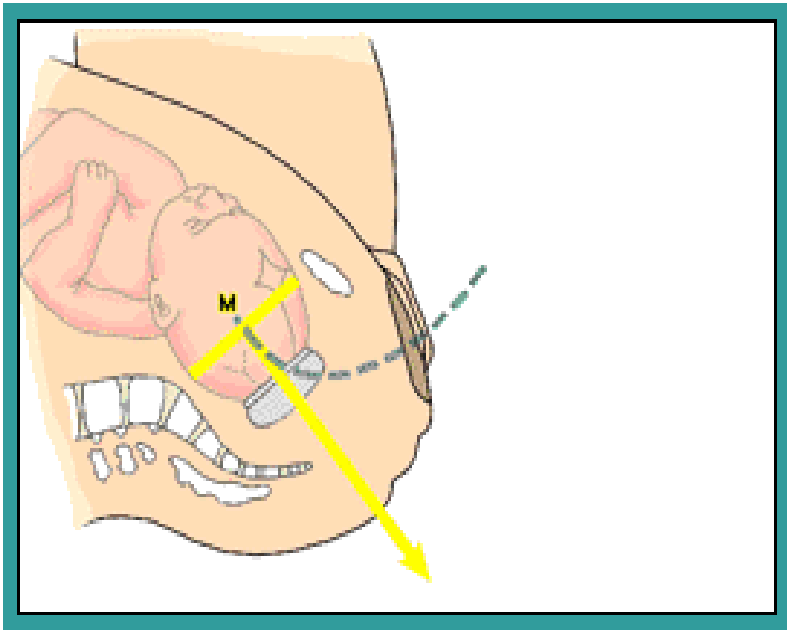


- **G – Gentle traction** ⇒ pull with contractions only\\

-

⇒ traction in the axis of the birth canal

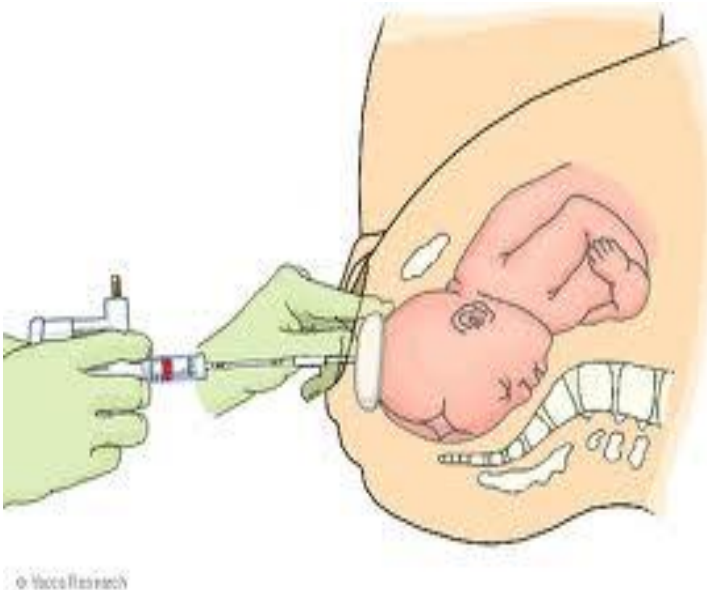
⇒ ask the mother to push during



- **H – Halt** ⇒ halt traction if no progress with three traction aided contractions

⇒ vacuum pops off three times

⇒ pulling for 30 min without significant



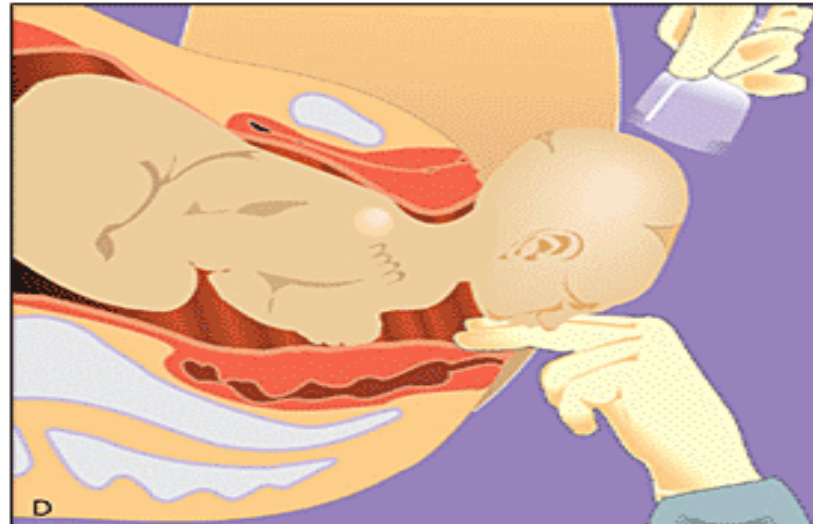
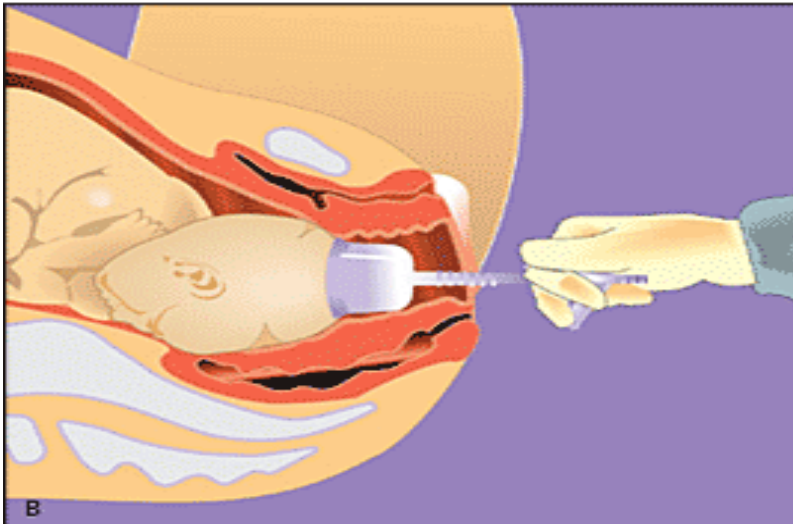
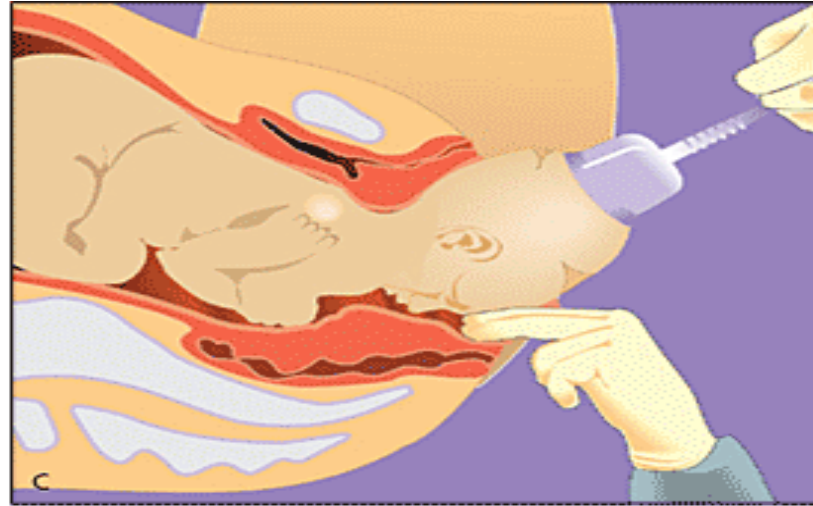
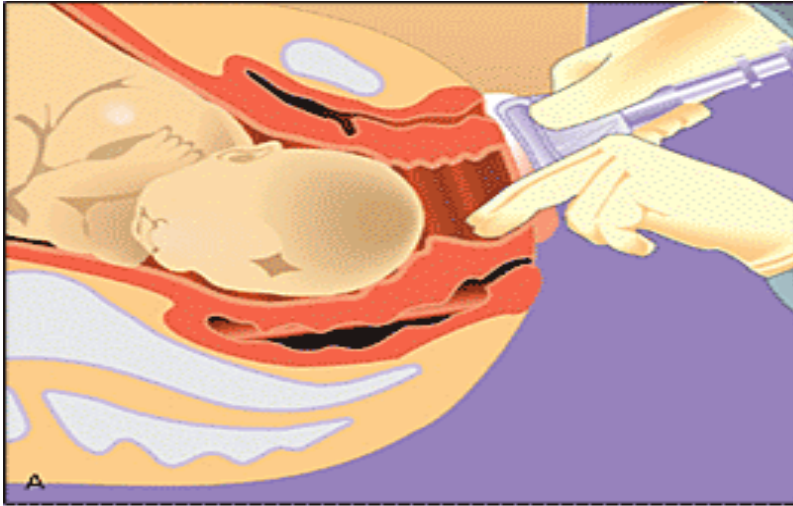
- **I – Incision** ⇒ consider episiotomy if laceration imminent



- **J – Jaw** ⇒ remove vacuum when jaw is reachable or delivery assured



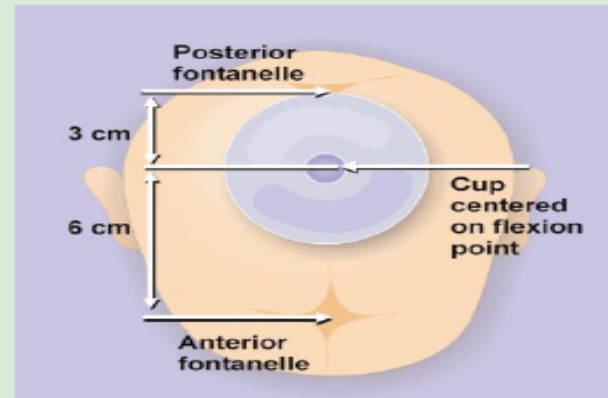
Steps of ventose application



VACUUM APPLICATION

A B C D E F G H I J

- A** Address the patient,
Ask for help,
Anesthesia adequate?
- B** Bladder empty?
- C** Cervix must be completely dilated
- D** Determine position of the head. Think of shoulder Dystocia. Review the Helperr mnemonic
- E** Equipment ready
Extractor ready
- F** Place cup in proper relation to Fontanelles on Flexion point



Proper cup placement on the flexion point is essential for safety and efficiency. Traction centered on the flexion point helps keep the neck flexed.

- G** Gentle traction following the pelvic curve, rising as head crowns



- H** Halt traction between contractions
Halt procedure if cup disengages 3 times
Halt if no progress in 3 pulls
Halt procedure after 20 minutes of use
 - ✓ Be prepared to abandon the procedure and move to cesarean.
 - ✓ Avoid prolonged use beyond the "Halt" guidelines.
 - ✓ Avoid pivoting and rocking motions.

- I** Incision: Evaluate for episiotomy when head is crowning. (May not be necessary)
- J** Remove the vacuum cup when the Jaw is reachable

Complications

- Vacuum –assisted delivery is less traumatic to the mother & fetus than forceps
- Ventouse should be the instrument of choice
- Maternal ⇒ Vaginal laceration due to entrapment of vaginal
mucosa between suction cup & fetal head

Fetal complications

- Scalp injuries $\Rightarrow$ chignon
 - $\Rightarrow$ abrasion & lacerations 12.6%
 - $\Rightarrow$ scalp necrosis 0.25-1.8%
- Cephalohematoma $\Rightarrow$ 25% $\Rightarrow$ jaundice /anemia
- Intracranial hemorrhage $\Rightarrow$ 2.5%
- Subgaleal hematoma



Fetal complications

- Birth asphyxia $\Rightarrow$ 2.6-12% $\Rightarrow$ related to extraction force & time

Some studies showed decrease birth asphyxia

- Retinal hemorrhage 50%
 Forceps 31%
 SVD 19%
- Neonatal jaundice

FORCEPS



Indications

MATERNAL

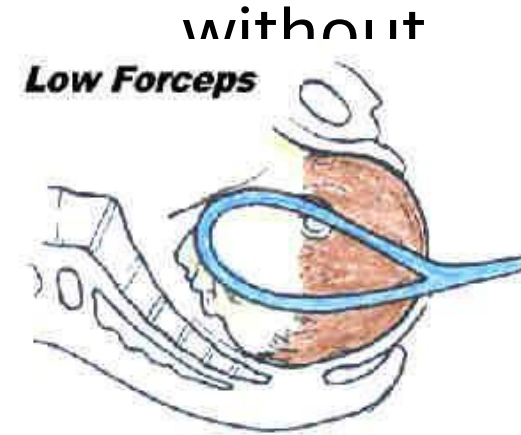
- Exhaustion
- Prolonged second stage
- Cardiac / pulmonary disease

FETAL

- Failure of the fetal head to rotate
- Fetal distress
- Control of the fetal head in vaginal breech delivery

Classification of forceps delivery

- **Outlet forceps** ⇒ Scalp visible at the vulva
without separating the labia



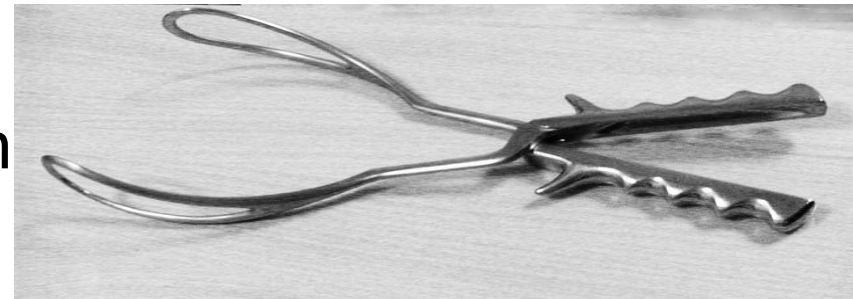
- **Low forceps** ⇒ Vertex at +2 station

- **Midforceps** ⇒ Head is engaged but leading part above +2 station
⇒ Sagittal suture not in of the mother



Classification Of Forceps Delivery

- Outlet ⇨ Wrigley's
- Outlet & low forceps ⇨ Simpson /Elliot
- Midforceps & outlet ⇨ Tucker Mclane
- Midforceps & rotation ⇨ Kielland
- After coming head in breech ⇨ Piper



After coming head in breech ⇨ Piper



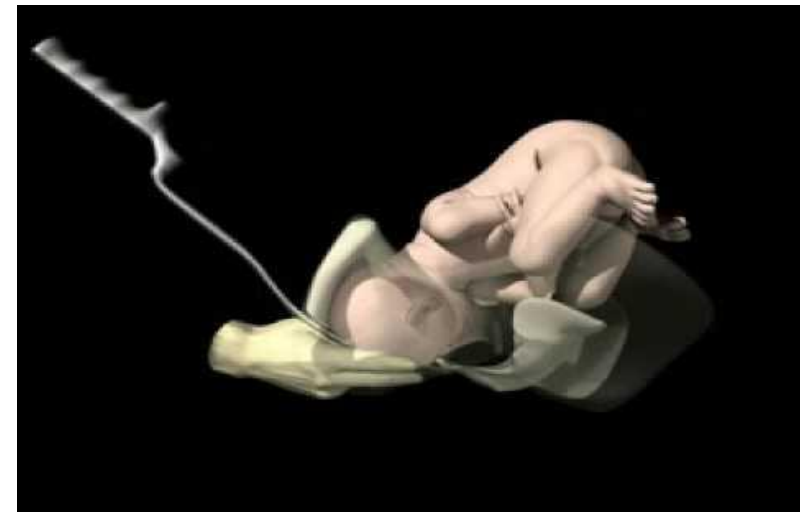
MNEMONIC

- A – Anesthesia ⇒adequate /epidural or pudendal
⇒ appropriate positioning & access
- B – Bladder ⇒ catheterization
- C – Cervix ⇒ fully dilated / membranes ruptured
- D – Determine ⇒ position, station, pelvic adequacy
- E – Equipment ⇒complete working forceps
⇒anesthesia support

- **F – Forceps** ⇒ phantom application

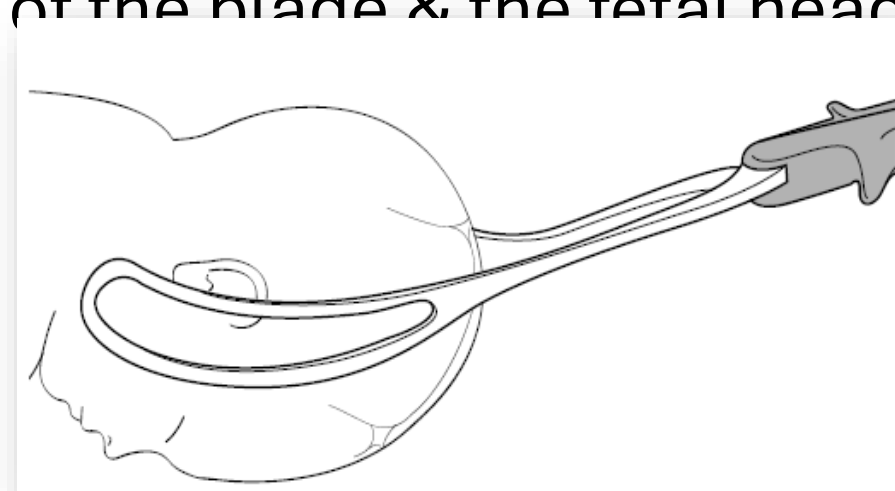
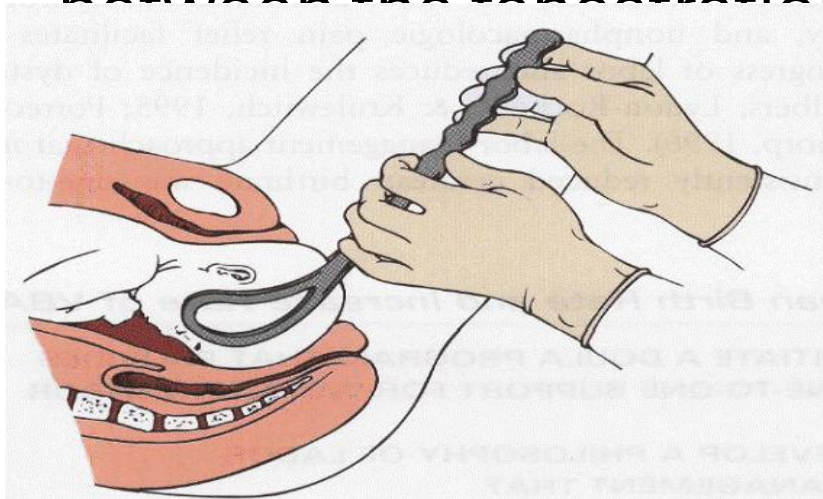
⇒ **Lt blade** , LT hand, maternal Lt side pencil grip & vertical insertion with Rt thumb directing blade

⇒ **Rt blade** , RT hand, maternal Rt side pencil grip & vertical insertion with Lt thumb directing blade



⇒ Check application:

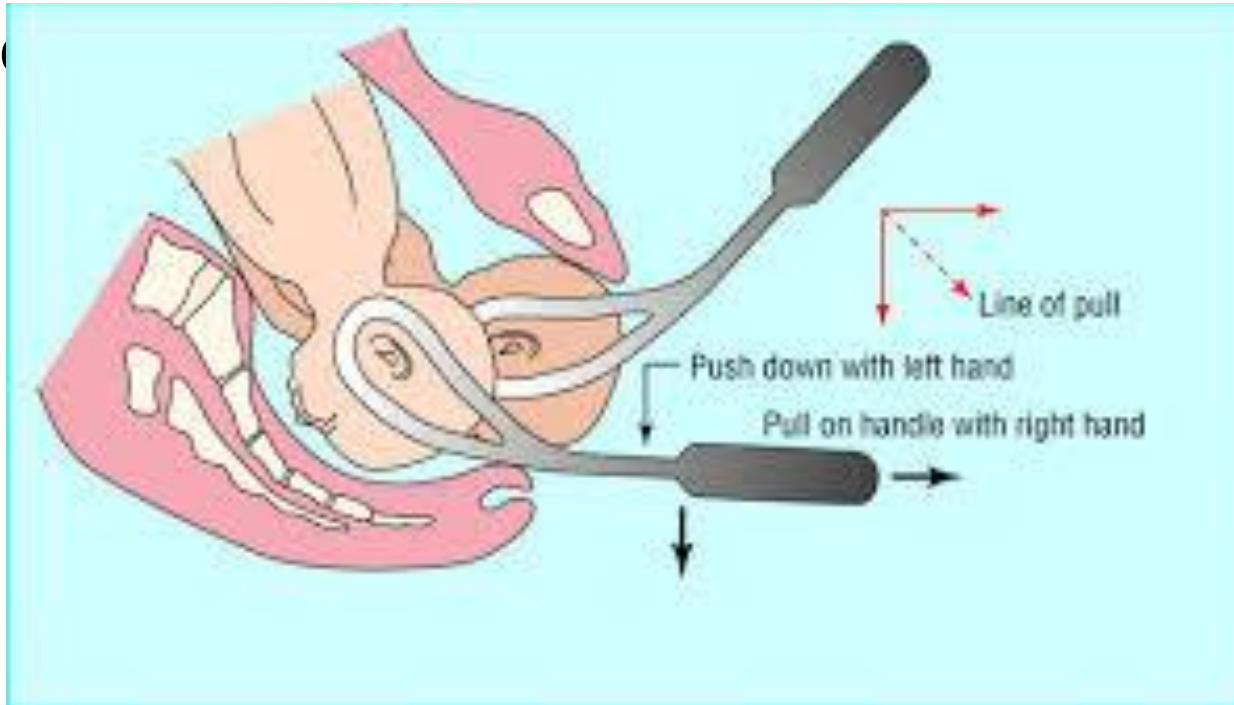
- Post fontanelle 1cm above the plane of the shanks
- Sagittal suture lies in the midline of the shanks /perpendicular to the plane of the shanks
- The operator can not place more than a fingertip between the fenestration of the blade & the fetal head



- **G – Gentle traction** ⇨ applied with contraction & maternal

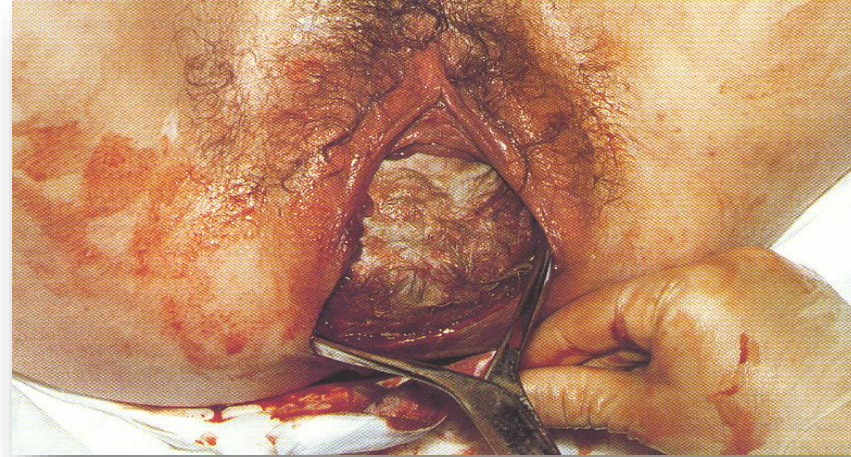
expulsive efforts

- **H – Handle elevated** ⇨ traction in the axis of the birth



le to early

- **I – Incision** ⇒ consider episiotomy if laceration imminent



- **J – Jaw** ⇒ remove for reachable or delivery a

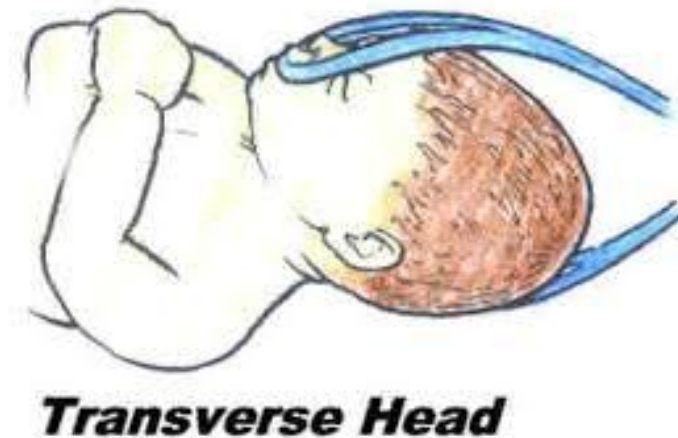
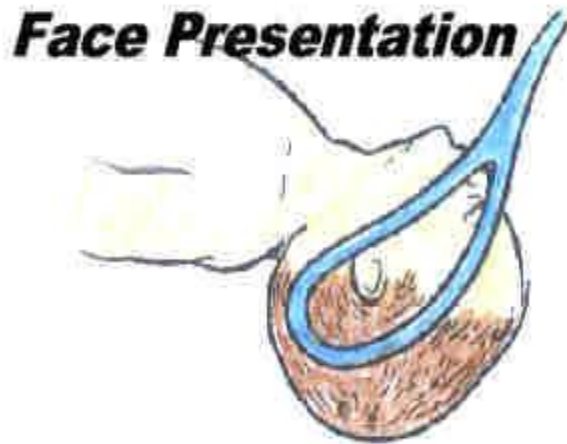


Complications

- Maternal ⇒ trauma to soft tissue ⇒ 3rd/4th degree
double the risk compared to ventouse
 - ⇒ bleeding from lacerations
 - ⇒ trauma to urethra & bladder ⇒ fistula
 - ⇒ Pain 17% ventouse 11%

Complications

- Fetal
 - ⇒ bruising & laceration to the face
 - ⇒ Injury to the fetal scalp
 - ⇒ cephalohematoma 9% Vent 25%
 - ⇒ retinal hemorrhage 30% Vent 50%
 - ⇒ skull fracture
 - ⇒ permanent nerve damage / Facial



CESAREAN SECTION

INDICATIONS FOR ELECTIVE CS

- Repeat CS
- Placenta previa
- VV fistula repair
- HIV (poor controlled)
- Active herpes
- Fetal macrosomia > 4500 gm
- Uterine surgery eg. Hystrotomy, myomectomy
- Severe IUGR
- Breech
- Multiple pregnancy
- Transverse lie
- Ca of the Cx/ TR obstructing the birth canal

INDICATIONS FOR EMERGENCY CS

- Severe PET
- Abruptio placenta (APH)
- Fetal distress
- Failure to progress in the first stage of labour
- Cord prolapse
- Obstructed labour
- Failed induction
- Malpresentation => brow, chin post, shoulder & compound presentations, breech
- Compromised fetus 2ry to DM, HPT, isoimmunization

COMPLICATIONS

INTRAOPERATIVE

- Bleeding & the need for bl transfusion
- Hysterectomy
- Complications of anaesthesia
- Damage to the bladder, ureter, colon , retained placental tissue
- Fetal injury

POSTOPERATIVE

- Paralytic ileus
- Wound dehiscence & infection
- Infections => UTI, pneumonia
- DVT & pulmonary embolism
- Fistula
- Death

Contraindications

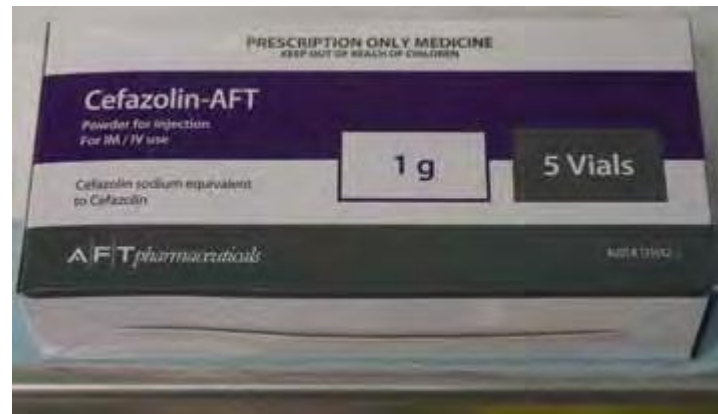
- Previous classical CS
- 2 or more previous CS
- Previous other uterine surgery
- Hx of scar rupture
- Placenta previa or transverse lie

Caesarian section steps

- *Preoperative steps: this patient was undergoing an emergency caesarean section. In this instance it is now our practice to prep the vagina with iodine solution. An IDC was already in situ as the patient had an epidural in labour. TED stockings have been placed on the patient for VTE prophylaxis.*



IV antibiotics: currently the Therapeutic Guidelines recommend 2 g
IV Cephazolin as routine prophylaxis 15–60 minutes prior to skin
incision



- *Skin preparation for caesarean section with chlorhexidine-alcohol prep, which should be allowed to dry prior to draping. It is important that solution does dry and doesn't pool underneath the drapes, as this is a fire risk and patient burn*



- *Skin incision: the Joel Cohen technique involves making a straight incision 3 cm below the level of the anterior superior iliac spines*



- *Entry technique: sharp entry through the skin, middle 3 cm of the subcutaneous fat and rectus sheath is demonstrated here.*

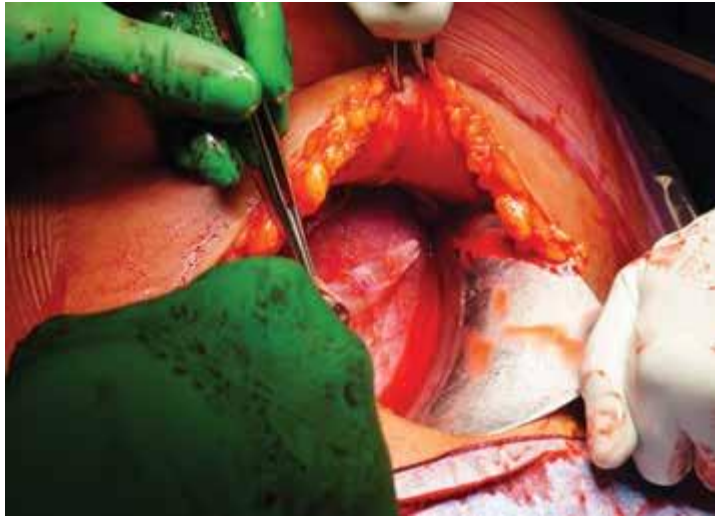


- *Blunt extension of the subcutaneous tissue and rectus sheath.*



Blunt entry into the peritoneal cavity is used as a part of the Joel Cohen caesarean section technique. This should be done high in order to avoid entering into the bladder, which may be high following prior CS or in the advanced stages of labour





Creation of a bladder flap. The loose utero-vesical peritoneum is identified. It should be opened approximately 2 cm below the level of its fixed attachment to the uterus in the midline and extended laterally each side. The peritoneum can then be picked up with forceps and the bladder gently separated from the lower segment bluntly with the forefinger, or in the presence of adhesions sharply reflected down.

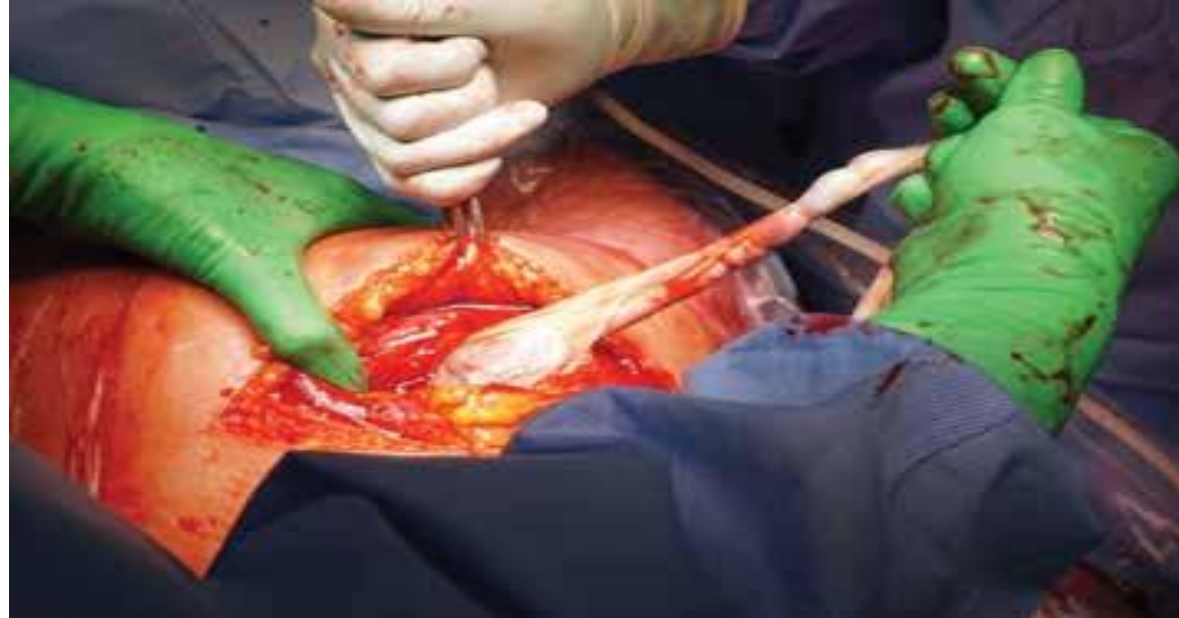
- *Uterine entry: this demonstrates cephalad-caudad blunt extension of the uterine incision performed after making a small 2–3 cm horizontal sharp incision on the lower uterine segment.*





Delivery of the fetal head is usually achieved with flexion and elevation of the fetal head toward the uterine incision, and then completed with the addition of the assistant giving fundal pressure (10b). In photo 10a, forceps delivery of the fetal head is demonstrated, which may be necessary in the elective setting where the fetal head can still be high.

- *Spontaneous delivery of the placenta: fundal massage and controlled cord traction are being used here to achieve spontaneous delivery of the placenta*



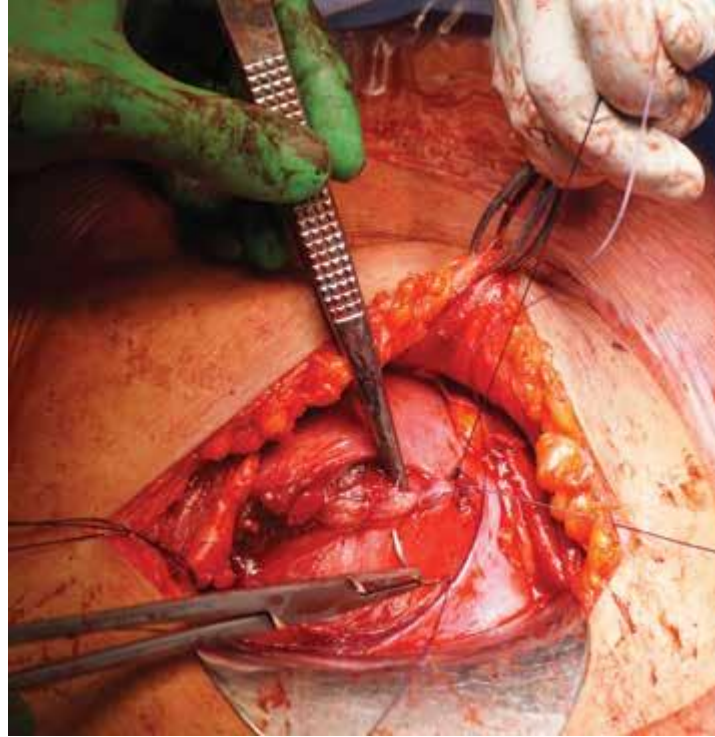
- *An oxytocic is usually given following the delivery of the baby to reduce the risk of PPH, and here is shown an oxytocin infusion, as is current evidence-based practice*



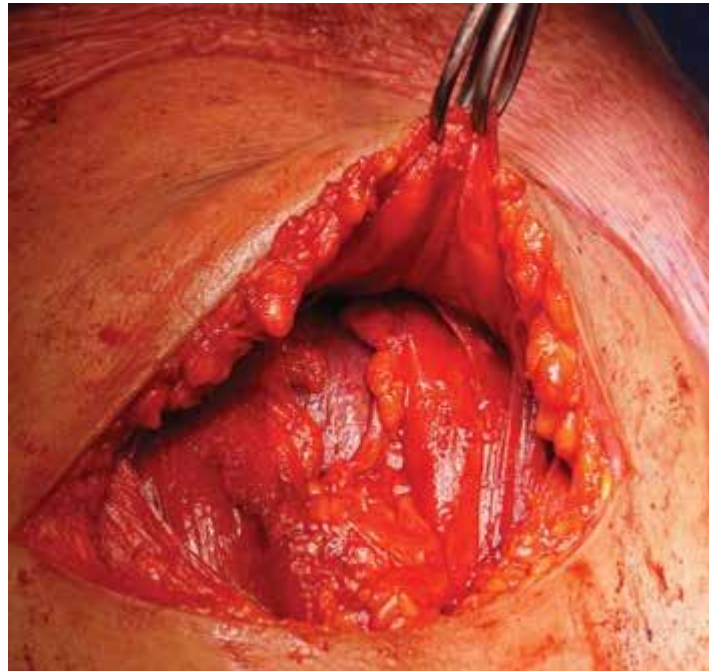
- *Identifying the uterine incision and uterine angles: it is useful to place Green-Armytage forceps on the upper and lower edges of the uterine incision to ensure they are identified correctly (particularly the lower edge, which can at times recede inferiorly and be difficult to identify because of bleeding, which has led some to mistake the posterior wall of the lower segment for the lower edge of the uterine incision). This practice also allows clear identification of the uterine angles that are often secured*



- *Closure of the uterus: here the uterus is closed with a double-layer, non-locking continuous monofilament (1 monocryl) suture. The first layer should include the cut edge of the myometrium and achieves haemostasis. The second layer pulls uncut myometrium together in order to cover*



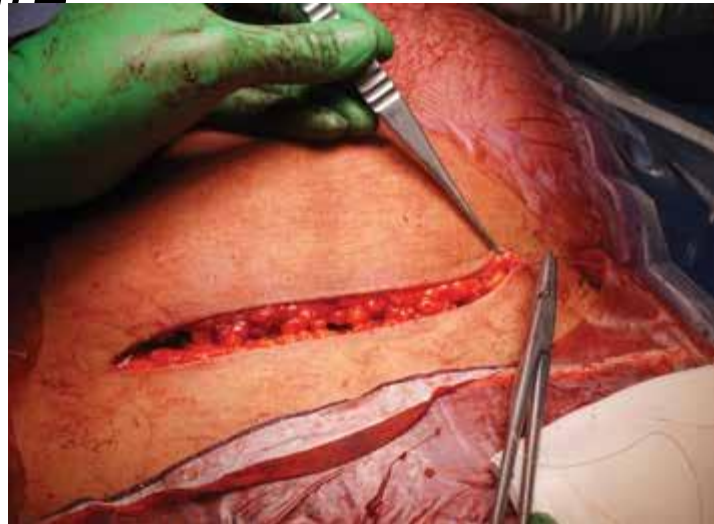
- . *Non-closure of the peritoneum. Haemostasis between the rectus sheath and muscle should be checked at this point because of the risk of injury to perforating vessels during entry.*



- *Rectus sheath closure is demonstrated here with a 1 PDS suture, using a continuous non-locking technique*



- *The subcutaneous fat in this instance has not been closed as it is less than 2 cm. A continuous subcutaneous suture is used for skin closure*





Thank you

